

Department of Physics and Astronomy
University of Heidelberg

Master Thesis in Physics
submitted by

Pranavandhan Upendranath

born in Coimbatore, Tamil Nadu, India

2026

Evolution of One-Dimensional Stellar Merger Products

This Master Thesis has been carried out by Pranavandhan Upendranath
at the
Heidelberg Institute for Theoretical Studies
under the supervision of
Priv.-Doz. Dr. Fabian Schneider
second examiner
Prof. Dr. Friedrich Roepke

Abstract

We investigate the evolution of one-dimensional stellar merger products formed in massive binary systems, focusing on the $M_{1,i} = 10.16 M_{\odot}$ slice of the binary grid of Henneco et al. (2024b). We identify 100 Case A, Case AB, and Case Be systems that reach contact and are expected to merge. The properties of the merger progenitors at merger differ substantially from those at the onset of mass transfer, both in mass and evolutionary state. We construct the merger products using two one-dimensional prescriptions, entropy sorting and PyMMAMS. These prescriptions differ qualitatively for post-main-sequence + main-sequence mergers. A three-dimensional smoothed-particle hydrodynamics calculation of one such configuration supports the entropy-sorting approach. We then evolve the merger products with MESA until central helium exhaustion.

We find two merger channels that populate the blue and yellow supergiant region in the Hertzsprung–Russell diagram (HRD) on a nuclear, rather than thermal, timescale. Post-main-sequence + main-sequence mergers ignite helium on the hot side of the HRD, with $\log_{10}(T_{\text{eff}}/\text{K}) > 3.7$, and remain there. In contrast, a subset of main-sequence mergers evolve to the cool side before helium ignition and later execute blue loops. These two channels are largely indistinguishable by their position in the HRD alone. In most cases, the merger alters the observable appearance of the star while leaving the carbon–oxygen core mass close to that of a single star of the same post-merger mass. Both channels plausibly contribute to the observed blue-supergiant population.

Zusammenfassung

Wir untersuchen die Entwicklung eindimensionaler Modelle von Sternverschmelzungsprodukten, die in massereichen Doppelsternsystemen entstehen. Dabei konzentrieren wir uns auf den Teil des Doppelsternrasters von Henneco et al. (2024b) mit einer anfänglichen Primärmasse von $M_{1,i} = 10.16 M_{\odot}$. Wir identifizieren 100 Systeme der Fälle A, AB und Be, die in Kontakt kommen und voraussichtlich verschmelzen. Die Eigenschaften der Verschmelzungsvorläufer zum Zeitpunkt des Kontakts unterscheiden sich sowohl hinsichtlich ihrer Masse als auch ihres Entwicklungszustands deutlich von denen zu Beginn des Massentransfers. Wir konstruieren die Sternverschmelzungsprodukte mit zwei eindimensionalen Verfahren: der Entropiesortierung und PyMMAMS. Diese Verfahren liefern für die Verschmelzung eines Sterns, der die Hauptreihe bereits verlassen hat, mit einem Hauptreihenstern qualitativ unterschiedliche Ergebnisse. Eine dreidimensionale Simulation einer solchen Konfiguration mittels geglätteter Teilchenhydrodynamik stützt den Ansatz der Entropiesortierung. Anschließend entwickeln wir die Sternverschmelzungsprodukte mit MESA bis zum Ende des zentralen Heliumbrennens.

Wir finden zwei Verschmelzungskanäle, die den Bereich der blauen und gelben Überriesen im Hertzsprung-Russell-Diagramm auf einer nuklearen statt auf einer thermischen Zeitskala besiedeln. Bei Verschmelzungen eines Sterns, der die Hauptreihe bereits verlassen hat, mit einem Hauptreihenstern setzt die Heliumfusion auf der heißen Seite des Hertzsprung-Russell-Diagramms bei $\log_{10}(T_{\text{eff}}/\text{K}) > 3.7$ ein. Die Verschmelzungsprodukte verbleiben anschließend dort. Im Gegensatz dazu entwickeln sich einige Verschmelzungsprodukte zweier Hauptreihensterne vor Beginn des Heliumbrennens zunächst in den kühlen Bereich des Hertzsprung-Russell-Diagramms und durchlaufen später eine blaue Schleife. Diese beiden Entwicklungskanäle lassen sich allein anhand ihrer Position im Hertzsprung-Russell-Diagramm kaum voneinander unterscheiden. In den meisten Fällen verändert die Verschmelzung das beobachtbare Erscheinungsbild des Sterns, während die Masse des Kohlenstoff-Sauerstoff-Kerns nahezu derjenigen eines Einzelsterns mit derselben Masse nach der Verschmelzung entspricht. Beide Kanäle stellen daher plausible Beiträge zur beobachteten Population blauer Überriesen dar.

Contents

1	Introduction	1
1.1	Stellar structure and evolution	2
1.1.1	The equations of stellar structure	2
1.1.2	Characteristic timescales	3
1.1.3	Energy transport, convective stability and mixing	4
1.1.4	Entropy as a structural variable	5
1.1.5	The evolution of a $10 M_{\odot}$ star	6
1.2	Binary stars	6
1.2.1	The Roche potential and the Roche-lobe radius	6
1.2.2	Mass transfer and its classification	7
1.2.3	Stability of mass transfer	8
1.2.4	Physical mechanisms leading to contact	9
1.2.5	From contact to a merged star	10
1.3	Constructing merger products in one dimension	10
1.4	The blue supergiant problem and the role of binary evolution	11
2	Methods	13
2.1	Binary grid setup	13
2.1.1	Termination of the binary calculations	13
2.1.2	Merger criteria and choice of progenitor models	14
2.1.3	Adopted stellar physics	15
2.2	One dimensional stellar merger prescriptions	15
2.2.1	(Python) Make Me A Massive Star	15
2.2.2	Entropy Sorting	18
2.3	Three-dimensional hydrodynamical simulation	19
2.3.1	Code and numerical configuration	19
2.3.2	Progenitor models	20
2.3.3	Relaxation and binary setup	20
2.3.4	Treatment of the artificial conductivity	22
2.3.5	Post-merger relaxation and merger product analysis	22
2.3.6	Computational cost	22
2.4	MESA	23
2.4.1	Engineering	23
2.4.2	Relax	24
2.4.3	Evolve	26
2.4.4	Classification of post-main-sequence evolution	26

3	Contact tracing and merger outcome classification	28
3.1	Formation of contact phases and the structures of stars at merger . . .	28
3.1.1	Binary systems that lead to a stellar merger	28
3.1.2	Merger progenitor properties	30
3.1.3	A system terminated by numerical failure	33
3.2	Initial evolutionary state of the merger products	33
3.2.1	Classification of merger products: passive scalar and origin of the merged star's core	33
3.2.2	Origin of the merged star's core	35
3.2.3	Evolutionary state of the merger product	38
4	Three-dimensional hydrodynamical simulations	39
4.1	In-spiral and merger	40
4.2	Structure of the merged star	41
4.3	Comparison with the one-dimensional prescriptions	42
4.4	Sensitivity to the artificial conductivity	44
5	Long-term evolution of the merger products	44
5.1	Case A mergers	45
5.2	Case Be mergers	47
5.3	Case AB mergers	49
5.3.1	MS + post-MS mergers	50
5.3.2	Post-MS + post-MS mergers	51
5.4	Occurrence of long-lived BSGs and blue-loop stars from stellar mergers	52
5.5	Distribution of merged stars across the HR diagram	55
5.6	Carbon–oxygen core masses	59
6	Discussion	63
6.1	Merger progenitors properties	63
6.2	Merger prescription choice	64
6.3	Three-dimensional verification of the prescriptions	66
6.4	MS + MS mergers: rejuvenation and the origin of the blue loops	68
6.5	Post-MS + MS mergers as long-lived blue and yellow supergiants	69
6.6	Post-MS + post-MS mergers	70
6.7	Two routes into the same part of the HR diagram	70
6.8	Distribution across the HR diagram and comparison with observations	71
6.9	The cores of the merged stars	72
6.10	Limitations	73
7	Conclusions	74

8 Acknowledgments	78
8.1 Tools used for analysis and visualization	79
Appendices	80
A Figures and tables	80
B Additional figures	84
C MESA inlist	87

1 Introduction

Most massive stars are not born alone. Observations over the past decade and a half have strongly established that most stars, especially those more massive than our Sun, don't live alone (Sana et al., 2012). This multiplicity fraction rises steeply with mass, so that by $\sim 10 M_{\odot}$ most stars have at least one companion (Moe and Di Stefano, 2017; Offner et al., 2023). Therefore, binary interaction is not a correction factor but a standard evolutionary path of a massive star (Langer, 2012; Marchant and Bodensteiner, 2024). Interpreting these stars' fates requires knowing how the companion has affected their evolution. The possible outcomes of an interaction span a wide range. If the stars are sufficiently close, the more evolved star can fill its Roche lobe and transfer mass to its companion through the inner Lagrange point, fundamentally altering the subsequent evolution of both stars. If mass transfer proceeds stably, the donor is stripped of most of its hydrogen-rich envelope, and the accretor spins up and is rejuvenated, leaving a binary whose components look nothing like single stars of their masses. If mass transfer is unstable, or if the accretor swells enough to fill its own Roche lobe, the two stars come into contact and share an envelope, and the system either ejects that envelope and survives as a much closer binary or fails to eject it and the two stars merge. Sana et al. (2012) infer that 20–30 % of massive binaries merge, and the population synthesis calculations of de Mink et al. (2014) predict that ~ 8 % of all early-type stars at any given moment are the products of a stellar merger.

Merger products have been used to explain blue straggler stars in clusters, whose masses exceed the main-sequence (MS) turn-off mass (Hills and Day, 1976; Sills et al., 1997; Glebbeek et al., 2008). The blue supergiant progenitor of SN 1987A (Podsiadlowski et al., 1990; Morris and Podsiadlowski, 2007) and, more generally, the population of long-lived blue supergiants (BSGs) that classical single-star evolution struggles to produce in the observed numbers can be explained by mergers (Menon and Heger, 2017; Menon et al., 2024; Schneider et al., 2024; Bellinger et al., 2024).

Rapid population synthesis codes determine whether a system merges based on the mass ratio evaluated at the onset of Roche-lobe overflow, and then construct the merged star using an analytic recipe applied to the pre-interaction masses (de Mink et al., 2013; Schneider et al., 2014). Detailed 3D hydrodynamical merger simulations exist (Gaburov et al., 2008; Schneider et al., 2019). Still, they are far too expensive to be run for a grid of systems and cannot be followed for more than a few dynamical timescales, so the long-term evolution of the merged star must still be computed in one dimension (1D). The step that connects the two is turning a pair of 1D progenitor models into a single 1D model that can be handed to a stellar evolution code to be evolved further.

The remainder of this chapter provides the background needed for what follows.

Sect. 1.1 describes the stellar structure and evolution, with an emphasis on the timescale hierarchy, the criteria for convective stability, and the role of entropy as a structural variable. Sect. 1.2 introduces binary evolution, the classification of mass transfer, and the mechanisms that drive a system into contact and towards a merger. Sect. 1.3 sets out the physical ideas behind the two 1D merger prescriptions used in this work, entropy sorting (ES) and PyMMAMS. Sect. 1.4 states the observational problem that motivates the thesis.

1.1 Stellar structure and evolution

1.1.1 The equations of stellar structure

A star in which spherical symmetry is assumed while rotation and magnetic fields are neglected is described by four coupled differential equations. Written in Lagrangian form, with the mass coordinate m enclosed by a radius sphere r as the independent variable, they are (Kippenhahn et al., 2012)

$$\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2 \rho}, \quad (1.1)$$

$$\frac{\partial P}{\partial m} = -\frac{Gm}{4\pi r^4} - \frac{1}{4\pi r^2} \frac{\partial^2 r}{\partial t^2}, \quad (1.2)$$

$$\frac{\partial L}{\partial m} = \varepsilon_{\text{nuc}} - \varepsilon_{\nu} - T \frac{\partial s}{\partial t}, \quad (1.3)$$

$$\frac{\partial T}{\partial m} = -\frac{GmT}{4\pi r^4 P} \nabla, \quad (1.4)$$

where ρ is the density, P the total pressure, T the temperature, L the luminosity, s the specific entropy, G the gravitational constant, ε_{nuc} the specific energy generation rate by nuclear burning, ε_{ν} the specific energy loss rate carried away by neutrinos, and

$$\nabla \equiv \frac{d \ln T}{d \ln P} \quad (1.5)$$

the actual temperature gradient of the stratification. Equation (1.1) is mass conservation, Eq. (1.2) is the momentum equation, whose second term on the right-hand side vanishes in hydrostatic equilibrium, Eq. (1.3) is energy conservation, and Eq. (1.4) describes energy transport.

The last term of Eq. (1.3) is the gravothermal energy generation rate, usually written as

$$\varepsilon_{\text{grav}} = -T \frac{\partial s}{\partial t}, \quad (1.6)$$

and its integral over the star gives the gravitational luminosity L_{grav} . A star in thermal equilibrium has $\varepsilon_{\text{grav}} \simeq 0$ and radiates exactly what it generates by nuclear burning. A

star that is contracting or expanding does not, and $|L_{\text{grav}}|$ measures how far it is from thermal equilibrium.

1.1.2 Characteristic timescales

The outcome of binary interactions, including mass transfer and stellar mergers, depends critically on how rapidly a star can restore hydrostatic equilibrium, thermal equilibrium, or evolve through nuclear burning. These processes are characterised by the dynamical, thermal, and nuclear timescales, respectively.

The *dynamical timescale* is the time it takes a star to restore hydrostatic equilibrium and is set by the free-fall or sound-crossing time.

$$\tau_{\text{dyn}} \simeq \sqrt{\frac{R_\star^3}{GM_\star}} \propto \bar{\rho}^{-1/2}, \quad (1.7)$$

where $M_\star$ and $R_\star$ are the stellar mass and radius and $\bar{\rho}$ the mean density. For a $10 M_\odot$ MS star, this is of the order of hours.

The *thermal* or Kelvin-Helmholtz *timescale* is the time in which a star radiates away its gravitational binding energy, and hence the time it needs to restore thermal equilibrium after being driven out of it,

$$\tau_{\text{KH}} \simeq \frac{GM_\star^2}{2R_\star L_\star}, \quad (1.8)$$

with $L_\star$ the stellar luminosity. For a $10 M_\odot$ MS star τ_{KH} is of the order of 10^4 – 10^5 yr, and it shortens as the star expands and brightens.

The *nuclear timescale* is the time over which the star can sustain its luminosity by burning the fuel available to it,

$$\tau_{\text{nuc}} \simeq \frac{q_{\text{nuc}} f_c X M_\star c^2}{L_\star}, \quad (1.9)$$

where X is the hydrogen mass fraction, $q_{\text{nuc}} \approx 0.007$ is the fractional rest-mass energy released in the conversion of hydrogen into helium, f_c is the fraction of the stellar mass that participates in the burning, and c is the speed of light. For a $10 M_\odot$ star the core-hydrogen-burning lifetime is a few times 10^7 yr.

The ordering

$$\tau_{\text{dyn}} \ll \tau_{\text{KH}} \ll \tau_{\text{nuc}} \quad (1.10)$$

holds throughout a massive star's evolution and cleanly separates the physics. The merger itself is a dynamical event, lasting tens of orbits. The subsequent readjustment of the merged star, during which it expands and then contracts, proceeds on τ_{KH} (Leonard and Livio, 1995; Schneider et al., 2019). Everything that follows, and

everything an observer is likely to catch the star doing, happens on τ_{nuc} .

1.1.3 Energy transport, convective stability and mixing

Energy transport governs a star's internal temperature gradient, and thus its pressure and density stratification, structure, evolution, and observable properties. If energy is transported by radiation alone, the temperature gradient defined in Eq. (1.5) takes the value

$$\nabla_{\text{rad}} = \frac{3}{16\pi acG} \frac{\kappa LP}{mT^4}, \quad (1.11)$$

with a the radiation constant. Where ∇_{rad} becomes large, because the opacity is high or because a large luminosity has to be carried through a small mass coordinate, the stratification becomes unstable to convection, and energy is transported by bulk motions instead. The classical criterion for the onset of convection, due to Schwarzschild (1958), compares ∇_{rad} with the adiabatic gradient $\nabla_{\text{ad}} \equiv (\partial \ln T / \partial \ln P)_s$; a layer is unstable if

$$\nabla_{\text{rad}} > \nabla_{\text{ad}}. \quad (1.12)$$

Equation (1.12) assumes that the gas is chemically homogeneous. It is not the case in an evolved star, where nuclear burning leaves behind a mean molecular weight μ that decreases outwards. A rising fluid element then carries heavy material into lighter surroundings, which restores it. Accounting for this stabilising effect gives the criterion of Ledoux (1947), according to which a layer is convectively unstable if

$$\nabla_{\text{rad}} > \nabla_{\text{ad}} + \frac{\varphi}{\delta} \nabla_{\mu}, \quad (1.13)$$

where

$$\nabla_{\mu} \equiv \frac{d \ln \mu}{d \ln P}, \quad \varphi \equiv \left(\frac{\partial \ln \rho}{\partial \ln \mu} \right)_{P,T}, \quad \delta \equiv - \left(\frac{\partial \ln \rho}{\partial \ln T} \right)_{P,\mu}. \quad (1.14)$$

For an ideal gas $\varphi = \delta = 1$. Since $\nabla_{\mu} \geq 0$ wherever burning has enriched the interior, the Ledoux criterion is more restrictive than the Schwarzschild criterion, and a steep composition gradient such as the one at the boundary of a helium core can stabilise a layer that Eq. (1.12) would declare convective. This distinction matters directly for merger products, whose interiors carry composition gradients inherited from two different progenitors.

Regions that are unstable by Schwarzschild but stable by Ledoux are not inert. They are subject to *semiconvection*, a slow, doubly diffusive mixing process whose efficiency is parametrised in stellar evolution codes by a dimensionless factor α_{sc} that is not well constrained (Langer et al., 1985; Schootemeijer et al., 2019). Semiconvection controls whether a convective core can grow across a pre-existing μ gradient, and therefore how completely an accreting or merged star can mix fresh hydrogen into its burning

core (Braun and Langer, 1995; Schneider et al., 2024). The mirror-image situation, in which μ increases outward in an otherwise stable region, drives *thermohaline mixing* (Kippenhahn et al., 1980). Both processes appear in the merger products studied here, since stacking material from two stars can produce either sign of ∇_μ .

Convective energy transport itself is treated with the mixing-length theory of Böhm-Vitense (1958) and Cox and Giuli (1968), in which the mean free path of a convective element is set to $\ell = \alpha_{\text{mlt}} H_P$, with H_P the local pressure scale height and α_{mlt} a free parameter of order unity. Convective elements do not stop abruptly at the formal boundary but instead overshoot into the stable region beyond it, extending the mixed core and thus the MS lifetime (Herwig, 2000). The extent of overshooting is likewise parametrised rather than derived. The choice of convective criterion, of α_{sc} and of the overshooting parameter choices alone can move the predicted effective temperature of a post-MS massive star across a large part of the HR diagram.

1.1.4 Entropy as a structural variable

The criteria for energy transport in the previous section can be recast in terms of specific entropy, which is what makes the 1D merger prescriptions of Sect. 1.3 possible at all. In a chemically homogeneous region, a stratification is stable against convection if and only if the specific entropy increases outwards,

$$\frac{ds}{dm} > 0, \quad (1.15)$$

and a convective region, being efficiently mixed, is nearly isentropic. A star in hydrostatic equilibrium has a monotonically increasing entropy profile, with flat plateaus wherever convection operates (Lombardi et al., 1996). Material with low specific entropy sits at the bottom, and material with high specific entropy sits on top.

For an ideal, fully ionised monatomic gas, the specific entropy follows from the Sackur-Tetrode relation and scales as

$$s \propto \frac{1}{\mu} \ln \left(\frac{T^{3/2}}{\rho} \right), \quad (1.16)$$

so that s decreases when the mean molecular weight increases and when the gas is compressed. Both happen in the core of an evolving star: hydrogen burning raises μ_c from ≈ 0.6 towards ≈ 1.3 , and the core contracts and heats as it does so. *The more evolved a star is, the lower the specific entropy of its core.*

In practice it is convenient to work not with s itself but with an entropic variable A , a quantity that is a monotonic function of the specific entropy and is cheaper to evaluate. For an ideal gas $A = P/\rho^{5/3}$; the generalisation used by Gaburov et al. (2008), which includes the contribution of radiation pressure, is given in Sect. 2.2.1,

together with the explicit expressions adopted by each of the two merger prescriptions. The essential point here is that ordering stellar material by A and ordering it by s amount to the same operation, and that both are statements about buoyancy.

1.1.5 The evolution of a $10 M_{\odot}$ star

A $10 M_{\odot}$ star of solar metallicity arrives on the zero-age main sequence (ZAMS) burning hydrogen through the CNO cycle. The strong temperature sensitivity of the CNO reactions concentrates the energy generation in a small central volume, which makes ∇_{rad} large there and produces a convective core surrounded by a radiative envelope. As hydrogen is converted into helium, μ_c rises and the core must contract and heat to maintain hydrostatic support, so the luminosity increases and the star moves upwards and slightly to the right in the HRD. The convective core recedes in mass coordinate during this process, leaving behind a composition gradient in the layers it abandons.

At core hydrogen exhaustion, the terminal-age main sequence (TAMS), the star loses its central energy source. The helium core contracts on the thermal timescale τ_{KH} , hydrogen ignites in a shell around it, and the envelope expands in response (mirror principle). The star crosses the HG (Hertzsprung Gap) at nearly constant luminosity, reaching effective temperatures of a few thousand kelvin, and it does so on a thermal timescale. Helium is ignited in the core once the central temperature reaches $\sim 10^8$ K, typically when the star is already close to the Hayashi limit, and core helium burning then proceeds on a nuclear timescale with the star sitting as a red giant with a deep convective envelope.

For some combinations of metallicity, mixing physics and input variables, the star does not stay *red* during core helium burning but undergoes a *blue loop*. During this phase, the star moves back to higher effective temperatures, spends a substantial fraction of its helium-burning lifetime there, and returns redward before central helium exhaustion. Whether a loop occurs, and how far *blueward* it extends, is still not very certain in predictions of stellar evolution theory, since it depends on the treatment of semiconvection, overshooting, the opacities and the metallicity (Walmswell et al., 2015).

1.2 Binary stars

1.2.1 The Roche potential and the Roche-lobe radius

Consider two stars of masses M_1 and M_2 in a circular orbit of separation a_{orb} , and suppose for the moment that they can be treated as point masses. In a frame co-

rotating with the orbit at the angular frequency

$$\Omega_{\text{orb}} = \sqrt{\frac{G(M_1 + M_2)}{a_{\text{orb}}^3}}, \quad (1.17)$$

a test particle at position $\vec{r}$ experiences the Roche potential

$$\Phi_{\text{R}}(\vec{r}) = -\frac{GM_1}{|\vec{r} - \vec{r}_1|} - \frac{GM_2}{|\vec{r} - \vec{r}_2|} - \frac{1}{2}\Omega_{\text{orb}}^2 d_{\text{rot}}^2, \quad (1.18)$$

where $\vec{r}_1$ and $\vec{r}_2$ are the positions of the two stellar centers and d_{rot} is the distance from the rotation axis through the center of mass. The last term is the centrifugal contribution.

The surfaces of constant Φ_{R} are the equipotentials of the system, and Φ_{R} has five stationary points, the Lagrange points L_1 to L_5 . The inner Lagrange point L_1 lies between the two stars and marks the saddle of the potential through which material can pass from one star to the other. The equipotential surface that passes through L_1 forms two teardrop-shaped lobes joined at their points at L_1 ; these are the *Roche lobes* of the two stars. The outer Lagrange points L_2 and L_3 lie beyond the less massive and the more massive star, respectively, and mark the surfaces beyond which material can escape the system altogether. A binary in which both stars are well inside their Roche lobes is called *detached*. If one star fills or overfills its Roche lobe, the system is *semi-detached*, and if both do so, the system is in *contact*.

A 1D stellar model is necessarily spherically symmetric, hence the comparison between a stellar radius and a Roche lobe has to be made through a volume-equivalent radius R_{RL} , defined as the radius of a sphere with the same volume as the Roche lobe. Several fitting formulae exist (Paczynski, 1971), of which the one most widely used is that of Eggleton (1983),

$$\frac{R_{\text{RL},i}}{a_{\text{orb}}} = \frac{0.49 q_i^{2/3}}{0.6 q_i^{2/3} + \ln(1 + q_i^{1/3})}, \quad q_i = \frac{M_i}{M_j}, \quad j \neq i, \quad (1.19)$$

which is accurate to better than 1% over the whole range of mass ratios. Two features of Eq. (1.19) are worth stating explicitly. First, $R_{\text{RL},i}$ depends only on a_{orb} and on the mass ratio, and not on the total mass of the system. Second, the mass ratio entering the formula is the mass of the star whose Roche lobe is wanted divided by the mass of its companion.

1.2.2 Mass transfer and its classification

More massive stars evolve faster and expand sooner, so in a binary that is close enough for the two stars to interact, it is the initially more massive component, the *primary*,

that first fills its Roche lobe and begins to transfer mass to the *secondary*. The star losing mass in a given episode is the *donor* and the star gaining it the *accretor*; note that these labels can and do swap during the evolution of a binary system, whereas “primary” and “secondary” are fixed by the initial masses. Mass transfer proceeds through the nozzle at L_1 .

Mass-transfer episodes are classified according to the evolutionary state of the donor at the moment the episode begins (Kippenhahn et al., 1967), because that state determines both how the donor responds to mass loss and what its internal structure looks like:

- **Case A:** the donor is still burning hydrogen in its core. It is driven to fill its Roche lobe by its slow nuclear-timescale expansion on the MS.
- **Case B:** the donor has left the MS but has not yet ignited helium in its core, so it is crossing the HG and expanding on a thermal timescale. Case B is subdivided further according to the state of the donor’s envelope. In **Case Be** (“e” stands for early) the envelope is still radiative or only shallowly convective, while in **Case Bl** (“l” stands for late) the donor has developed a deep convective envelope.
- **Case C:** the donor has already ignited helium in its core.
- **Case AB:** A system can undergo more than one episode. A binary that first transfers mass while the primary is on the MS and then again after the primary has left the MS.

The distinction between Case Be and Case Bl is vital since the donor with a radiative envelope shrinks in response to mass loss, whereas one with a deep convective envelope tends to expand, which makes the two regimes behave in opposite ways with respect to stability (Sect. 1.2.3). A Case Bl or Case C donor also has a well-developed core-envelope boundary, with an envelope that is loosely bound and a compact core, and when such a star engulfs its companion, the outcome is a classical common envelope (CE) rather than a contact binary (Röpke and De Marco, 2023).

1.2.3 Stability of mass transfer

Whether mass transfer remains under control or runs away is decided by a competition between the response of the donor’s radius to mass loss and the response of its Roche lobe. Both responses are quantified by logarithmic mass-radius exponents (Webbink, 1984),

$$\xi_{R_1} \equiv \frac{d \ln R_1}{d \ln M_d} \quad \text{and} \quad \xi_{RL} \equiv \frac{d \ln R_{RL}}{d \ln M_d}, \quad (1.20)$$

and ξ_{ad} is defined as the value of ξ_{R_1} for the adiabatic response of the donor, reached on τ_{dyn} , and ξ_{th} its value once thermal equilibrium has been restored, on τ_{KH} . Comparing the three quantities gives the classical stability criteria:

- $\xi_{\text{RL}} \leq \xi_{\text{th}}$: the donor stays inside its Roche lobe in both hydrostatic and thermal equilibrium, and mass transfer proceeds on the nuclear timescale of the donor.
- $\xi_{\text{th}} < \xi_{\text{RL}} \leq \xi_{\text{ad}}$: the donor remains in hydrostatic but not in thermal equilibrium, and mass transfer proceeds on τ_{KH} at correspondingly higher rates.
- $\xi_{\text{RL}} > \xi_{\text{ad}}$: the Roche lobe fails to contain the donor even on a dynamical timescale, and the situation runs away, with mass loss driving further Roche-lobe overflow and an ever-increasing $\dot{M}_{\text{trans}}$.

As a rule of thumb, donors with radiative envelopes (Case A and Case Be) contract upon mass loss and tend to be stable, while donors with deep convective envelopes (Case B1 and Case C) expand and are prone to instability.

1.2.4 Physical mechanisms leading to contact

Unstable mass transfer is not the only way into a contact phase. Following the classification of Henneco et al. (2024b), we distinguish the following mechanisms.

Runaway mass transfer. If the donor’s radius response outruns that of its Roche lobe, in the sense $\xi_{\text{RL}} > \xi_{\text{ad}}$ of Sect. 1.2.3, the mass-transfer rate grows without bound. The donor overfills its Roche lobe by an increasing margin, and the system is driven into contact on a timescale approaching τ_{dyn} . For a Case A or Case Be donor, the result is a contact binary; for a Case B1 or Case C donor, whose core-envelope boundary is well defined and whose companion is at least an order of magnitude smaller in radius, the result is a classical CE phase (Paczynski, 1976; Röpke and De Marco, 2023).

Accretor expansion. These binary systems enter the contact phase due to the accretor expanding during mass transfer. This can happen either because the accretor is out of thermal equilibrium due to mass transfer and expands on the thermal timescale, or because it expands on the nuclear timescale. The subsequent behavior depends on the timescale of the expansion: an accretor that expands on its nuclear timescale, having already regained thermal equilibrium, produces a contact phase that is likely to be stable, whereas an accretor whose thermal expansion continues past the point of contact produces one that is not.

L_2 overflow. A contact binary can eventually fill the equipotential surface passing through the outer Lagrange point L_2 . In such cases, material then leaves the system through L_2 , carrying with it a large amount of angular momentum (Marchant et al., 2021). The orbit shrinks in response, the components are pushed deeper into contact, and the process is self-reinforcing. L_2 overflow is therefore widely taken as a sufficient condition for a merger.

More than one of these mechanisms can be satisfied by the same system, and the boundaries between them in the initial mass ratio-separation plane are not sharp. Section 3.1.1 presents this classification for the $M_{1,i} = 10.16 M_\odot$ slice of the grid and states which regions we take to merge.

1.2.5 From contact to a merged star

Once a contact phase becomes unstable, the two stars spiral in and merge. Binary evolution is not the only route to a merger: in dense clusters, dynamical encounters produce direct collisions. These differ from evolution-driven in-spiral mergers since the progenitors arrive with much higher relative kinetic energy in direct/head-on collisions (Hills and Day, 1976; Portegies Zwart et al., 1999). A head-on collision on a parabolic orbit deposits considerably more entropy in the merged material than a slow spiral-in driven by binary evolution does.

During the merger, the material of the two stars is mixed turbulently, shocks alter the thermal structure, part of the mass is ejected dynamically, and seed magnetic fields can be amplified by orders of magnitude (Schneider et al., 2019; Ryu et al., 2024). What is left is a single star with an internal structure that a single-star evolutionary track cannot produce: a composition profile assembled from two progenitors and a thermal structure far from equilibrium.

1.3 Constructing merger products in one dimension

Buoyancy sorting. The central assumption is that the merged star, once it has settled into hydrostatic equilibrium, must satisfy the stability condition of Eq. (1.15): its entropy must increase outwards. Material from the two progenitors therefore cannot be placed arbitrarily. It has to be arranged in order of increasing entropic variable, with the lowest-entropy material at the center and the highest-entropy material at the surface. This is the buoyancy sorting principle, introduced for stellar collisions by Lombardi et al. (1996) and Lombardi et al. (2002) and developed further by Gaburov et al. (2008) and Glebbeek and Pols (2008). By Eq. (1.16), the more evolved of the two stars has the lower central entropy, so it is the more evolved star that supplies the core of the merged star.

Shock heating. A merger is not adiabatic. The two stars collide, shocks propagate through the merged material, and the irreversible dissipation raises the specific entropy of the gas. How much entropy is deposited, and where, depends on the encounter. A head-on collision at high relative velocity converts far more kinetic energy into thermal energy than a slow spiral-in in which the two stars have already been brought into near-contact by binary evolution.

Dynamical mass loss. Not all of the combined progenitor mass ends up in the merged star. During the merger, a fraction of the material acquires enough energy to become unbound and is ejected, and this dynamical mass loss has to be imposed by hand in a 1D prescription. The ejected fraction is largest for equal-mass mergers, where the encounter is most violent relative to the binding energy of the system, and it falls off towards extreme mass ratios where the secondary is simply absorbed. The expression adopted in this work, which is applied identically in both prescriptions and for every system in the grid, is given in Eq. (2.6). For a merged star of mass M_{rmn} and radius R_{rmn} , the ejected mass M_{ej} leaves with approximately the escape velocity $v_{\text{esc}} = \sqrt{2GM_{\text{rmn}}/R_{\text{rmn}}}$, and the corresponding kinetic energy has to be subtracted from the energy available to the product.

1.4 The blue supergiant problem and the role of binary evolution

Spectroscopic surveys of Galactic and Magellanic Cloud massive stars find a substantial population of B-type supergiants occupying the region of the HRD that single-star models predict should be traversed rapidly (de Burgos et al., 2023; Bernini-Peron et al., 2023). Explaining this unexpectedly large population of blue supergiants constitutes the blue supergiant problem.

The Binarity at LOw Metallicity (BLOeM) campaign, a multi-epoch VLT/FLAMES survey of massive stars in the Small Magellanic Cloud, reports an observed spectroscopic binary fraction of $23 \pm 3\%$ for its early B-type supergiants against $50 \pm 3\%$ for the early B-type dwarfs and giants in the same survey (Shenar et al., 2024). Although this result does not address the overabundance of blue supergiants itself, it provides independent evidence that a substantial fraction of blue supergiants have experienced binary interaction. Since most massive stars are born in binary systems, the comparatively low binary fraction among blue supergiants suggests that many have lost their companions through binary evolution, for example as a result of stellar mergers.

Stellar mergers provide a natural explanation for the low binary fraction of blue supergiants, since the merger product is single by construction. They may also help resolve the blue-supergiant problem if the merger remnant spends a significant fraction

of its lifetime burning helium in the blue part of the Hertzsprung–Russell diagram on a nuclear, rather than thermal, timescale. An alternative pathway is through blue-loop evolution, in which a star returns to higher effective temperatures during core helium burning. Whether either of these channels operates depends on the internal structure of the merger product.

The remainder of the thesis is organised as follows. Sect. 2 describes the binary grid, the two merger prescriptions, the three-dimensional SPH in-spiral merger simulation setup, and the three-stage procedure by which a merged star is imported into **MESA** and evolved. We then present the results, beginning with the identification of the merging systems, construction and classification of the merger products (Sect. 3), continuing with the three-dimensional SPH merger simulation comparison to our 1D prescriptions (Sect. 4), and then the merger product’s long-term evolution and comparison with the BLOeM sample (Sect. 5). Sect. 6 discusses the results, and Sect. 7 summarises the conclusions.

2 Methods

2.1 Binary grid setup

The binary grid used in this thesis is from the work of Henneco et al. (2024b), which is of binary star models with primary masses $M_{1,i}$ ranging from $0.8 M_{\odot}$ to $20 M_{\odot}$, a wide range of initial orbital separations a_i from $7 R_{\odot}$ to $3000 R_{\odot}$ and mass ratios $q_i = M_{2,i}/M_{1,i}$ ranging from 0.1 to 0.9 with a linear spacing of 0.1 and an additional mass ratio $q_i = 0.97$ to model systems with stars of almost equal mass (twin systems). Here, $M_{2,i}$ is the initial mass of the secondary star. The grid of around 6000 one-dimensional (1D) binary evolution models was computed using the binary module of MESA (12778 release). The binary systems are for solar metallicity ($Z_{\odot}$), with $Y = 0.2703$ and $Z = 0.0142$ (Asplund et al., 2009). In this thesis, only the binary systems with an initial primary mass of $10 M_{\odot}$ are considered. We classify the binary models into different cases based on the evolutionary stage of the primary star at the onset of mass transfer (see Sect. 1.2.2).

2.1.1 Termination of the binary calculations

The binary calculations are not evolved to a merger. They are terminated at the point where the 1D treatment ceases to be meaningful, and the stars' profiles saved at that point are what we use as merger progenitors (Sect. 2.1.2). Henneco et al. (2024b) describe the conditions in full. We summarize those that are reached by systems in the $M_{1,i} = 10.16 M_{\odot}$ slice.

A binary calculation can terminate for several reasons, and which applies depends on how mass transfer is computed at the time. MESA offers two schemes. The **contact** scheme (Marchant et al., 2016) solves for the mass-transfer rate when both components overfill their Roche lobes, and can therefore follow a system through a contact phase. The **kolb** scheme (Kolb and Ritter, 1990) is formulated for a semi-detached configuration with a single donor, and breaks down once the second star also overfills its Roche lobe (i.e., contact), so a calculation using it has to be stopped at that point. Consequently, the two schemes have different termination conditions.

The choice between them is made every timestep, from the central hydrogen mass fraction of the star that is currently the donor. A donor still burning hydrogen in its core is computed with the **contact** scheme, and a donor that has left the MS is computed with the **kolb** scheme. Because the test is applied to the donor and not to the primary, a system whose mass ratio has reversed, so that the post-MS secondary has become the donor, also runs with the **kolb** scheme. This rule is overridden while the system is in contact: a contact configuration is always computed with the **contact** scheme, whatever the donor's evolutionary state, and the donor-based choice resumes

if the system detaches again.

Under the `kolb` scheme, the calculation stops once the accretor overfills its Roche lobe by more than its own Roche-lobe radius, that is, once

$$\frac{R_a - R_{\text{RL},a}}{R_{\text{RL},a}} > 1, \quad (2.1)$$

with R_a the radius of the accretor and $R_{\text{RL},a}$ its volume-equivalent Roche-lobe radius from Eq. (1.19). Under the `contact` scheme, this condition is disabled, and the calculation stops instead when the envelope overflows the equipotential surface through the outer Lagrange point L_2 , using the volume-equivalent L_2 radii of Marchant et al. (2016) (i.e., L2-overflow). Independently of the scheme, the calculation is terminated once the mass-transfer rate reaches $\dot{M}_{\text{trans}} = 10 M_{\odot} \text{ yr}^{-1}$, at which point mass transfer proceeds on a timescale approaching τ_{dyn} and can no longer be followed hydrostatically (i.e., Runaway mass transfer). Or once the primary reaches the end of core carbon burning. Our merger progenitors never reach the latter, since we consider only Cases A and Be systems.

Unlike Henneco et al. (2024b), who end their contact tracing at the onset of reverse mass transfer, we retain the reverse mass-transfer phases that the models compute. This matters for certain Case AB systems, in which the mass ratio reverses during the first phase of mass transfer and the primary regains mass before the calculation terminates (Sect. 3.1.2).

2.1.2 Merger criteria and choice of progenitor models

The conditions above record where a calculation ended, but they do not identify a merger on their own. We therefore state separately which binaries we assume to merge and which profiles we take as their progenitors. The application of these criteria to the grid and the justification for each are given in Sect. 3.1.1. We assume a system to merge if it establishes contact through one of the mechanisms of Sect. 1.2.4, and if the primary is still an MS or HG star when it does. The first requirement selects systems that reach contact at all; the second excludes systems whose primary has developed a clear core-envelope boundary, whose outcome is set by envelope ejection rather than by just merging the progenitors (see Sect. 3.1.2).

For each such system we take the structure profiles of both components from the last model computed, that is, from the point at which one of the conditions of Sect. 2.1.1 was met. The consequences of this choice for the progenitor masses are quantified in Sect. 3.1.2. Some binary computations encountered numerical difficulties before reaching any of these stopping conditions, so we cannot use their final profiles. For these, we step back to the last numerically sound profile (by hand) and ask whether the system had already satisfied a more lenient merger criterion then. If it had, we merge

the stars using that profile. If it had not, we discard the system, and it does not enter the grid. In our merging parameter space, this affects only a single system, which we examine individually in Sect. 3.1.3. We find that it satisfies our lenient merger criteria before the failure occurs and retain it, taking its progenitor profiles from the last converged model.

2.1.3 Adopted stellar physics

Each of the binary components was initialized from a precomputed ZAMS model at solar metallicity of $Y = 0.2703$ and $Z = 0.0142$ (Asplund et al., 2009). The mass transfer scheme adopted by Henneco et al. (2024b) for their binary grid is explained in detail in Sect. 2.1.1. The stellar physics adopted below is also used consistently in the post-merger evolution grid. The entire physics inlist settings are given in Appendix C. MESA’s `approx21` nuclear network was used. Convective regions were set using the mixing length theory (Böhm-Vitense, 1958; Cox and Giuli, 1968) with a mixing length parameter set to $\alpha_{\text{mlt}} = 2.0$ (Paxton et al., 2013), the efficiency of semi-conductive mixing is set to $\alpha_{\text{sc}} = 10.0$ (Schootemeijer et al., 2019), and thermohaline mixing efficiency of $\alpha_{\text{th}} = 1.0$ (Marchant et al., 2021). The Ledoux criterion was used to determine the regions in the star that are unstable to convection. For a detailed description of the mass-loss prescription, see Henneco et al. (2024b).

2.2 One dimensional stellar merger prescriptions

2.2.1 (Python) Make Me A Massive Star

“Make Me a Massive Star” is a standalone merger routine written in C++, originally presented in Gaburov et al. (2008) and currently available in the “AMUSE” framework (Portegies Zwart et al., 2009, 2013; Pelupessy et al., 2013; Portegies Zwart and McMillan, 2018; Portegies Zwart et al., 2023). This one-dimensional code approximates entropy generation (shock heating) during head-on collisions, in contrast to the in-spiral mergers driven by binary evolution that we focus on here. In this thesis we use the Python version of the code, Python Make Me A Massive Star (referred to as “PyMMAMS” in the following), presented in Heller et al. (2025).

The code constructs a merger product by shock-heating the progenitors and then assembling a new star from their shells. The steps are summarised below.

1. The progenitor’s MESA profile is loaded and the buoyancy A (entropic variable) of each shell is computed using

$$A = \frac{\beta_{\text{gas}} P}{\rho^{5/3}} \exp \left[\frac{8}{3} \frac{1 - \beta_{\text{gas}}}{\beta_{\text{gas}}} \right], \quad (2.2)$$

where ρ is the density and β_{gas} is the ratio of the gas pressure P_{gas} to the total pressure P . Writing the specific entropy of a mixture of an ideal gas and radiation in terms of β_{gas} and substituting for the temperature shows that Eq. (2.2) is related to the specific entropy s by

$$A \propto \frac{\mathcal{R}}{\mu} \exp \left[\frac{2}{3} \frac{\mu}{\mathcal{R}} s \right], \quad (2.3)$$

with $\mathcal{R}$ the gas constant and μ the mean molecular weight. The chemical composition enters A , through its effect on β_{gas} . Two shells with the same A at the same pressure have different specific entropies whenever their compositions differ. Equation (2.2) provides the entropy (or buoyancy) label for each shell, from which the shell's density at any pressure can be determined through the equation of state.

2. The progenitors are shock-heated by modifying the buoyancy of each shell according to

$$\log_{10} \left(\frac{A_{n,f}}{A_{n,i}} \right) = a' + b \log_{10} \left(\frac{P_{n,i}}{P_{c,i}} \right), \quad (2.4)$$

where $A_{n,i}$ and $A_{n,f}$ are the buoyancy of the n^{th} shell before and after shock heating, $P_{n,i}$ is that shell's pressure before shock heating, and $P_{c,i}$ is the initial central pressure of the progenitor to which the shell belongs. The parameter a' is given by

$$a' = a + \log_{10}(f_{\text{heat}}), \quad (2.5)$$

where f_{heat} is a free parameter adjusted so that energy is conserved (step 4).

The parameters a and b are taken from Gaburov et al. (2008), who obtained them by fitting Eq. (2.4) to SPH simulations of head-on stellar collisions. They are tabulated separately for the primary and the secondary progenitor, and depend on two quantities only: the mass ratio q , for which the values $q = 0.1, 0.2, 0.4$ and 0.8 were computed, and the evolutionary stage of the progenitors, for which two cases were computed: half-age main sequence (HAMS) and terminal-age main sequence (TAMS). Both calibration points therefore lie on the MS, and the fit does not explicitly depend on progenitor mass. Outside $q = 0.1$ – 0.8 , the original code disables shock heating altogether, but PyMMAMS instead interpolates the tabulated values (by default, nearest-neighbor in q) or extrapolates them, depending on user input (Heller et al., 2025).

Our Cases Be and AB merger progenitors consist of a post-MS primary and a

much less evolved secondary, a configuration lying outside the range over which Eq. (2.4) was fitted, so that applying PyMMAMS to them amounts to an extrapolation in evolutionary stage. The consequences are examined in Sect. 3.2.2.

3. Once both progenitors have been shock-heated, the shells of each are ordered so that A increases monotonically outwards, which allows A to be interpolated as a function of mass coordinate. The merger product is then constructed by integrating the hydrostatic equilibrium equation outwards from the center. The central pressure is initially estimated as the sum of the progenitors' central pressures, and the outer boundary condition is zero pressure at the surface.

At each integration step, the buoyancies of both progenitors are evaluated at the current mass coordinate, and together with the merger product's local pressure, the density each candidate shell would have at that pressure is computed. The denser of the two is placed next in the merger product. This is Archimedes' principle applied shell by shell, and the ordering criterion is density at a common pressure, not the buoyancy variable itself. The procedure is repeated until the merger product reaches the combined mass of the progenitors minus the mass lost dynamically, $M_{\text{prod}} = M_1 + M_2 - M_{\text{loss}}$, after which the surface pressure is used to correct the central pressure, and the integration is repeated until the outer boundary condition is satisfied. In this thesis, the dynamical mass loss is set using Eq. (2.6) for both ES and PyMMAMS,

$$\frac{M_{\text{loss}}}{M_1 + M_2} = 0.03 \frac{q}{(1 + q)^2}. \quad (2.6)$$

Because the selection is made on density rather than on entropy, a PyMMAMS merger product is guaranteed to have a monotonically decreasing density profile but is *not* guaranteed to have a monotonically increasing entropy profile, and entropy inversions do occur (Gaburov et al., 2008; Heller et al., 2025). Entropy sorting, which orders material by specific entropy, provides the opposite guarantee. The two prescriptions therefore differ not only in the amount of shock heating they apply but also in the variable by which they arrange the material, and the two orderings coincide only where the progenitors have similar mean molecular weight profiles, by Eq. (2.3).

4. The code then checks energy conservation,

$$(E_1 + E_2) - (E_{\text{rnn}} + E_{\text{ej}}) = 0, \quad (2.7)$$

where E_1 and E_2 are the total energies (thermal plus gravitational) of the primary and secondary progenitors, E_{rnn} is the total energy of the merger product, and

E_{ej} is the kinetic energy carried away by the dynamically ejected mass, assumed to leave with the escape velocity,

$$E_{\text{ej}} = \frac{1}{2} M_{\text{ej}} v_{\text{esc}}^2, \quad \text{with} \quad v_{\text{esc}} = \sqrt{\frac{2GM_{\text{rmn}}}{R_{\text{rmn}}}}, \quad (2.8)$$

where M_{rmn} and R_{rmn} are the mass and radius of the merger product. Because the code assumes a head-on collision of two stars on a parabolic orbit, the orbital energy vanishes and does not appear in Eq. (2.7). If energy is not conserved, f_{heat} is adjusted and the code returns to step 2; the iteration stops once the relative energy difference falls below 0.1%.

5. Finally, the code removes double-valued regions¹ using the same “remeshing” scheme as entropy sorting, in which steep chemical composition gradients are separated, and adjacent shells are mixed according to the requested number of output shells (Sect. 2.2.2) (Heller et al., 2025).

2.2.2 Entropy Sorting

In ideal cases, stars have an entropy profile increasing outwards, except in convective regions, where the profile is approximately flat (Lombardi et al., 1996). The most basic approximation of a merger product is therefore to stack the progenitor shells in order of increasing specific entropy, placing the lowest-entropy material at the center and the highest-entropy material in the envelope. Entropy sorting (referred to as “ES” in the following) is a 1D merger prescription that works on this principle. The entropy profiles of the progenitors are not modified during the merger, in contrast to PyMMAMS, where shock heating is applied to the progenitors’ buoyancy (Sect. 2.2.1). We then apply dynamical mass loss by removing mass from the merger product’s surface.

Since the central entropy of a star decreases as it evolves (Sect. 1.1.4), ES assigns the core of the merger product to the more evolved progenitor. In a merger between a post-MS primary and a less evolved secondary, ES places the primary’s helium core at the center, and the merger product retains the post-MS character of its progenitor. This is the origin of the Type-P classification discussed in Sect. 3.2.2.

During stacking, ES can place adjacent shells with vastly different chemical compositions next to each other, making the merger product’s hydrogen mass fraction very spiky. These double-valued regions are problematic when the model is imported into a stellar evolution code, and are removed by a mixing procedure. The user specifies how many shells the merger product should have after mixing. Dividing the total mass of the merger product by this number yields the average shell mass, and new mass frac-

¹PyMMAMS can place shells of greatly different composition next to each other, so-called “double-valued” profiles, terminology used in Gaburov et al. (2008).

tions are computed by averaging the mass fractions of the combined shells, weighted by shell mass.

2.3 Three-dimensional hydrodynamical simulation

The one-dimensional prescription PyMMAMS has been calibrated against three-dimensional hydrodynamical calculations for main-sequence mergers, but no equivalent calibration exists for a post-main-sequence primary merging with a main-sequence secondary. In order to assess which of the two prescriptions (ES and/or PyMMAMS) provides the more faithful representation of such a merger, we performed three-dimensional simulations of the in-spiral merger of a post-main-sequence + main-sequence binary. The three-dimensional calculation serves as an independent reference against which we can compare the one-dimensional merger product structures.

2.3.1 Code and numerical configuration

The simulation was carried out with the smoothed particle hydrodynamics (SPH) code PHANTOM (Price et al., 2018), version 2026.0.1. We compiled the code with the k -D tree self-gravity solver and individual-particle timesteps enabled. Densities and smoothing lengths are obtained self-consistently from the M_4 cubic-spline kernel, with the smoothing length set by $h = h_{\text{fact}}(m/\rho)^{1/3}$ and $h_{\text{fact}} = 1.2$, iterated to a relative tolerance of 10^{-4} . We set the tree opening criterion to 0.5. We set the timestep using the more restrictive of two criteria. The first is a Courant-like condition, $\Delta t_C = C_{\text{cour}}(h/v_{\text{sig}})$, where h is the smoothing length and v_{sig} is the signal velocity between interacting particles. The second is a force condition, $\Delta t_F = C_{\text{force}}(h/|\mathbf{a}|)^{1/2}$, where $\mathbf{a}$ is the particle acceleration. We adopt $C_{\text{cour}} = 0.3$ and $C_{\text{force}} = 0.25$, so that the timestep is given by $\Delta t = \min(\Delta t_C, \Delta t_F)$.

Shock capturing follows the standard PHANTOM implementation. The artificial shock viscosity parameter is allowed to vary between $\alpha_{\text{AV}}^{\text{min}} = 0$ and $\alpha_{\text{AV}}^{\text{max}} = 1$ under the Cullen and Dehnen (2010) switch, with the quadratic term fixed at $\beta_{\text{AV}} = 2$ to suppress particle interpenetration in strong shocks. Both P d V heating and shock heating are active. No radiative cooling and no physical (Navier–Stokes) viscosity are created or present at any point in the calculation. The treatment of the artificial conductivity parameter α_u is non-standard in this run and is described separately in Sect. 2.3.4.

Gas thermodynamics are treated with an ideal gas of non-uniform mean molecular weight, supplemented by a radiation pressure contribution, which is the appropriate choice for the envelope of an evolved massive star, where radiation pressure is substantial. The composition is carried as a per-particle quantity, so the mean molecular weight entering the equation of state varies with position rather than remaining a single global value.

2.3.2 Progenitor models

We constructed both stars directly from one-dimensional **MESA** single-star models using the same input physics. The primary is a $10 M_{\odot}$ star that has just left the main sequence and is crossing the Hertzsprung gap, with a radius of $10 R_{\odot}$ and a helium core mass of $\approx 3.3 M_{\odot}$. The secondary is a $2 M_{\odot}$ zero-age main-sequence star. The mass ratio is therefore $q = 0.2$. The structure and composition of both progenitors at the point at which they enter the three-dimensional calculation are shown in Fig. 12.

The primary is represented by 10^6 particles and the secondary by 2×10^5 particles, for a total of 1.2×10^6 particles. This partition gives both stars the same particle mass, $m_{\text{part}} = 10^{-5} M_{\odot}$. In simulations of giant stars, the dense stellar core is often excised and replaced by a softened point mass to avoid resolving the steep pressure gradient near the core, which would require many small cells or particles and correspondingly small timesteps. Here, however, this approximation would remove the very region whose interaction we need to follow. Hence, resolving the helium core with gas is essential for determining whether the secondary penetrates the primary’s core.

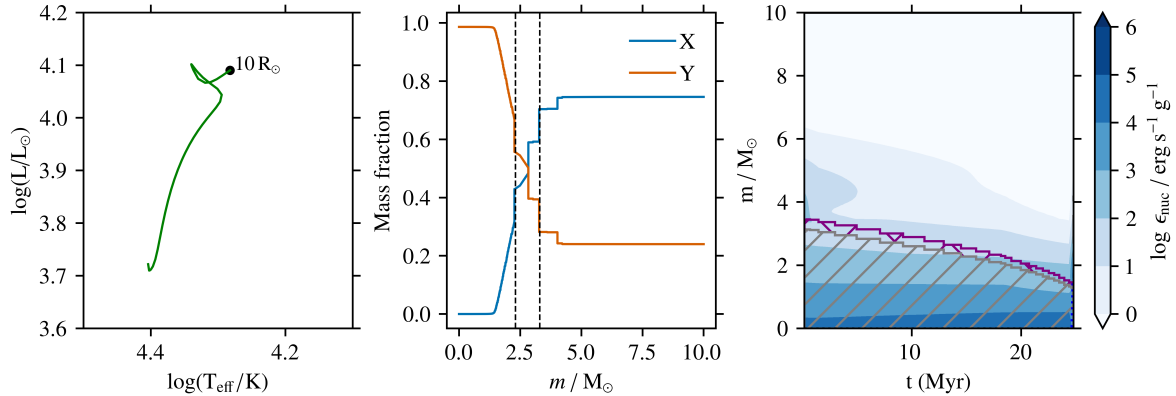


Figure 1: Structure and evolutionary state of the primary progenitor. *Left:* Hertzsprung–Russell diagram with the evolutionary track of the $10 M_{\odot}$ primary until it reaches $10 R_{\odot}$. *Middle:* hydrogen and helium mass fractions against mass coordinate at the extraction point. The primary’s pure helium core sits within $\approx 2.3 M_{\odot}$, marked by the first vertical dashed line, and the core-envelope boundary is marked by the second vertical dashed line at $\approx 3.3 M_{\odot}$. *Right:* Kippenhahn diagram with hatching indicating convective regions.

2.3.3 Relaxation and binary setup

Each star was relaxed in isolation before being placed in the binary, so that the mapping from the one-dimensional **MESA** structure onto the particle distribution does not introduce spurious oscillations that could be mistaken for merger-driven mixing. Particles were first placed in a random but point-symmetric sphere and then stretch-mapped in the radial direction to reproduce the tabulated density profile of the **MESA** model,

after which the specific internal energy of each particle was assigned from the same model through the `ieos = 12` equation of state with the local composition. The resulting configuration was then relaxed under the standard **Phantom** iterative scheme (using the method in Appendix C of Lau et al. (2022)), in which the internal energy profiles are driven towards the one-dimensional target, with the iteration terminated once the ratio of kinetic to potential energy fell below 10^{-7} . For the primary, the relaxation began from $E_{\text{therm}} = 18.4$ and $E_{\text{pot}} = -31.5$ in code units. For the secondary, the relaxation began from $E_{\text{therm}} = 2.88$ and $E_{\text{pot}} = -4.94$ in code units. The final $E_{\text{kin}}/E_{\text{pot}}$ actually reached by each star is 1.015×10^{-7} for the primary and 1.004×10^{-7} for the secondary. The code unit conversion is given in the following table:

Table 1: Code units adopted in the PHANTOM SPH simulation, with $u_{\text{mass}} = 1 M_{\odot}$ and $u_{\text{dist}} = 1 R_{\odot}$ as the two independent base units.

Quantity	Symbol	Value (cgs)
Mass	u_{mass}	$1.989 \times 10^{33} \text{ g}$
Length	u_{dist}	$6.960 \times 10^{10} \text{ cm}$
Time	u_{time}	$1.594 \times 10^3 \text{ s}$
Density	u_{ρ}	5.901 g cm^{-3}
Energy	u_E	$3.793 \times 10^{48} \text{ erg}$
Velocity	u_v	$4.367 \times 10^7 \text{ cm s}^{-1}$

Both stars were given a rigid-body spin of $\Omega_{\text{spin}} = 2.5 \times 10^{-5} \text{ rad s}^{-1}$ before the construction of the binary. This corresponds to a spin period of $\approx 2.8 \text{ d}$, or $\Omega_{\text{spin}}/\Omega_{\text{orb}} \approx 0.49$ at the initial separation. We perform the calculation in the inertial frame with no co-rotation enforced. We set the spin value assuming synchronization at the onset of Roche-lobe overflow.

The binary was initialized on a circular orbit with a separation of $a_i = 12 R_{\odot}$. No constraint of circularity is imposed thereafter; the orbit is free to evolve under the mutual gravitational and hydrodynamical forces. The choice of a_i is dictated by computational cost, since starting the calculation at a wider separation requires following many orbits before the dynamically interesting phase begins. To verify that this choice does not bias the outcome, trial calculations were performed from both wider and narrower initial separations ($a_i = 11, 12, 13$ and $13.6 R_{\odot}$) at a lower resolution of 1.2×10^5 particles. The resulting passive scalar distribution in the merger product was insensitive to a_i . Only the unbound mass showed any dependence, and the variation remained $< 0.83\%$. The passive scalar is initialized to unity for all particles belonging to the primary and to zero for all particles belonging to the secondary, so that its mass-averaged profile in the product gives directly the local fraction of material originating from the primary.

2.3.4 Treatment of the artificial conductivity

Artificial thermal conductivity is required in SPH to remove the spurious pressure discontinuities that arise at contact discontinuities, and is therefore important for the correct treatment of the composition and entropy interfaces that develop once the secondary is disrupted inside the envelope of the primary (Price and Monaghan, 2007; Price et al., 2018). In this calculation, however, enabling the conductivity from the start of the run proved impossible. During the detached phase, before the stars come into contact, the secondary’s core is hotter and denser than its outer layers, so with $\alpha_u > 0$ active, thermal energy is transported from the core outward into the cooler envelope rather than only smoothing genuine shock fronts. The core consequently loses thermal energy it needs for pressure support, contracts, and collapses, terminating the simulation. This is a numerical artifact of applying artificial conductivity before contact and not a physical instability of the secondary.

We therefore adopt a two-stage treatment. The calculation starts with $\alpha_u = 0$ and evolves through the early phases where the stars deform smoothly and the orbit decays under tides. Once the secondary has penetrated a significant distance into the primary’s envelope, the run continues with $\alpha_u = 1$. Tests were conducted at a resolution of 1.2×10^5 particles to determine an appropriate threshold separation above which conductivity may be switched off. An orbital separation of $4 R_\odot$ was chosen as the point at which the secondary remained stable against the introduction of conductivity. A convergence test was performed with $\alpha_u = 0, 0.1$ and 1.0 applied after the switching point, and the outcome is reported in Sect. 4.4.

2.3.5 Post-merger relaxation and merger product analysis

We continued the calculation well beyond the in-spiral and merger itself, until the merger product settled into a quasi-hydrostatic configuration. We determine the end point of the relaxation phase from the merger product’s passive scalar: we continued the calculation until the mass-binned passive scalar profile no longer changed appreciably between successive snapshots. This was found to be ~ 44.28 hrs (100 code time units) after merger. We analyze the merger product structure in one dimension by first identifying the highest-density particle, which defines the merger product center, then constructing radial bins outward from that point, each containing equal particle mass. Within each bin, we mass-average the passive scalar and chemical composition.

2.3.6 Computational cost

We ran the simulation on a single node with an AMD EPYC 9554 64-core processor at 3.10 GHz and 1024 kB cache, using OpenMP parallelization across all 64 cores. Since the distributed-memory parallelization of PHANTOM does not scale satisfactorily for

this problem, the calculation is confined to a single node, which sets a hard ceiling on the accessible particle number. The complete run, including the pre-merger inspiral, the coalescence, and the post-merger relaxation, required approximately 240 wall-clock hours.

2.4 MESA

After obtaining our merger product, we can now import this into the stellar evolution code **MESA** (Jermyn et al., 2023; Paxton et al., 2011, 2013, 2015, 2018, 2019) version r12778 to simulate the evolution of the merger product. We do this in three phases, hereafter referred to as the “engineering”, “relax” and “evolve” phases, and we use this terminology consistently throughout this thesis. All **MESA** inlist physics settings used in this thesis are provided in Appendix C. The physical and numerical parameters are mostly consistent with those of Henneco et al. (2024b). Neither PyMMAMS nor ES tracks rotation (i.e., angular momentum). As a result, we assume the merger product is non-rotating at the time of import. However, we allow the merger product to rotate during the evolution phase by setting the initial angular velocity to 5% of the critical angular velocity Ω_{crit} (the reasoning behind this value is discussed in Sect. 2.4.3). We also allow the merger product to lose mass via stellar winds during the evolution phase, using the mass-loss prescription of Henneco et al. (2024b).

2.4.1 Engineering

The merged star is imported into **MESA** using the engineering routine, referred to as “model relaxation” in Paxton et al. (2018), which takes the temperature-density/entropy and composition profiles of the merger product from PyMMAMS or ES and builds a **MESA**-compatible model of a star from it. The engineering phase is carried out in two distinct phases.

1. In this first phase, **MESA** tries to match the composition profile by a pseudo-evolution of a model star in which angular momentum transport, mixing, stellar winds and changes in the composition due to nuclear burning are turned off. At the same time, the chemical composition profile is iteratively adjusted to the desired value within a specified tolerance. Hydrostatic equilibrium is imposed throughout this phase. The starting model is a ZAMS (zero-age main sequence, the time at which hydrogen is ignited in the core) star with $Z = 0.0142$. The composition profile is matched with the input model after this first phase.
2. In the second phase, shells are cooled or heated depending on whether their entropy is higher or lower than that of the input model. For some models, we see

the error `do1_energy_eqn`² at this stage, for which the option `num_steps_to_relax_composition` is varied from 100 to 500 until that model converges (usually this option is set to 200 since it works the best for most models). When importing from PyMMAMS, we ask MESA to extract the entropy with the help of the EOST module since the entropy output from PyMMAMS is in terms of density and temperature³. Whereas ES outputs the entropy of each shell. For the entropy adjustment phase, `max_steps_to_relax_entropy` is set at 1000 since values below this has a risk of incomplete cooling/heating of the stellar model. This means we may not reach the target entropy profile, and the resulting merger will have the wrong thermal structure.

2.4.2 Relax

The engineering phase returns a MESA model whose composition and entropy profiles match those of the merger product, but it does not guarantee that the model satisfies the equations of stellar structure. The relax phase exists to enforce that, and everything in this section is a diagnostic of how far the model still is from doing so. The quantity we use to measure this is the energy residual $\overline{E_{\text{residual}}}$, the mass-weighted average over the star of the local imbalance in the energy equation, Eq. (1.3). A model that obeys the stellar structure equations has $\overline{E_{\text{residual}}} \rightarrow 0$, and a large value means the solver has not found a solution that conserves energy locally. The transient convection zones that appear if the model is evolved directly without this “relax” step is a symptom of the imbalance. They occur in regions where the Ledoux criterion, Eq. (1.13), forbids convection, i.e., the steep μ gradient at the boundary of the primary’s helium core should stabilize the layer. MESA nonetheless returns convection there, which is a sign that the energy equation is not being satisfied locally.

A merger product is far from thermal equilibrium and contracts rapidly, and we find specific energy generation rates from gravitational contraction of $\sim 10^{15}$ erg g⁻¹ s⁻¹ against $\sim 10^6$ erg g⁻¹ s⁻¹ from hydrogen burning. The energy equation is therefore dominated by a term that the solver resolves poorly, and the resulting local energy imbalance inflates the radiative gradient enough to trigger transient convection zones. The consequence is not confined to the first few timesteps. A transient convection zone spanning the envelope to core carries fresh hydrogen into the helium core. In post-MS merger products, this rejuvenates the star and can return it to the MS, which the merger itself does not produce.

Non-relaxed models follow a different track for the remainder of the computation and, at central helium exhaustion, differ from an otherwise identical model that was

²This error occurs when the energy equation cannot be solved for a particular shell.

³Therefore, we set `get_entropy_for_relax_from_eos` to `eosDT` for PyMMAMS and for ES, this setting is not used.

relaxed first. They have a larger CO core mass because they burned hydrogen in a core that was artificially replenished. It has a different total mass, because the altered luminosity and radius change the integrated wind mass loss. None of these differences can be identified numerically from the final model alone, which is why the problem must be addressed before the evolution begins rather than diagnosed afterward.

We therefore insert a pseudo-evolution phase, referred to as the “relax” phase, during which energy generation and convective energy transport remain active, so the model can shed the energy imbalance, but chemical mixing and composition changes from nuclear burning are switched off by setting `mix_factor` and `dxdt_nuc_factor` to zero for the merged star. Stellar winds are also switched off, since the relax phase does not correspond to a physical phase of the star’s life. Because the composition is frozen, no rejuvenation can occur, no matter how long the phase lasts.

The phase duration is bounded from both sides. It must be long enough for $\overline{E_{\text{residual}}}$ and the absolute gravitational luminosity $|L_{\text{grav}}|$ to fall to values at which the model obeys the stellar structure equations, and for the transient convection zones to disappear. It must be short enough that the star does not evolve thermally during it. Energy generation and transport are active, so over a time approaching the Kelvin-Helmholtz timescale the model would contract and restructure, and we would no longer be importing the merger product itself, but a thermally evolved version of it with its nuclear clock stopped.

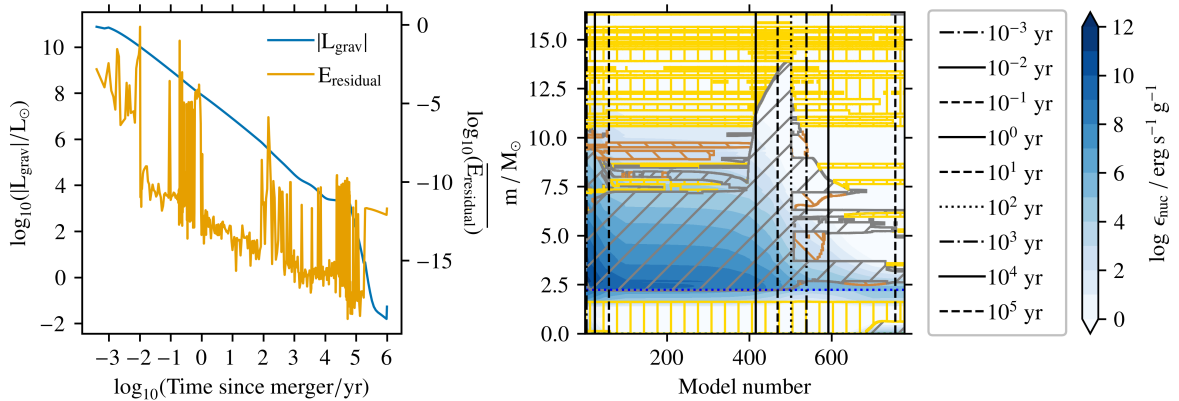


Figure 2: *Left:* Time evolution of the absolute gravitational luminosity, $|L_{\text{grav}}|$, and the average energy residual, $\overline{E_{\text{residual}}}$, during the relaxation phase of the merged star. *Right:* Kippenhahn diagram of the same product over 10^6 yr of relaxation. Vertical reference lines mark logarithmically spaced times after the merger, separated by one dex.

Figure 2 shows both diagnostics over a relax phase of 10^6 yr, together with the corresponding Kippenhahn diagram, for the merger product of the $M_p = 10.16 M_{\odot}$, $q_i = 0.7$, $a_i = 14.98 R_{\odot}$ system. $|L_{\text{grav}}|$ falls from $\log_{10}(|L_{\text{grav}}|/L_{\odot}) \approx 11$ at the start of the phase to ≈ 6 after 100 yr, and $\log_{10}(\overline{E_{\text{residual}}})$ falls from ≈ -2.5 to ≈ -15 over the same interval. The grey hatched regions in the Kippenhahn diagram, which mark

convection, have largely disappeared by 100 yr and the star is radiative over most of its mass.

We adopt 100 yr for all models. Relaxing for ≈ -3 yr instead leaves $\log_{10}(\overline{E_{\text{residual}}})$ at ≈ -1 for most of the merged stars, and the transient convection zones are still present. Hence, the subsequent evolution mixes hydrogen into the core. Relaxing for 10^6 yr removes the convection zones but is a substantial fraction of the merger product's $\tau_{\text{KH}} \approx 270$ yrs. At 100 yr, the residual and $|L_{\text{grav}}|$ have already dropped below the levels at which transient convection appears. We also stopped the relax phase at logarithmically spaced times after the merger, separated by one dex to see the subsequent evolution in the HR diagram (see Appendix B for a few exemplary relax times of the same merger product). This exercise showed that the subsequent evolution is highly relax-time sensitive and must be treated with care.

2.4.3 Evolve

Neither PyMMAMS nor ES tracks angular momentum, so the merger product is non-rotating when it is imported into MESA and any rotation has to be imposed by hand. We set the initial angular velocity to $0.05 \Omega_{\text{crit}}$, where Ω_{crit} is the angular velocity at which the centrifugal acceleration at the equator cancels the effective gravitational acceleration there,

$$\Omega_{\text{crit}}^2 = \left(1 - \frac{L}{L_{\text{Edd}}}\right) \frac{GM}{R_{\text{e}}^3}, \quad (2.9)$$

with R_{e} the equatorial radius and L_{Edd} the Eddington luminosity. At $\Omega = \Omega_{\text{crit}}$, the effective gravity at the equator vanishes and material there is no longer bound to the star, so Ω_{crit} is an upper limit on the rotation rate a star can sustain.

Our choice is guided by what is known about the rotation of merger products. Schneider et al. (2019) follow the $17 M_{\odot}$ merger product of a 3D magneto-hydrodynamical simulation through its post-contraction phase⁴ and report an angular velocity of $\approx 0.1 \Omega_{\text{crit}}$. Schneider (2025) notes that the rotation rates of merger products once they have fully relaxed remain poorly constrained, and that a merger product that is initially a rapid rotator can spin down efficiently, so that merged stars may be observed as genuine slow rotators. We therefore adopt $0.05 \Omega_{\text{crit}}$, a value below that reported by Schneider et al. (2019) and consistent with merger products being slow rotators, while retaining some rotation and the rotational mixing that comes with it.

2.4.4 Classification of post-main-sequence evolution

We classify the post-main-sequence evolution of the merger products according to their effective temperature at core helium ignition and their subsequent evolution during core

⁴The phase after the merger product has reached its maximum radius and begins contracting towards the main sequence.

helium burning. For this purpose, we define the threshold

$$\log_{10}(T_{\text{eff}}/\text{K})_{\text{threshold}} = 3.7, \quad (2.10)$$

which is slightly hotter than the effective temperature at core helium ignition of single stars with masses between 10 and 20 $M_{\odot}$.

A merger product is classified as a *blue-loop* (BL) star if it ignites core helium on the cool side of the threshold, $\log_{10}(T_{\text{eff}}/\text{K})_{\text{He,ign}} < 3.7$, subsequently evolves to higher effective temperatures, and later returns to the cool side before the end of core helium burning. A merger product is classified as a *yellow/blue supergiant* (Y/BSG) if it ignites core helium on the hot side of the threshold, $\log_{10}(T_{\text{eff}}/\text{K})_{\text{He,ign}} > 3.7$, and remains there for a significant fraction of its core helium-burning lifetime. We adopt a residence-time criterion rather than one based solely on position in the Hertzsprung–Russell diagram, since only evolutionary phases lasting substantially longer than the Kelvin–Helmholtz timescale, τ_{KH} , are expected to contribute appreciably to the observed population (Sect. 1.4). Merger products that ignite core helium on the cool side of the threshold and remain near the Hayashi limit throughout core helium burning are classified as *red supergiants* (RSGs). These three classes are mutually exclusive, so each merger product is assigned to exactly one evolutionary category.

Previous studies have adopted higher effective-temperature thresholds to distinguish blue from cool supergiants, for example $\log_{10}(T_{\text{eff}}/\text{K}) = 3.9$ in Schneider et al. (2024) and 4.0 in Menon et al. (2024). Our objective, however, is not to distinguish yellow from blue supergiants, but rather to separate merger products that undergo a blue loop from those that remain on the hot side of the Hertzsprung–Russell diagram throughout core helium burning. We therefore adopt the lower threshold of $\log_{10}(T_{\text{eff}}/\text{K}) = 3.7$.

3 Contact tracing and merger outcome classification

Before we can evolve the merger products, three things must be settled: which binary systems in the grid merge, at what point in their evolution we take the progenitors, and what structure the merger produces. Section 3.1 identifies the region of the $M_{1,i} = 10.16 M_{\odot}$ grid slice in which contact phases form, and quantifies how much the masses of the progenitors change between the onset of mass transfer and the moment we deem the system to merge. Section 3.2 then uses the passive scalar to determine where the material in the core of the merger product comes from and whether the merger product starts its further evolution as a main-sequence or a post-main-sequence star, separately for the two merger prescriptions. The long-term evolution of these merger products follows in Sect. 5.

3.1 Formation of contact phases and the structures of stars at merger

3.1.1 Binary systems that lead to a stellar merger

In this section, we present the results of the contact tracing analysis of the $10.16 M_{\odot}$ initial primary mass binary grid (a slice of the full binary grid described in Sect. 2.1). Identifying which binary systems ultimately undergo a merger is non-trivial, as it depends on the interplay between the orbital evolution of the binary and the internal structure of the two stars (see Sect. 1.2.4). In Fig. 3, the colored regions represent different physical mechanisms leading to contact: ‘Runaway MT’, ‘L2-overflow’, and ‘Accretor expansion’ (see Sect. 1.2.4 for details). In some regions of the figure, binary systems satisfy more than one mechanism, indicated by hatchings in the corresponding colors.

The blue shaded region represents binary systems classified as ‘No contact’, i.e., systems that undergo at least one phase of mass transfer but do not reach contact by the end of the simulation. The grey shaded region marks systems flagged as ‘Numerical Issues’, i.e. systems for which the simulation was unable to reach the imposed stopping conditions due to numerical failure.

In this thesis, we focus exclusively on mass transfer Cases A and Be systems as merger progenitors. Systems undergoing runaway mass transfer in Cases Bl and C are excluded because they represent a physically distinct regime, i.e. in these systems, the primary has already evolved into the giant phase and possesses a well-defined core-envelope boundary at the onset of mass transfer. Consequently, such systems are expected to enter the “classical common envelope” phase rather than leading to

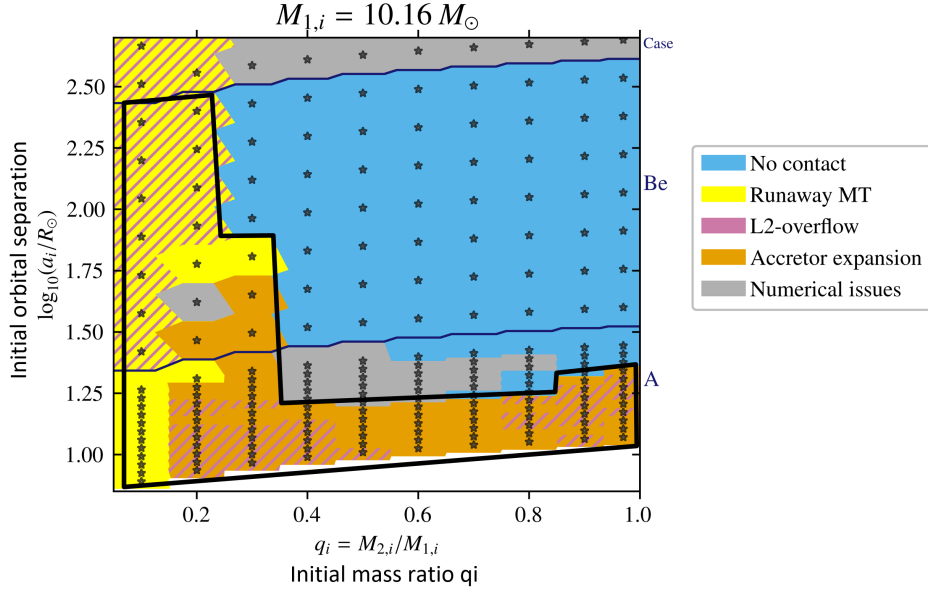


Figure 3: Occurrence of contact phases in the initial mass ratio (q_i) vs. initial orbital separation ($\log_{10}(a_i/R_\odot)$) plane for mass transfer Cases A and Be binary systems with initial primary mass $M_{1,i} = 10.16 M_\odot$. The dark blue quasi-horizontal lines separate the mass-transfer cases and mark, for each q_i , the initial separation at which the primary first fills its Roche lobe at the corresponding evolutionary stage; the case is labeled on the right-hand side of each boundary. The bold black contour encloses the systems that we consider to merge. Each star marks a computed binary model.

a stellar merger. Restricting the analysis to Case A and Case Be systems reduces the parameter space to ~ 100 binary systems, corresponding to the orange (‘Accretor expansion’) and yellow (‘Runaway MT’) shaded regions enclosed within the Case A and Case Be boundaries in Fig. 3.

Looking at Fig. 3, we can see that binary systems with initial mass ratios $q_i = 0.15 - 1.00$ and Case A (initially the closest orbital separations) go into contact because of accretor expansion (except for one system with $q_i = 0.2$ and $\log_{10}(a_i/R_\odot) \sim 1.26$, which goes into runaway mass transfer, marked with yellow filling). Some of these systems have a hatching indicating that they experience both accretor expansion and L2-overflow. This implies that the system experiences mass loss through the L2 point, which carries away orbital angular momentum and subsequently leads to a stellar merger. However, it is not certain that systems with accretor expansion but no L2-overflow will not lead to a merger; hence, we include all systems with accretor expansion as potential merger progenitors. The Case A systems with $q_i \leq 0.15$ go into runaway mass transfer (yellow filling) and are expected to lead to a merger. The Case Be systems with $q_i = 0.15 - 0.25$ and $\log_{10}(a_i/R_\odot) \sim 1.30 - 1.34$ also go into runaway mass transfer (yellow filling) and are expected to lead to a merger.

3.1.2 Merger progenitor properties

Practically all population-synthesis models, such as `binary_c` as used in de Mink et al. (2013) and Schneider et al. (2014), and also more detailed binary grids, decide whether a system merges from the properties of the binary at the onset of Roche-lobe overflow. Having identified the regions of the $M_{1,i} = 10.16 M_{\odot}$ grid that are expected to lead to a stellar merger (Fig. 3), we now examine how much the primary’s mass changes over the course of the mass-transfer phase(s) that precede the point at which we deem the system to merge, and what this implies for the choice of merger progenitor mass and the evolutionary stages. In our models, we record the primary’s mass at the time the simulation terminates as $M_{\text{p,final}}$. We compare this to the primary’s mass at the onset of the (first) mass-transfer phase, $M_{\text{p,mt,i}}$, and define

$$\Delta M \equiv M_{\text{p,mt,i}} - M_{\text{p,final}}, \quad (3.1)$$

to quantify how much mass evolution is missed if we were to merge the systems at the onset of mass transfer.

A small number of studies in the literature examine the properties of the binary system at the onset of mass transfer and decide whether to merge. de Mink et al. (2013), in their standard treatment of contact formation, determine whether a binary system merges by evaluating the mass ratio $q = M_2/M_1$ at the onset of Roche-lobe overflow alone: if q falls below a critical value, the system is classified as a merger without evolving the mass-transfer phases for that system, and for post-main-sequence donors an analogous onset-only criterion with $q_{\text{crit,HG}} = 0.4$ is used. Schneider et al. (2014) adopt the same onset-based criterion and additionally construct the merger product directly from the pre-interaction masses, assuming a fixed 10% mass loss and 10% envelope mixing rather than further evolving them.

Notably, de Mink et al. (2013) themselves caution that reaching contact does not necessarily imply an immediate merger, since a star that only marginally overfills its Roche lobe may detach again once thermal equilibrium is restored. This tells us that the onset-only criterion does not capture the full picture. This is precisely the limitation we set out to quantify since neither study evolves the primary through the mass-transfer phase(s) itself, so neither can capture how much mass the primary has already lost (or, in some cases, regained) by the time contact is actually reached. To make this quantitative for our own grid, Fig. 4 shows ΔM across the same q_i vs $\log_{10}(a_i/R_{\odot})$ plane, restricted to the Case A and Case Be systems bounded by the dark blue lines, i.e. the same merger-progenitor parameter space identified in Fig. 3.

Case AB is a special case with 2 phases of mass transfer. The first phase occurs when the primary is on the main sequence, and the second phase occurs when the primary has left the main sequence (Case A + Case B). Across most of the accretor-

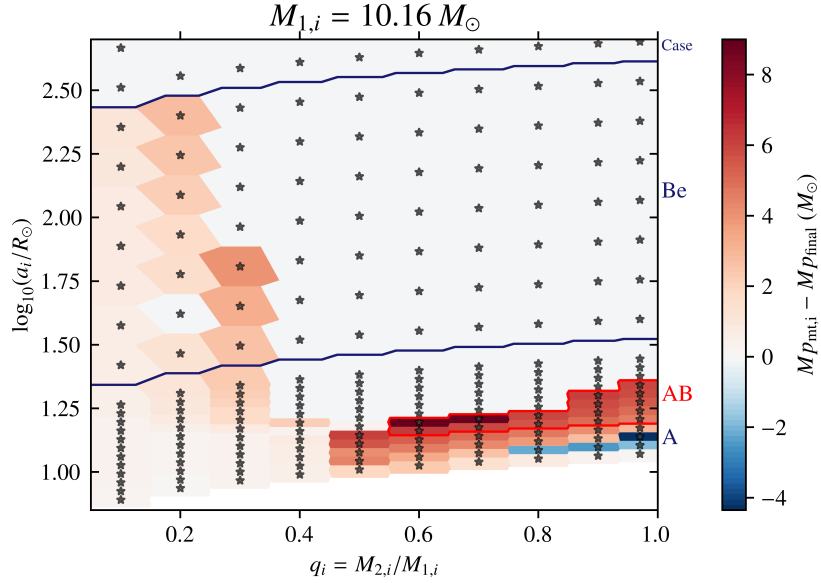


Figure 4: Difference $\Delta M = M_{p,mt,i} - M_{p,final}$ between the primary’s mass at the onset of mass transfer, $M_{p,mt,i}$, and the primary’s final mass, $M_{p,final}$ (“final” is the point at which the simulation terminates, which is also the point at which we merge the system), for the same $M_{1,i} = 10.16 M_{\odot}$ grid slice as in Fig. 3. Positive regions indicate substantial net mass loss by the primary over the mass-transfer phase(s) preceding contact; negative regions indicate net mass regain by the primary due to a reverse mass-transfer phase, discussed below. The AB contour marks a special mass transfer case explained below, and dark blue lines mark the same Case A/Case Be boundaries as in Fig. 3.

expansion part of the grid with $q_i \geq 0.45$, ΔM is more significant and reaches up to ~ 6 to $8 M_{\odot}$ and ~ -2 to $-4 M_{\odot}$, showing that the primary continues to lose or gain substantial mass throughout the mass-transfer phase compared to the lower mass ratio regions. This partition is an indicator of the timescales at which accretor expansion is reached. Systems with $q_i \leq 0.45$ reach accretor expansion on a thermal timescale, whereas systems with $q_i \geq 0.45$ reach it on nuclear timescales.

The binary systems with ΔM becoming negative (down to $\sim -4 M_{\odot}$) are the ones in which the primary is *more* massive before merger than it was at the onset of mass transfer. This arises from mass-ratio reversal: after the primary transfers mass to the secondary in the first mass-transfer phase, the secondary can become the more massive star and, in a subsequent phase, itself overflow its Roche lobe and transfer mass back onto the primary. By the time the computation is terminated, the primary has regained mass through this reverse mass-transfer episode, making $M_{p,final}$ larger than $M_{p,mt,i}$. We checked the binary evolution of these systems individually and find that the reverse mass-transfer episode always takes place while the system is in a contact phase, i.e. while MESA’s contact scheme is active and both components overflow their Roche lobes. The mass flip is therefore not a detached, semi-detached reverse Roche-lobe overflow but rather a consequence of the contact scheme redistributing mass between the two

stars once the secondary has become the more massive component.

Therefore, taking $M_{p,mt,i}$ as the merger progenitor mass, as in the onset-based prescriptions as done in de Mink et al. (2013) and Schneider et al. (2014), would therefore overestimate/underestimate the true primary and secondary mass at the time of merger. It would, moreover, get their interior structures wrong, and those structures determine the structure of the merger product. This motivates our choice throughout this thesis to define merger progenitor properties at the termination rather than at the onset of mass transfer. Concretely, we deem a system to merge at the end of its binary calculation and take both progenitor profiles from the last available model, so that the masses and interior structures we feed into the merger prescriptions are those the stars have when the binary evolution can no longer be followed. The only exception is the single system discussed in Sect. 3.1.3, which terminates on numerical failure rather than on one of the stopping conditions of Sect. 2.1.1.

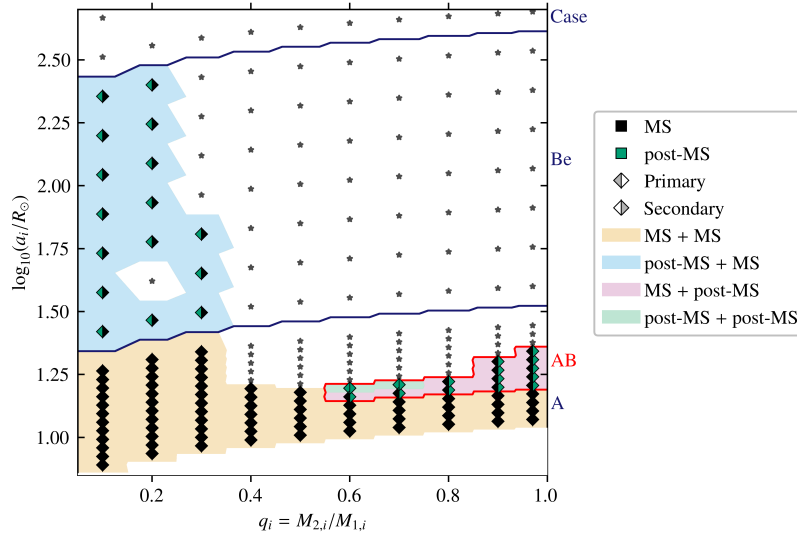


Figure 5: Evolutionary state of the primary (left half of each marker) and the secondary (right half) at the moment of merger in the initial mass ratio–separation plane for the 100 merging systems. Black marks a star still burning hydrogen in its core (MS) and teal marks a star that has exhausted core hydrogen (post-MS). The background shading indicates the four resulting (primary + secondary) combinations, listed in the legend. Systems marked with a plain star are computed binary models that do not merge. Dark blue lines mark the mass-transfer Case Boundaries and the red contour the Case AB region, as in Fig. 3.

In Fig. 5, we present the evolutionary stages of the progenitor stars of the binary systems at the point of their computation when we deem them to merge, in the style of Fig. C.1 of Henneco et al. (2024a). Every Case A system merges two main-sequence stars, and every Case Be system merges a post-main-sequence primary with a main-sequence secondary. In the Case AB strip, we see that most of the system merges a main-sequence primary with a post-main-sequence secondary, making an exception for

the only 2 binary systems which are post-main-sequence primary with a post-main-sequence secondary in our grid (at $q_i = 0.6$, $a_i = 15.68 R_\odot$ and $q_i = 0.7$, $a_i = 16.20 R_\odot$).

3.1.3 A system terminated by numerical failure

Of the ~ 100 merging systems identified in Sect. 3.1.1, one system does not reach any of the termination conditions of Sect. 2.1.1 and instead stops on numerical failure. This is the $q_i = 0.6$, $a_i = 15.68 R_\odot$ Case AB system, and because it is the only such case in our merging region, we examine it individually rather than applying an automated criterion. Its evolution is shown in Fig. 31.

The system undergoes two mass-transfer episodes. The first begins while the primary is on the MS and reverses the mass ratio, leaving the secondary as the more massive component (Figs. 31d and 31e). The second begins after the primary has left the MS. Contact is established during this second episode, marked in Fig. 31a, at which point both components overfill their Roche lobes in Fig. 31b. The mass-transfer rate rises to $\log_{10}(\dot{M}_{\text{trans}}) \approx -4 M_\odot \text{yr}^{-1}$ over the same interval (Fig. 31c), which exceeds the thermal-timescale rate $\dot{M}_{\text{KH}} = M_1/\tau_{\text{KH},1} \approx -4.23 M_\odot \text{yr}^{-1}$. The calculation fails shortly afterwards, before $\dot{M}_{\text{trans}}$ can reach the imposed ceiling of $10 M_\odot \text{yr}^{-1}$.

The system therefore satisfies our merger criteria before the numerical failure occurs. It has established contact, and it is doing so while the mass-transfer rate is rising faster than the primary can adjust to it. We consequently retain it as a merger progenitor and take the progenitor profiles from the last converged model, and we note that on the timescale over which the failure occurs the primary's mass changes by less than $10^{-7} M_\odot$, so the choice of exactly which model to use does not affect the merger outcome. This is the only system in the grid to which this treatment is applied.

3.2 Initial evolutionary state of the merger products

Now that we know which systems to merge and when to merge them, we must choose the appropriate merger prescriptions to apply to the selected systems.

3.2.1 Classification of merger products: passive scalar and origin of the merged star's core

To characterize the merger products, we track two properties of the merger product: its evolutionary state (MS or post-MS) and the origin of the material that ends up in its core. The latter is understood using a *passive scalar*. Before merging, every mass element (or shell) of the primary is assigned a passive scalar value of 1 and every mass element of the secondary a value of 0. After the merger prescription has redistributed the material, the passive scalar in a given mass shell is the mass-weighted fraction of that shell originating from the primary: a value of 1 marks material purely from the

primary, 0 marks material purely from the secondary, and intermediate values mark mixtures of the two.

Both prescriptions used here, entropy sorting (ES) and PyMMAMS (Sect. 2.2), build the merger product by ordering the progenitor material by its entropic variable with and without shock heating the material, respectively. As Glebbeek and Pols (2008) and Glebbeek et al. (2013) show, stellar evolution lowers the entropic variable in the core of the primary faster than in the core of the secondary, so an evolved primary tends to retain the lowest-entropy core.

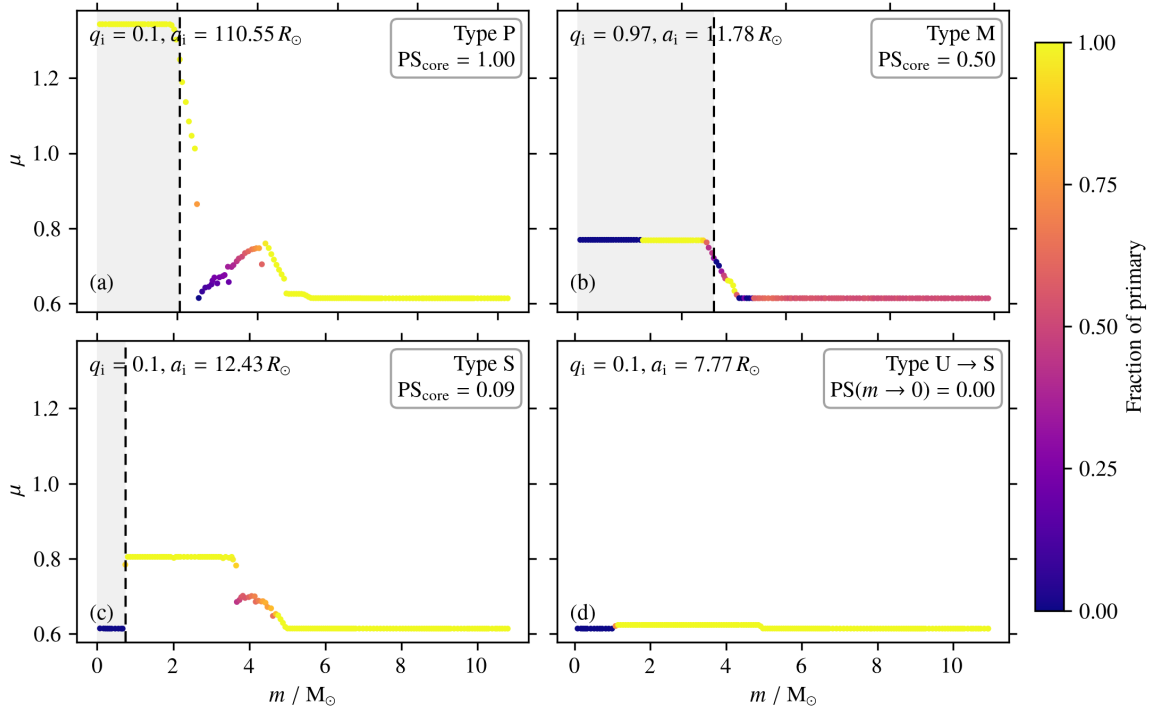


Figure 6: Mean molecular weight μ as a function of enclosed mass m for four merger products, one for each type. Each point is a mass shell of the merger product and is colored by its passive scalar. Yellow marks material purely from the primary and dark blue material purely from the secondary. The dashed vertical line marks the core boundary, defined as the innermost shell at which μ has changed by more than 5% relative to its central value, and the shaded area is the core region over which PS_{core} is averaged. The class and the value of PS_{core} are given in the upper right of each panel, and the binary system is identified in the upper left. *Top left:* Type P, *top right:* Type M and *bottom left:* Type S. *Bottom right:* a system with an undefined core, for which the flat inner μ profile means that no boundary can be located and no vertical line is drawn. For these systems, we assign the type by hand (here, Type S). All of these merged stars were constructed using ES.

Firstly, we define the core of the merger product as the innermost region out to the mass where the mean molecular weight μ has changed by 5% relative to its central value, and we take the average passive scalar over this region, PS_{core} . Following the case classification of Glebbeek and Pols (2008), we label the merger product as:

- **Type P** (primary-dominated core) if $\text{PS}_{\text{core}} > 0.8$;
- **Type M** (mixed core) if $0.2 \leq \text{PS}_{\text{core}} \leq 0.8$;
- **Type S** (secondary-dominated core) if $\text{PS}_{\text{core}} < 0.2$.

Figure 6 shows one merger product of each type. The core boundary, marked by the dashed line, sits where μ falls from its central value to the 95% value, and the color of the shells inside it is what sets the type: the core is built almost entirely from primary material in Fig. 6a, almost entirely from secondary material in Fig. 6c, and from a mixture of the two in Fig. 6b.

In a small number of systems, the mean molecular weight does not rise by 5% anywhere in the interior, so a core boundary cannot be located and PS_{core} is undefined. We flag these systems as *undefined core* and classify them by hand from the innermost passive-scalar value. An example is the $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.1$, $\log_{10}(a_i/R_{\odot}) \approx 0.89$ system shown in Fig. 6d, whose flat inner μ profile leaves no clear core boundary and whose central passive scalar is ~ 0 , so it is hand-marked as Type S.

3.2.2 Origin of the merged star’s core

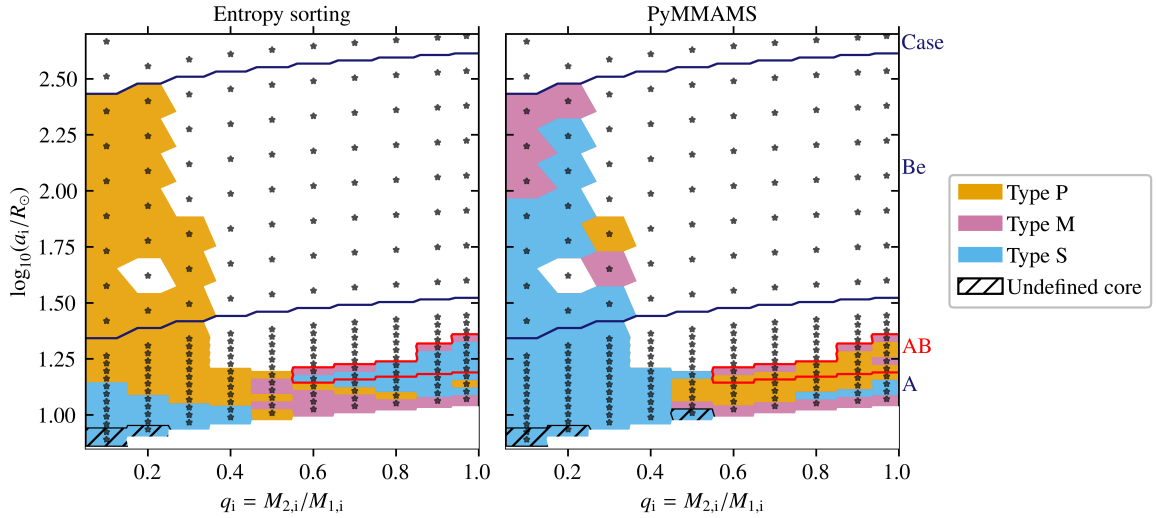


Figure 7: Core classification (Type P/M/S; see Sect. 3.2.1) of the merger products produced by entropy sorting and PyMMAMS for the grid. Hatching marks systems with an undefined core.

Figure 7 shows the core classifications (Type P, M, or S) of the merger products produced by ES and PyMMAMS. In the close-separation, low-mass-ratio ($q_i < 0.45$) Case A region, both prescriptions predict Type-S merger products. The two prescriptions differ most at wider initial separations. In Fig. 7, ES predicts a broad Type-P region, indicating that the evolved primary contributes the central material of

the merger product. In contrast, PyMMAMS (Fig. 7) produces Type-S and Type-M merger products predominantly over the same region.

The transition from Type S to Type P shown by ES can be understood from the entropy structure of the progenitor stars. Figure 8 compares the entropy profiles of two binaries with the same initial mass ratio ($q_i = 0.1$) but different initial orbital separations ($\log_{10}(a_i/R_\odot) \approx 0.89$ and ≈ 1.25), corresponding to locations in the Type-S and Type-P regions of Fig. 7a, respectively. Closer and wider initial orbits alter the evolutionary stage of the progenitor stars. The closer system contains the less evolved primary star, which is still close to the ZAMS at the onset of mass transfer, while the wider system contains a much more evolved primary star, close to the TAMS. The secondary’s entropy profile changes little in either case because its nuclear timescale is much longer than the primary’s. For the closer binary, the primary’s core has a higher specific entropy than the secondary’s core, implying that the lower-entropy secondary material is expected to sink to the center of the merger product, producing a Type-S core. In contrast, for the wider binary, the primary’s core has lower specific entropy and is therefore expected to occupy the center of the merger product, resulting in a Type-P core. This behavior follows the buoyancy sorting principle, in which lower-entropy material preferentially sinks during the merger.

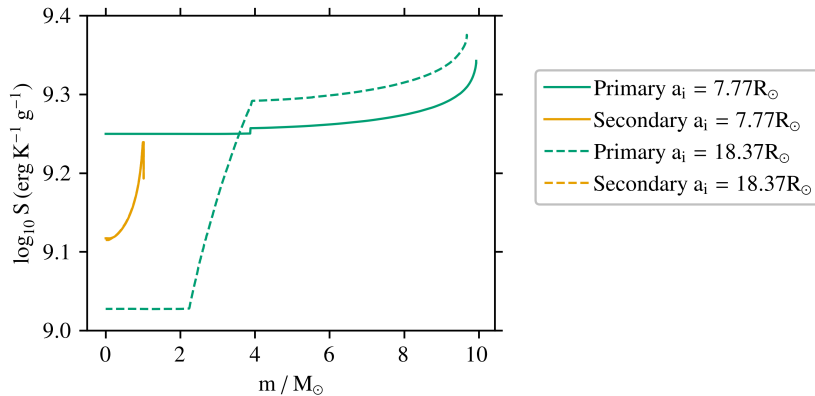


Figure 8: Specific entropy S (in units of $\text{erg K}^{-1} \text{g}^{-1}$) as a function of enclosed mass m for the primary and secondary at the end of the mass transfer phase, for two binary systems with the same mass ratio ($q_i = 0.1$) but different initial separations, $a_i = 7.77 R_\odot$ ($\log_{10}(a_i/R_\odot) \sim 0.89$) and $a_i = 18.37 R_\odot$ ($\log_{10}(a_i/R_\odot) \sim 1.25$).

Approximating the stellar material as an ideal gas, the Sackur–Tetrode equation gives the specific entropy as

$$s = \frac{\mathcal{R}}{\mu} \ln\left(\frac{T^{3/2}}{\rho}\right) + \text{const.}, \quad (3.2)$$

and combining this with the central scaling relation $T_c \propto M^{2/3} \rho_c^{1/3}$ yields

$$S_c \propto \frac{1}{\mu_c} \ln \left(\frac{M^2}{T_c^{3/2}} \right). \quad (3.3)$$

Hydrogen burning raises the mean molecular weight of the core, so the pre-factor μ_c^{-1} falls. At the same time, the core contracts as it burns, so T_c rises and the logarithmic term falls with it. μ_c roughly doubles between ZAMS and TAMS, from ≈ 0.62 for a hydrogen-rich core to ≈ 1.3 once the core is helium-rich, whereas the logarithmic term changes only by a few per cent. The central entropy of a star therefore drops as it crosses the main sequence. This is what sets the specific entropy profile in Fig. 8. In the wider system, the primary is close to the TAMS and has built up a helium core, so its central entropy has fallen below that of the secondary, which has barely evolved on its much longer nuclear timescale and still has its ZAMS composition. For the closer binary, neither star has evolved appreciably, so μ_c is nearly the same in both, and the ordering is set instead by the mass dependence of Eq. (3.3). Hydrostatic equilibrium gives $\rho_c \propto M/R^3$, which falls with mass for the main-sequence relation $R \propto M^{0.6-0.8}$, so substituting $T_c \propto M^{2/3} \rho_c^{1/3}$ into Eq. (3.3) gives $S_c \propto \mu_c^{-1} \ln(M/\rho_c^{1/2})$, in which the central density decreases with stellar mass. The low-mass secondary therefore has the lower central entropy and sinks to the center of the merger product for closer orbital separation.

The disagreement with PyMMAMS therefore does not arise from the buoyancy argument itself, but from its treatment of shock heating during the merger. ES applies no shock heating, so it preserves more of the progenitor’s evolved structure and retains the post-main-sequence character of systems whose progenitor had already left the main sequence at merger. Whereas PyMMAMS applies a shock-heating prescription (as mentioned in Sect. 2.2.1) that raises the entropic variable of the material during a merger and promotes mixing of fresh hydrogen into the deep interior, rejuvenating the merger product and driving it back onto the main sequence even if the progenitor had already left the main sequence. This leads PyMMAMS to predominantly mark Type-S and Type-M merger products in a region where ES predicts Type-P merger products. PyMMAMS over-heats evolved primaries since PyMMAMS was fine-tuned using SPH head-on MS+MS merger simulations. The excess entropy deposited in the primary’s dense helium core raises its entropic variable, so that the core no longer has the lowest buoyancy and no longer sinks to the center. Secondary material takes its place, and the merger product is (incorrectly, as we show below) classified as Type S or Type M rather than Type P. This discrepancy is very clearly seen in the Case Be systems.

Another discrepancy we see in Fig. 7 is in the higher mass ratio systems, namely the Case A and Case AB systems at $q_i \gtrsim 0.6$. In these systems the primary fills its Roche lobe first and transfers mass to the secondary. The primary is stripped, and the

accreting secondary rejuvenates, so that by the time we deem for the system to merge, the secondary is the more evolved of the two. Fig. 5 shows the extreme form of this in the Case AB strip, where the secondary has left the main sequence altogether while the stripped primary has not, but the effect does not require the secondary to be a post-main-sequence. Throughout the high mass ratio part of the Case A region, both stars are still burning hydrogen in the core while the secondary has nonetheless been carried further along its main-sequence evolution than the primary. Either way the ordering of the entropic variable is the reverse of the Case Be situation, and the two prescriptions respond to it in the way already described. Entropy sorting leaves the progenitor entropies untouched, so the more evolved secondary retains the lower central entropy, sinks, and produces the Type S and Type M cores seen in Fig. 7a across this part of the grid. PyMMAMS instead shock-heats preferentially the dense core of the more evolved star, which here is the secondary, raising its entropic variable until it no longer has the lowest buoyancy. The less evolved primary then takes the center, and PyMMAMS marks these systems predominantly Type P in Fig. 7b. The disagreement between the two prescriptions in this region is therefore the same overheating of evolved cores that we identified for the Case Be systems, appearing with the opposite picture because mass transfer has reversed which progenitor is the more evolved one.

3.2.3 Evolutionary state of the merger product

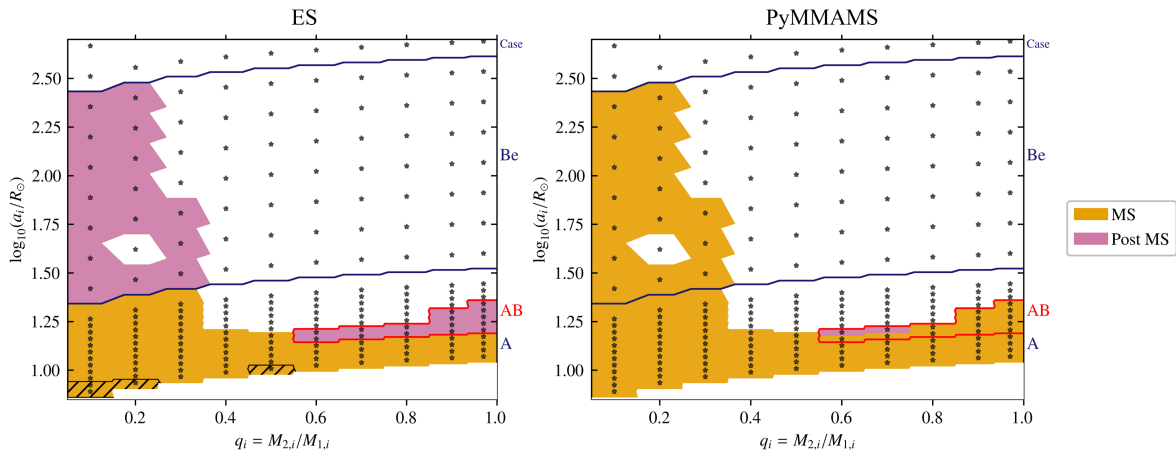


Figure 9: Evolutionary state (main sequence or post-main-sequence) of the merger product produced by ES (left) and PyMMAMS (right). Black hatching marks systems with an undefined core.

Fig. 9 shows whether the merger product is a main-sequence or a post-main-sequence star across the grid, for ES (Fig. 9a) and PyMMAMS (Fig. 9b) based on the PS_{core} 's chemical composition (i.e., the hydrogen mass fraction of the core). The two prescriptions give clearly different outcomes. Under ES (Fig. 9a), the Case Be and Case AB systems produce post-MS merger products, while the Case A systems

produce main-sequence merger products. In contrast, PyMMAMS (Fig. 9b) yields almost the entire merging region as main-sequence merger products, with only 2 systems producing post-main-sequence merger products. Upon closer examination, we found that these two systems are the only two post-MS + post-MS mergers in the grid (see Fig. 5). This suggests that PyMMAMS is better suited to MS + MS mergers, or even post-MS + post-MS mergers (since both progenitors would have a similar mean molecular weight μ profile). We verify whether ES is a good fit for post-MS + MS mergers in Sect. 4 by comparing the 1D merger products with our 3D SPH merger products for representative systems.

4 Three-dimensional hydrodynamical simulations

In this section, we present the outcome of the three-dimensional hydrodynamical calculation described in Sect. 2.3 and use it as an independent reference against which we assess the ES and PyMMAMS prescriptions. The hydrodynamical calculation is done only at one mass ratio, and its role here is to provide a self-consistent benchmark without making any assumptions in 1D prescriptions for one specific merger configuration, rather than to survey the parameter space.

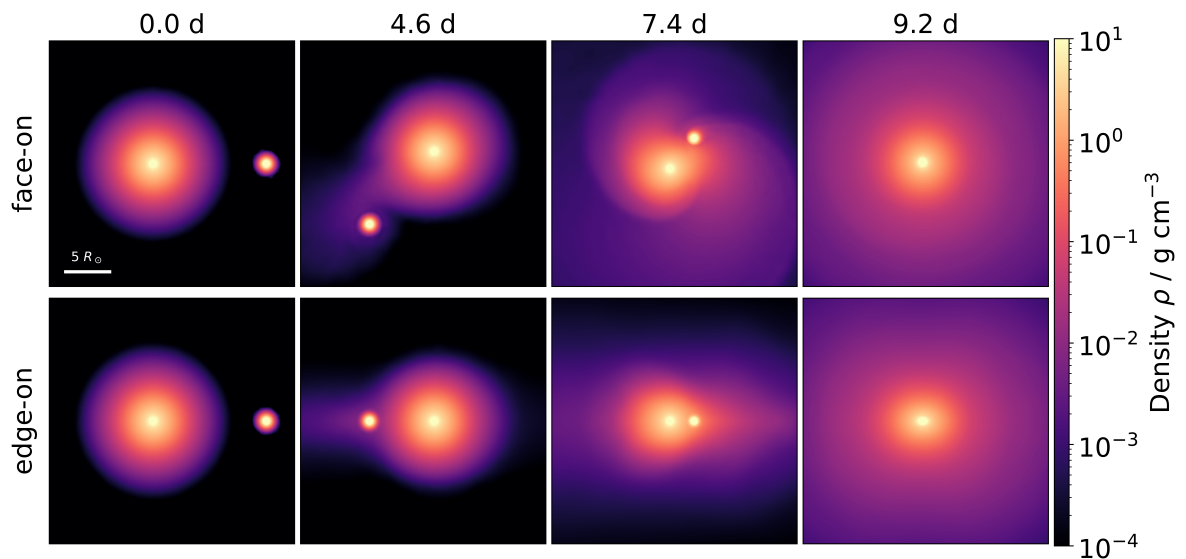


Figure 10: Face-on (top row) and edge-on (bottom row) SPH logarithmic density renders of the hydrodynamical simulation of a $10 M_{\odot}$ Hertzsprung-gap (HG) primary merging with a $2 M_{\odot}$ zero-age main-sequence (ZAMS) secondary, shown at four times spanning the simulation from the initial binary setup to the relaxed merger product. From left to right: the system at the start of the simulation (0.0 d); an early stage of the orbital inspiral (4.6 d); a render of a few snapshots before the dynamical merger (7.4 d); and the merger product after it has relaxed (9.2 d). Color bar encodes $\log_{10}(\rho/\text{g cm}^{-3})$.

Figure 10 shows the density evolution of the SPH merger simulation across four representative snapshots. At the start of the simulation (first column), the HG primary

and the ZAMS secondary are set up on a binary orbit and are still well separated and largely spherical. During the in-spiral (second column), the orbit has decayed further, and the primary's envelope begins to show tidal distortion. By the third column, taken a few snapshots before the merger, both stars are strongly deformed. By the fourth column, the two stars have merged, and the resulting merger product has relaxed into a single, centrally concentrated object with no visible signature of the original binary.

4.1 In-spiral and merger

Starting from a circular orbit at $a_i = 12 R_\odot$, the primary begins transferring mass to the secondary in the first orbits. The mass transfer is dynamically unstable, as expected for a radiative-envelope donor at this evolutionary stage and mass ratio, and the orbit contracts rapidly. The secondary cannot accrete the transferred material on an orbital timescale and instead plunges into the primary's envelope. The evolution of the orbital separation is shown in Fig. 11a.

The calculation was followed to $t_{\text{max}} = 14.58$ days ($t_{\text{max}} \approx 350$ hrs), corresponding to approximately an order of ten initial orbital periods, which is far longer than the duration of the dynamical phase itself and allows the merger product to be relaxed before it is analyzed.

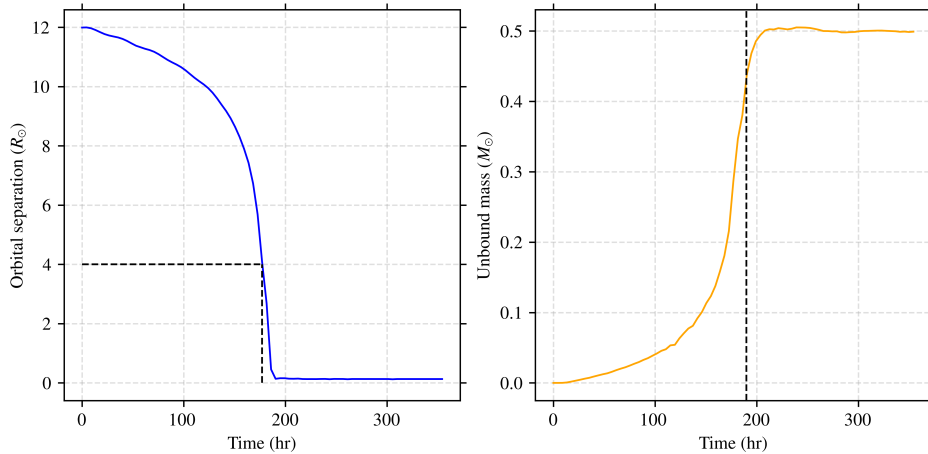


Figure 11: *Left:* Evolution of the orbital separation between the centers of the two stars during the in-spiral. The dashed vertical line marks the point at which the artificial conductivity is activated ($a = 4 R_\odot$, see Sect. 2.3.4). *Right:* Unbound mass during the in-spiral. The vertical dashed line marks the point in the simulation when the progenitor cores can no longer be distinguished.

A fraction of the envelope becomes unbound during the plunge, driven by the transfer of orbital energy to the envelope through the drag on the in-spiralling secondary. The evolution of the unbound mass is shown in Fig. 11b.

Adopting the criterion that a particle is unbound if its specific kinetic energy exceeds the modulus of its specific gravitational potential energy, $\frac{1}{2}v^2 + \Phi > 0$, a total of

$M_{\text{unb}} \approx 0.45 M_{\odot}$ becomes unbound, corresponding to 3.75% of the total system mass.

The unbound mass is the only quantity that showed any measurable dependence on the initial separation in the trial calculations (Sect. 2.3.3), and the variation there remained $< 0.83\%$. Since the unbound mass is small compared with the total mass of the system, this sensitivity does not affect the structure of the bound merger product, which is the quantity of interest here.

4.2 Structure of the merged star

The helium core of the primary has a mass of $\approx 3.3 M_{\odot}$, and the secondary material in the final merged star after relaxation is deposited no deeper than a mass coordinate of $\approx 3.3 M_{\odot}$ from the core. The secondary is disrupted, and its material is deposited in the primary's hydrogen-rich envelope, leaving the helium core chemically intact.

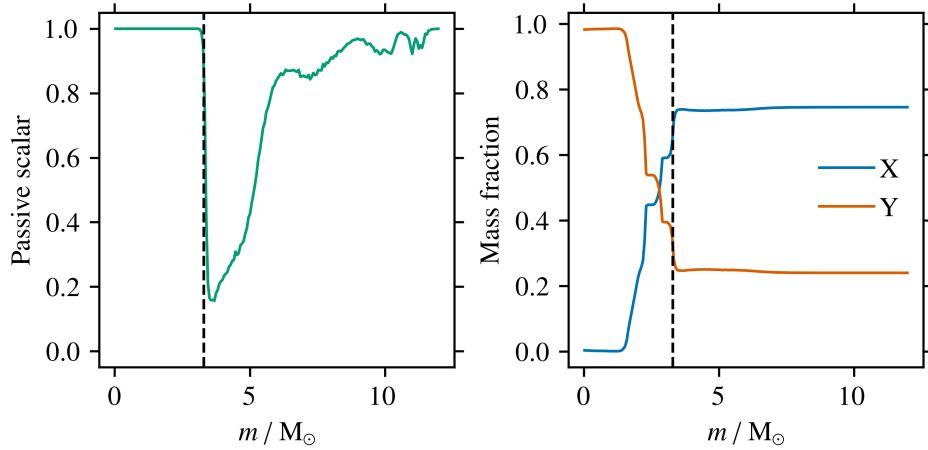


Figure 12: *Left:* Passive scalar of the merged star plotted against mass coordinate. *Right:* Hydrogen (X) and helium (Y) mass fractions against mass coordinate. Dashed vertical line at $\approx 3.3 M_{\odot}$ on both plots is the base of the H-rich envelope of the primary.

This behavior follows mainly from the entropy stratification of the two stars. The $2 M_{\odot}$ zero-age main-sequence secondary has a lower specific entropy than the extended envelope of the Hertzsprung-gap primary, and it therefore sinks through that envelope. It does not, however, have an entropy low enough to displace the helium core. The core's higher mean molecular weight, a consequence of its helium enrichment, makes it *more* buoyant at fixed entropy, since $s \propto \mu^{-1} \ln(P/\rho)$ in the gas-pressure-dominated limit implies a lower ρ at the same s and P for higher μ . The core remains centrally concentrated because its specific entropy, set by its nuclear-burning history, is low enough that the resulting density excess dominates this compositional buoyancy. The merger product is thus a Type P merger product, i.e., the core forms almost entirely from primary material.

4.3 Comparison with the one-dimensional prescriptions

Figure 13 compares the passive scalar profile of the merged star obtained from the SPH simulation with those obtained from ES, applied to the same pair of progenitor models.

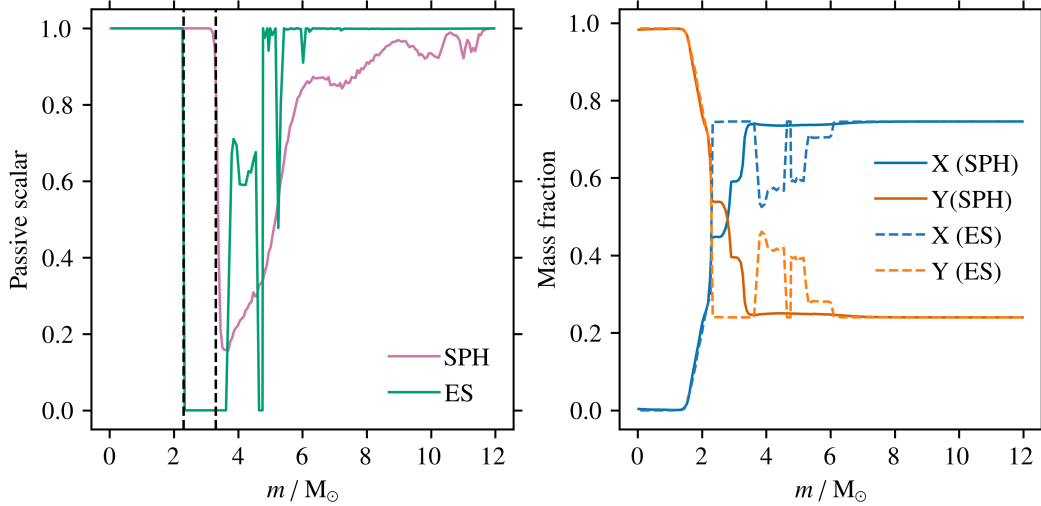


Figure 13: Comparing the three-dimensional SPH calculation with ES. *Left:* Passive scalar against mass coordinate in the merger products. The passive scalar is unity for material originating from the primary and zero for material originating from the secondary. The vertical dashed lines mark the passive scalar drop given by ES (at $2.3 M_{\odot}$) and SPH (at $3.3 M_{\odot}$). *Right:* Hydrogen (X) and helium (Y) mass fractions against mass coordinate in the merger product, comparing the three-dimensional SPH calculation (presented as solid lines) with ES (presented as dashed lines).

The hydrodynamical calculation and the ES prescription agree closely. Both place the secondary material entirely outside the primary’s helium core and therefore classify the merger product as Type P. The ES prescription deposits the secondary immediately outside the pure helium core, at $\approx 2.3 M_{\odot}$, whereas in the three-dimensional calculation the secondary comes to rest around one solar mass further out, at $\approx 3.3 M_{\odot}$. The two profiles also differ in detail at higher mass coordinates, where the three-dimensional merger product shows a broader and less sharply defined transition than the one-dimensional prescription.

These discrepancies are expected and do not undermine the agreement in classification. ES redistributes material according to the specific entropy of the progenitor models and, by construction, does not account for the entropy generated during the merger itself. In the SPH calculation, shock heating during the in-spiral raises the entropy of the primary and the secondary material, which is otherwise conserved due to the adiabatic nature of the simulations. This puts the secondary’s material at a larger mass coordinate than a purely adiabatic sorting of the initial profiles would predict. Additional processes are absent from the one-dimensional treatment, in particular the

hydrodynamical mixing driven by the shear between the in-spiralling secondary and the envelope, and the mass loss associated with the unbound material.

PyMMAMS places material from the secondary at the center of the merger product and returns a Type S merged star, in direct contradiction with the three-dimensional calculation (see Fig. 14). The two one-dimensional prescriptions predict different objects. Because PyMMAMS applies an entropy increase to the progenitor mass shells to mimic the effect of shock heating based on calibration from MS+MS head-on collision mergers, it overheats the primary’s core to a higher entropic variable than the secondary, sorting the shells so the secondary ends up in the central region of the merged star. The hydrodynamical calculation does not reproduce this sorting for our configuration.

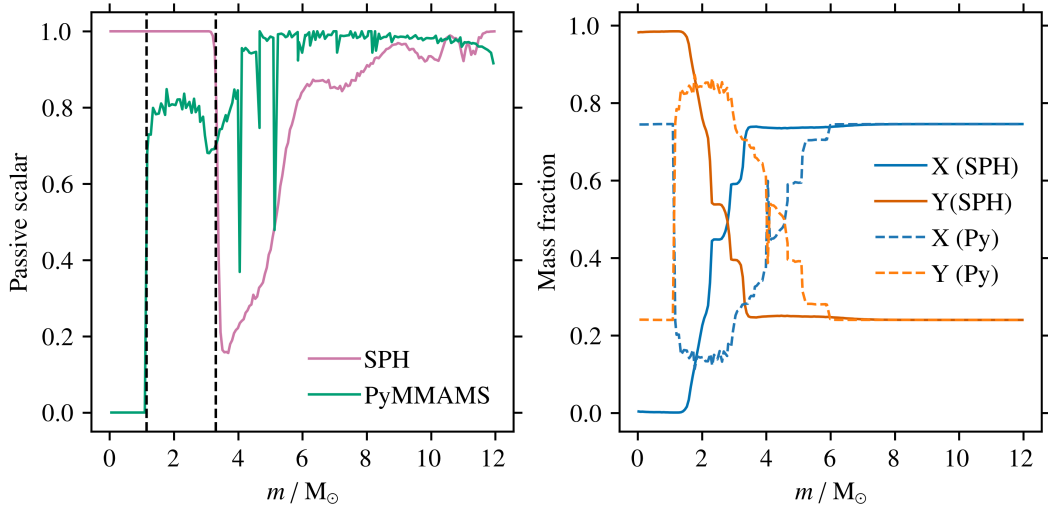


Figure 14: Comparing the three-dimensional SPH calculation with PyMMAMS. *Left:* Passive scalar against mass coordinate in the merger products. The vertical dashed lines mark the passive scalar drop given by PyMMAMS (at $1.15 M_{\odot}$) and SPH (at $3.3 M_{\odot}$). *Right:* Hydrogen (X) and helium (Y) mass fractions against mass coordinate in the merger product, comparing the three-dimensional SPH calculation (presented as solid lines) with the PyMMAMS (presented as dashed lines).

In both SPH and ES merger products, the primary’s helium core survives with its composition profile intact, and the merger product retains a core mass close to that of the primary progenitor while acquiring a rejuvenated, hydrogen-rich envelope. This structure gives rise to the blue supergiant merger products discussed in Sect. 5.2. The PyMMAMS-generated merger product, with a significant amount of hydrogen carried to the center, has a different chemical composition and would be brought back to the MS.

4.4 Sensitivity to the artificial conductivity

We check the sensitivity of the adopted value of α_u to see if it alters the merger product. Figure 15 shows the passive-scalar profile obtained with $\alpha_u = 0, 0.1$ and 1.0 where $\alpha_u = 0$ is when we turn off any artificial conductivity.

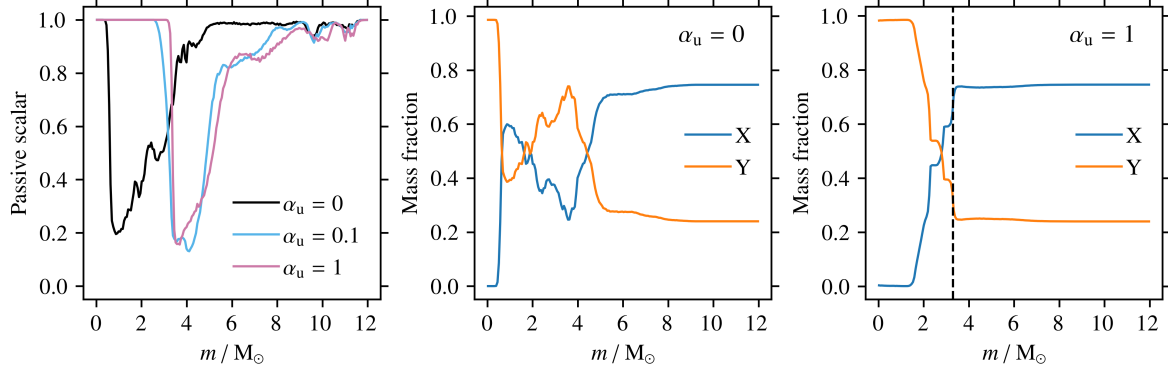


Figure 15: Passive scalar against mass coordinate in the merger product for three values of the artificial conductivity parameter, $\alpha_u = 0, 0.1$ and 1.0 , applied after the switching point. With $\alpha_u = 0$ the secondary is carried spuriously deep into the helium core of the primary.

Without artificial conductivity, the thermal energy discontinuities are not smoothed out, and the secondary is carried spuriously into the helium core. With active conductivity, the secondary sits outside the helium core once the merged star has relaxed. The results for $\alpha_u = 0.1$ and $\alpha_u = 1.0$ are consistent with one another, which indicates that the Type P classification reported above is a converged result and does not depend on the adopted α_u value.

5 Long-term evolution of the merger products

In Sect. 3 we showed which binary systems merge and when, and how to build the merger product’s chemical and thermal structure. We now take those merger products, engineer their structure, relax them, and evolve them with MESA until core helium exhaustion. We study the further evolution of the merger products and contrast it with the evolution of single stars of comparable mass, focusing particularly on X, Y , and Z , where the star burns hydrogen and helium, and how long it spends in each region of the HR diagram. The Case A merger products are built with PyMMAMS because both progenitors are still on the main sequence when they merge, and the Case AB and Case Be merger products are built with entropy sorting (see Sect. 3.2.3).

Section 5.1 describes the long-term evolution of Case A (MS+MS) merger products as MS stars and blue-loop stars. Section 5.2 describes the long-term evolution of merged stars from the Case Be (post-MS+MS) region of the grid. Section 5.3 describes the

long-term evolution of merged stars from the Case AB region of the grid containing MS+post-MS and post-MS+post-MS systems. We track the occurrence of long-lived blue supergiants and blue-loop stars in our grid of merged stars (see Sect. 5.4). We look at the HR diagram distribution of the merged stars and compare it with the BLOeM survey in Section 5.5. Then we look into the carbon–oxygen core masses of the merger products at core helium exhaustion in Section 5.6.

5.1 Case A mergers

We start with the Case A merger products, which are mergers of two main-sequence stars, and we use the profiles built with PyMMAMS since they agree better with SPH simulations for this type of merger (as presented in Heller et al. (2025)). Fig. 16 shows the evolution of the merger product of the binary system with $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.2$, and $a_i = 13.8 R_{\odot}$.

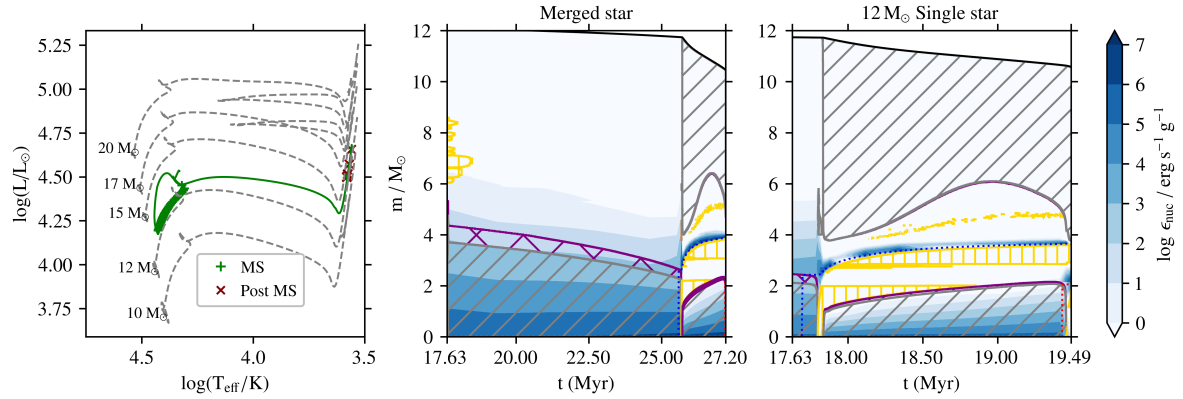


Figure 16: Evolution of the merger product of a Case A merger with $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.2$, and $a_i = 13.8 R_{\odot}$. *Left:* HR diagram evolution track of the merger product given in green, plotted over grey dashed single-star tracks of masses 10, 12, 15, 17 and $20 M_{\odot}$. Plus signs mark models that are still burning hydrogen in the core (MS), and dark red crosses mark models after core hydrogen exhaustion (post-MS), spaced by 2×10^5 yrs. The dense clump of green markers is the rejuvenated main-sequence phase. *Center:* Kippenhahn diagram of the merged star. *Right:* the same diagram for a $12 M_{\odot}$ single star. In both Kippenhahn panels, the color bar indicates the specific nuclear energy generation rate $\log_{10} \epsilon_{\text{nuc}}$, grey hatching marks convective regions, purple hatching marks convective overshooting, and yellow hatching marks thermohaline mixing. The red dotted line is the CO core mass, the blue dotted line marks the helium core boundary, and the black line marks the stellar surface.

The merger product is a $12 M_{\odot}$ star after the merger. It initially goes through a relaxation phase, which is shown in the figure (hook-like feature near $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.4$), and then settles back onto the main sequence. The green markers in the left panel are tightly packed near $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.3$, which is where the merger product spends most of the computed evolution, and the crossing of the Hertzsprung gap afterward takes less than the marker intervals (since it is a thermal timescale process), which

is why we do not see any markers in this region. The Kippenhahn diagram of the merger product shows a large convective hydrogen-burning core that is present from the moment the star settles into thermal equilibrium after the merger and recedes slowly in mass, just as it would in a single star.

The Kippenhahn diagrams start at the time of merger, i.e., at around $t \approx 17.63$ Myr. This means the binary system has already spent 17.63 Myr on the main sequence before merging, and the merger product has a fresh supply of hydrogen in its core and a rebuilt convective core around it. The right panel shows the Kippenhahn diagram of a $12 M_{\odot}$ single star, which is the same mass as the merger product at the time of merger. The single star exhausts central hydrogen at around 17.8 Myr, while the merger product keeps burning until roughly 26 Myr. This is the usual rejuvenation of main-sequence merger products (Braun and Langer, 1995; Glebbeek et al., 2008; Schneider et al., 2016).

The quantity we compare is the *remaining* main-sequence lifetime, $t_{\text{MS,rem}} = t(X_{\text{c}} \leq X_{\text{c,exh}}) - t_{\text{merge}}$ where $t(X_{\text{c}} \leq X_{\text{c,exh}})$ is the time when the core hydrogen abundance falls below the exhaustion value and t_{merge} is the time of the merger. The comparison is against a single star of the same mass evaluated at the merger age, not against a freshly formed ZAMS star. At the moment of the merger the nuclear clock of the merger product is partly reset.

How complete that reset is depends on the mixing physics. Braun and Langer (1995) showed, and Schneider et al. (2024) confirmed, that it is semiconvection which decides whether the convective core can grow through the mean molecular weight gradient left by the previous evolution, so the efficiency parameter α_{sc} effectively controls how far the rejuvenation goes. We use $\alpha_{\text{sc}} = 10$ (Appendix C), which is a very efficient value, so we expect our Case A merger products to rejuvenate essentially completely and to behave from then on like single stars of their new mass. This expectation matters for what follows because it allows us to compare the Case A merger products directly against the single-star tracks. Overshooting also overrides this. If overshooting exceeds the semi-convective layer, it will simply mix through and erase the semi-convective layer.

Once the merger product has left the main sequence, it crosses the HG and ignites helium at $\log_{10}(T_{\text{eff}}/\text{K}) \approx 3.6$, which is on the cool side of the threshold defined in Sect. 2.4.4. The merger product climbs the RGB and stays at a cool T_{eff} as it burns helium in the core and hydrogen in the shell, with a convective envelope. The merger product and the single star behave comparably in terms of time spent in the core helium-burning phase and have a comparable CO core at the end of the core helium-burning phase ($M_{\text{CO}} \approx 2.1 M_{\odot}$).

Blue loops. A subset of the Case A merger products evolves through blue loops as shown in Fig. 17. The model crosses the Hertzsprung gap, ignites helium close

to the Hayashi limit at $\log_{10}(T_{\text{eff}}/\text{K}) \approx 3.6$, then moves back towards higher effective temperature, spends a long time near $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.0$, and finally returns redward. The Kippenhahn diagram shows that the convective helium-burning core is present throughout, so the whole excursion occurs during core helium burning rather than after it, and the envelope is radiative during the hot part of the loop and becomes convective again as the model returns.

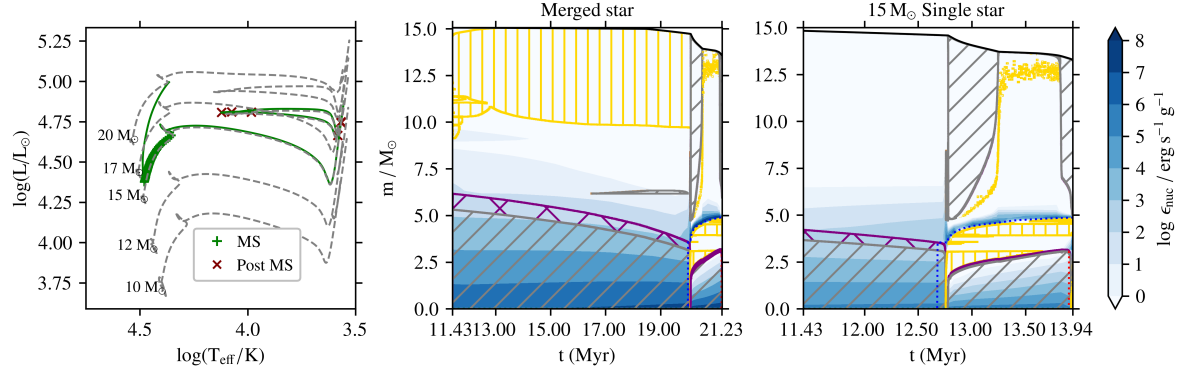


Figure 17: Evolution of the merger product of a Case A merger with $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.5$, and $a_i = 11.94 R_{\odot}$, which evolves through a blue loop. *Left:* HR diagram evolution track with markers as in Fig. 16. The model ignites helium at $\log_{10}(T_{\text{eff}}/\text{K}) \approx 3.6$, moves blueward to $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.0$, where the dark red markers pile up, and returns redward close to core helium exhaustion. *Middle:* Kippenhahn diagram of the merger product against time t (Myr). *Right:* Kippenhahn diagram of a single star against time t (Myr), with colors and hatchings as in Fig. 16.

Blue loops are not a merger-specific phenomenon. Our own single-star models of 15 and $17 M_{\odot}$ also show them, and they spend a substantial part of core helium burning in the loop $t_{\text{BL}}/t_{\text{He}} = 0.49$ and 0.61 , respectively. t_{BL} is the time spent by our model in the blue loop, burning helium in the core, and t_{He} is the time spent by our model burning helium in the core. What is specific to the mergers is which systems end up in the mass range where loops happen at all, and we return to that in Sect. 5.4.

5.2 Case Be mergers

We now turn to the merger products of mergers in which the primary had already left the main sequence (post-MS + MS mergers), which we build with entropy sorting (see Sect. 3.2.3). Fig. 18 shows the $q_i = 0.3$, $a_i = 44.87 R_{\odot}$ Case Be system merger product together with a $10 M_{\odot}$ single star. After the merger, the merger product undergoes a thermal relaxation phase roughly on its Kelvin-Helmholtz timescale, during which it first expands and then contracts as presented in Schneider et al. (2019). The model ignites helium already at $\log_{10}(T_{\text{eff}}/\text{K})|_{\text{He ign}} > \log_{10}(T_{\text{eff,BSG}}/\text{K}) = 3.7$ and then crosses the blue and yellow part of the HR diagram extremely slowly, accumulating markers all the way from $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.1$ down to 3.7 before finally reaching near the Hayashi

limit at the time of central helium exhaustion. The star lives in this temperature range for much longer than τ_{KH} : the single $10 M_{\odot}$ star track given as a grey dashed line passes through the same region on a thermal timescale and ignites helium already as a red giant and stays in the same temperature regime ($\log_{10}(T_{\text{eff}}/\text{K}) \approx 3.6$) till the next phase of evolution, i.e., core helium exhaustion. Looking at the Kippenhahn diagrams, the merger product has a convective helium-burning core and a radiative envelope throughout the core helium-burning phase, while the single star develops a deep convective envelope while burning helium in the core.

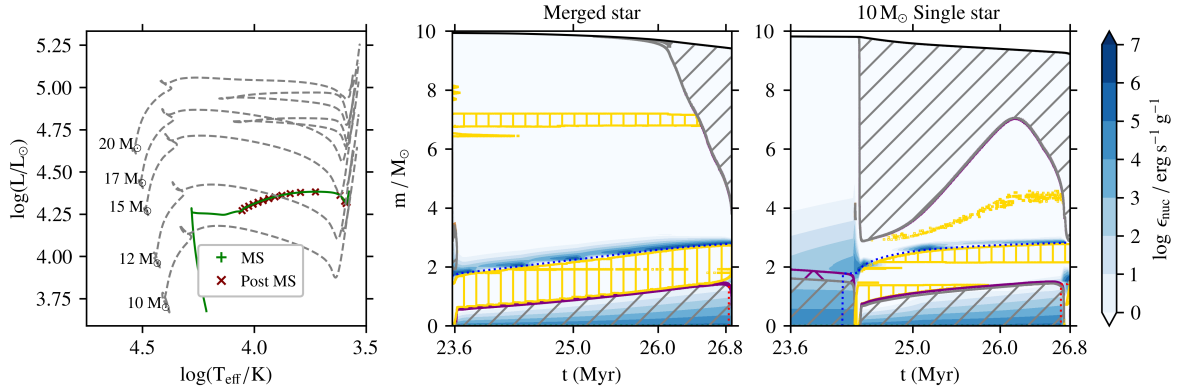


Figure 18: Evolution of the Case Be merger product of the $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.3$, $a_i = 44.87 R_{\odot}$ system, built with entropy sorting. *Left:* HR diagram evolution track with markers as given in Fig. 16, where the part with dark red markers spaced by 2×10^5 yrs is the core-helium-burning evolution. *Center:* Kippenhahn diagram of the merged star. *Right:* the same for the $10 M_{\odot}$ single star. Colors and hatchings as in Fig. 16. The single star develops the deep convective envelope typical of a red supergiant, which the merger product does not when it burns helium in the core.

In the literature, the quantity that controls this is the fractional core mass. The classical result of Podsiadlowski et al. (1992) is that a small $\xi_c = M_c/M_{\text{total}}$ favors a compact, blue configuration while a large one drives the star redward, and a merger with an evolved primary is an efficient way of producing a small ξ_c : the secondary is added entirely to the envelope while the primary’s helium core, already decoupled from it, is left untouched (see Sect. 3.2.2). This is the same channel that has been used to explain long-lived blue supergiants since Podsiadlowski et al. (1990), and more recently by Menon and Heger (2017) and Schneider et al. (2024).

Our merger products do not have quite the same internal structure as the long-lived blue supergiants of Schneider et al. (2024). In their models, the hydrogen-burning shell itself becomes convective once the temperature at the base of the hydrogen-rich envelope, which scales as

$$T_{\text{base}} \propto \frac{\mu_{\text{env}} M_{\text{env}}}{R_{\star}}, \quad (5.1)$$

exceeds roughly $0.02 M_{\odot} R_{\odot}^{-1}$, and such a star is best pictured as a relatively unevolved main-sequence star sitting on top of a helium star, with the convective hydrogen-

burning shell playing the role of the main-sequence convective core. Our merger products reach the blue and stay there without a convective hydrogen-burning shell. Either our models lie below that threshold and are blue for the classical fractional core mass reason alone, or the threshold does not transfer directly to this mass range.

The lowest mass ratios. The $q_i = 0.1$ systems behave differently from the rest of the post-MS + MS grid. Fig. 19 shows the evolution of the $q_i = 0.1$, $a_i = 110.55 R_\odot$ binary system’s merger product. Most of the Case Be mergers at this mass ratio evolve similarly to this merger product.

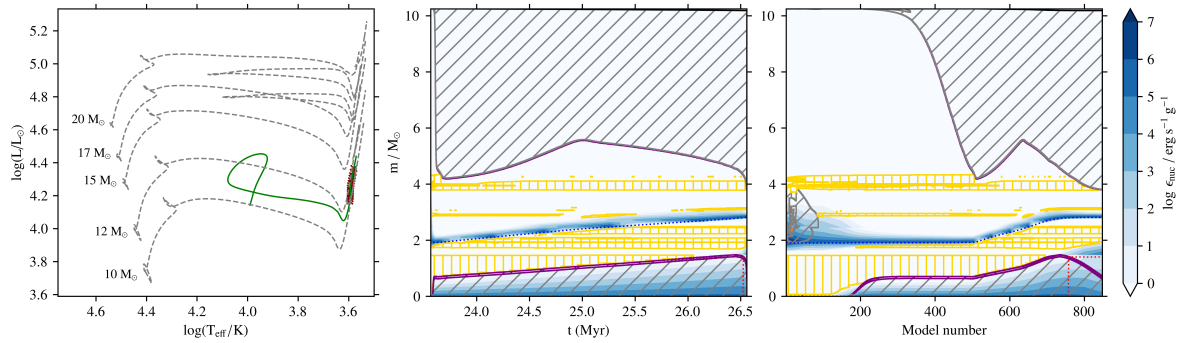


Figure 19: Evolution of the Case Be merger product of the $M_{1,i} = 10.16 M_\odot$, $q_i = 0.1$, $a_i = 110.55 R_\odot$ system, built with entropy sorting. *Left:* HR diagram evolution track with markers as given in Fig. 16. The excursion at $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.0$ is the initial relaxation of the merger product and not a blue loop, after which the model settles near the Hayashi limit and burns helium there, with the dark red markers piling up at $\log_{10}(T_{\text{eff}}/\text{K}) \approx 3.6$. *Center and right:* Kippenhahn diagrams against stellar age and against model number, respectively, with colors and hatchings as in Fig. 16. The deep convective envelope that develops after the relaxation is what a typical post-main-sequence star looks like while burning helium in the core.

The excursion visible in the left panel is not a blue loop. It is the initial relaxation of the merger product, after which the merger product simply behaves like a typical post-main-sequence star. The reason is that the secondary is only about $1 M_\odot$, so the merger barely changes the core-to-envelope mass ratio of the merger product and none of the structural conditions described above is met. These systems accordingly appear as neither blue loop nor hot-side ignition in the grid of Fig. 22. This also sets a rough lower limit on how much mass must be added before a merger changes the evolutionary outcome at all.

5.3 Case AB mergers

The Case AB systems merge a stripped, less evolved primary with a secondary that has evolved past the main sequence by the preceding mass transfer (Fig. 5), so that the evolved star entering the merger is the accretor rather than the donor.

5.3.1 MS + post-MS mergers

Fig. 20 shows the merger product of the $q_i = 0.6$, $a_i = 14.5 R_\odot$ system, built with entropy sorting, together with a single star of comparable mass ($15 M_\odot$).

After the relaxation phase, the merger product is a $15.1 M_\odot$ star. It ignites helium on the hot side of the threshold defined in Sect. 2.4.4 and then crosses the blue and yellow part of the HR diagram slowly at $\log_{10}(L/L_\odot) \approx 4.9$, accumulating markers from $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.3$ down to 3.6 before reaching the Hayashi limit close to central helium exhaustion. It remains a blue/yellow supergiant for essentially the entire core helium-burning phase.

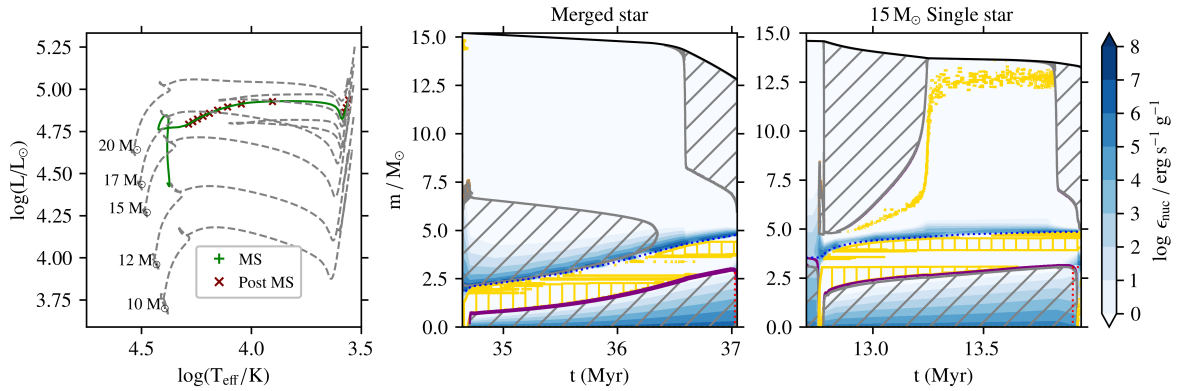


Figure 20: Evolution of the merger product of a Case AB merger with $M_{1,i} = 10.16 M_\odot$, $q_i = 0.6$, and $a_i = 14.5 R_\odot$, built with entropy sorting. *Left:* HR diagram evolution track of the merger product in green, plotted over grey dashed single-star tracks. *Center:* Kippenhahn diagram of the merged star against stellar age. *Right:* the same for a $15 M_\odot$ single star. Colors and hatchings as in Fig. 16. The convective region on top of the helium core between $m \approx 2.5$ and $6.7 M_\odot$ is the convective hydrogen-burning shell, which the single star does not develop; it disappears at $t \approx 36.3$ Myr as the deep convective envelope forms and the star moves to the Hayashi limit.

The internal structure behind this is different from that of the Case Be merger products of Sect. 5.2. The Kippenhahn diagram shows a convective region sitting directly on top of the helium core, extending from $m \approx 2.5$ to $6.7 M_\odot$ and coinciding with the hydrogen-burning shell, with a radiative envelope above it. This is precisely the configuration described by Schneider et al. (2024), in which the hydrogen-burning shell becomes convective, and the star is best pictured as a relatively unevolved main-sequence star sitting on top of a helium star, with the convective shell playing the role of the main-sequence convective core. The Case AB merger products therefore reach the blue by the same structural route as the long-lived blue supergiants of Schneider et al. (2024), whereas the Case Be merger products of Sect. 5.2 reach it with a radiative shell and by the classical fractional core mass argument.

The convective shell survives until the star approaches the Hayashi limit at $t \approx 36.4$ Myr. At that point, a deep convective envelope develops from the surface inwards,

reaching into $m \approx 7.5 M_\odot$, and the hydrogen-burning shell becomes radiative. From then on, the merger product is a red supergiant with the same envelope structure as the single star. The merger product also burns helium for considerably longer than the single star. Core helium burning lasts ≈ 2.5 Myr in the merger product against ≈ 1 Myr in the $15 M_\odot$ single star, so the merger both moves the star into the blue and keeps it there for longer than a single star of the same mass would spend burning helium at all. The carbon–oxygen core at central helium exhaustion, $M_{\text{CO}} \approx 3 M_\odot$, is nonetheless close to the single-star value of $3.05 M_\odot$ (Table 2).

5.3.2 Post-MS + post-MS mergers

Only two systems in the grid merge two stars that have both left the main sequence, at $q_i = 0.6$, $a_i = 15.68 R_\odot$ and $q_i = 0.7$, $a_i = 16.20 R_\odot$, both in the Case AB region (Fig. 5). These are also the only systems for which the two merger prescriptions agree on the evolutionary state of the merger product (Sect. 3.2.3). Fig. 21 shows the $q_i = 0.7$ merger product obtained from PyMMAMS and ES alongside the same $15 M_\odot$ single star. The Kippenhahn diagram given in Fig. 21b is for the evolution of the merged star obtained from PyMMAMS.

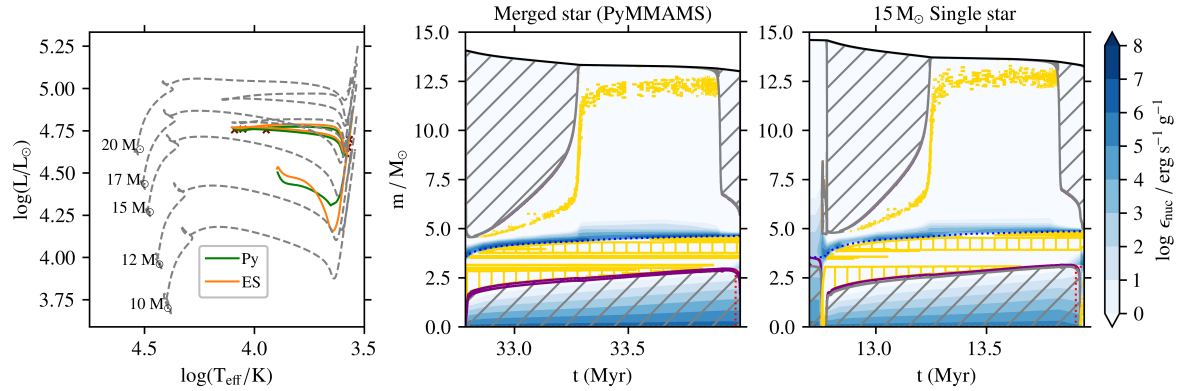


Figure 21: Evolution of the merger product of a Case AB post-MS + post-MS merger with $M_{1,i} = 10.16 M_\odot$, $q_i = 0.7$, and $a_i = 16.20 R_\odot$, built with entropy sorting and PyMMAMS. Panels, colors, and hatchings as in Fig. 20. The merger products and the $15 M_\odot$ single star follow nearly identical tracks in the HR diagram and have nearly identical interior structures throughout core helium burning.

The two merger products are almost indistinguishable. Both ignite helium close to the Hayashi limit and execute a blue loop that reaches $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.1$ at $\log_{10}(L/L_\odot) \approx 4.75$ before returning redward, both burn helium in a convective core that grows to $m \approx 2.9 M_\odot$, both show the same thermohaline region above the helium core boundary at $m \approx 3.5 - 4.5 M_\odot$, and in both the deep convective envelope retreats as the model moves blueward at $t \approx 33.3$ Myr and returns at $t \approx 33.9$ Myr as it comes back. The core helium burning phase spans the same ≈ 1.2 Myr in the two models and they end with the same carbon–oxygen core mass of $\approx 3.05 M_\odot$.

This is the agreement anticipated in Sect. 3.2.3. The disagreement between the two prescriptions everywhere else in the grid comes from the shock heating that PyMMAMS applies to the dense core of the more evolved progenitor, which raises its entropic variable and changes which progenitor ends up at the center. When both progenitors are evolved and carry similar mean molecular weight profiles, that heating is applied to both cores alike, the ordering of the entropic variable is preserved, and the two prescriptions build the same star. The post-MS + post-MS systems are thus the one regime in which the choice between entropy sorting and PyMMAMS is immaterial.

5.4 Occurrence of long-lived BSGs and blue-loop stars from stellar mergers

Fig. 22 is a contour plot that shows the occurrence of blue loops and hot-side ignitions (BSGs) in the initial mass ratio-orbital separation plane. Instead of a simple flag, we color each system by the fraction of its core-helium-burning lifetime spent in the relevant configuration, because a flag on its own would give the same visual weight to a 0.05 Myr crossing (thermal timescale crossing) and to a 2.6 Myr supergiant phase (nuclear timescale crossing).

Three things stand out. The Case AB strip just above the Case A boundary is uniformly dark red, meaning that these merger products are hot supergiants for essentially the whole of their core helium burning phase. The Case Be region at $q_i = 0.2$ – 0.3 and $\log_{10}(a_i/R_\odot) \approx 1.45$ – 2.4 is also mostly red, except for the mass ratio $q_i = 0.1$ where it behaves similar to a typical post-main-sequence star (see Sect. 5.2). In the Case A region at the closest initial separations, there is a compact group of blue cells at $q_i \approx 0.35$ – 0.85 , which are the blue loop merger products, some of them spending up to $t_{\text{BL}}/t_{\text{He}} \approx 0.74$ of core helium burning in the hot side. Of the 100 merging systems, 15 are flagged as blue-loop models and 20 as BSGs, with the remaining 65 falling into neither category.

The clustering of the blue loops in one region of the Case A part of the grid was initially puzzling, but the single-star models in Fig. 22 suggest a straightforward explanation. Our 15 and 17 $M_\odot$ single stars perform blue loops and spend 49% and 61% of core helium burning in them, while the 10, 12 and 20 $M_\odot$ models do not. In other words, at the physics and metallicity we adopt, there is a mass window in which blue loops occur.

Since the Case A merger products rejuvenate essentially completely (Sect. 5.1), they should behave from the end of the relaxation phase onwards like single stars of their post-merger mass. The merger product mass rises monotonically with q_i at fixed separation (Fig. 30). A first reading of Fig. 22 is therefore that the merger simply delivers the star into the mass range in which a single star would loop anyway. This

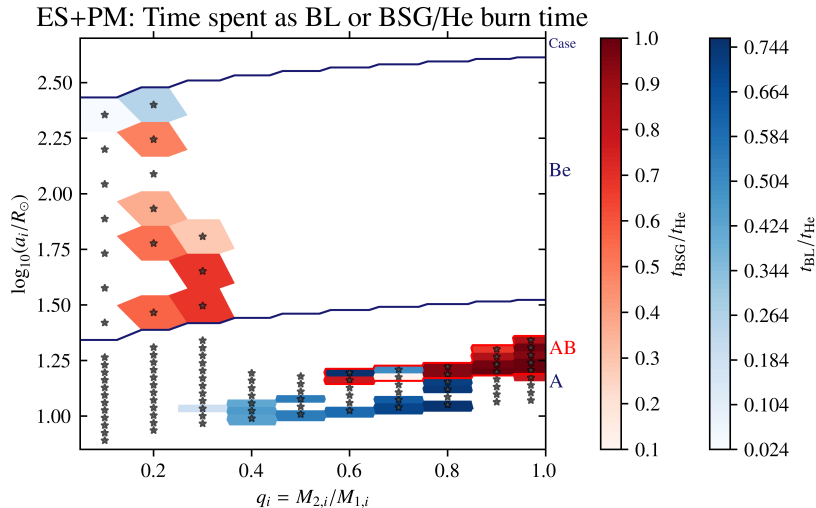


Figure 22: Fraction of the core-helium-burning lifetime spent as a blue supergiant ($t_{\text{BSG}}/t_{\text{He}}$, red scale) or in a blue loop ($t_{\text{BL}}/t_{\text{He}}$, blue scale), in the initial mass ratio–separation plane. The Case A part of the grid is built with PyMMAMS and the Case AB and Case Be parts with entropy sorting. Dark blue lines mark the Case Boundaries and the red contour the Case AB region, as in Fig. 3. Each star represents the computed merger product evolution, and the colored regions are nearest-neighbor interpolations between them. The two categories are mutually exclusive, so no cell carries both colors, and uncolored cells with a star are models flagged as neither. Note that the two color bars have different ranges.

cannot be the whole story since several merger products fall inside the mass range spanned by our looping single stars and nonetheless do not loop, so the post-merger mass is necessary but not sufficient.

Following Lauterborn et al. (1971) and Fricke and Strittmatter (1972), a blue loop is triggered when the hydrogen-burning shell burns outwards through the composition gradient left behind by the retreating convective core, removing the excess helium that sits above the shell and that keeps the star red and extended. Walmswell et al. (2015) isolated the three ways in which that excess helium acts, by giving opacity, the nuclear energy generation, or the mean molecular weight of helium in turn, and showed that the mean molecular weight is the decisive one. The opacity change favors looping only weakly, the increased fuel supply actively suppresses it, and the raised μ immediately above the burning shell drives the star redward. Whether a loop occurs then becomes a competition between the shell burning out through the gradient and the growth of the core, which drives the star back towards the Hayashi line.

Both prescriptions redistribute material when they build the merger product. Both therefore carry helium-rich material from the progenitors’ deep interiors out into the region that will later sit above the hydrogen-burning shell. Two merger products of the same mass can accordingly begin core helium burning with quite different amounts of excess helium above the shell, and on the argument above the one that carries more

should be the redder of the two and should loop less, or not at all. On this reading, the post-merger mass sets which part of the grid can loop, and the μ profile inherited from the merger decides which systems within that region actually do.

Figure 23 tests this directly for a matched pair of Case A merger products at $q_i = 0.6$ with essentially the same post-merger mass (16.05 and $16.01 M_\odot$ at $a_i = 10.6$ and $11.47 R_\odot$ respectively). The former performs an extensive blue loop, spending 60.6% of core helium burning above the $\log_{10}(T_{\text{eff}}/\text{K}) = 3.7$ threshold and reaching $\log_{10}(T_{\text{eff}}/\text{K}) = 4.14$; the latter barely leaves the Hayashi line, spending only 0.4% of core helium burning above the threshold and reaching no hotter than $\log_{10}(T_{\text{eff}}/\text{K}) = 4.07$. We locate the hydrogen-burning shell at helium ignition as the mass coordinate of peak ϵ_{nuc} within the hydrogen-rich region, and measure the excess helium sitting above it as $\int (Y(m) - Y_{\text{env}}) dm$, following Walmswell et al. (2015). Immediately above the shell, the two merger products are indistinguishable, both reaching $Y_{\text{He4}} \approx 0.96$, but integrated further out they diverge. The looping system carries an excess helium mass of at least $0.97 M_\odot$ against at least $1.09 M_\odot$ for the non-looping system, over the $3 M_\odot$ above the shell that we searched.

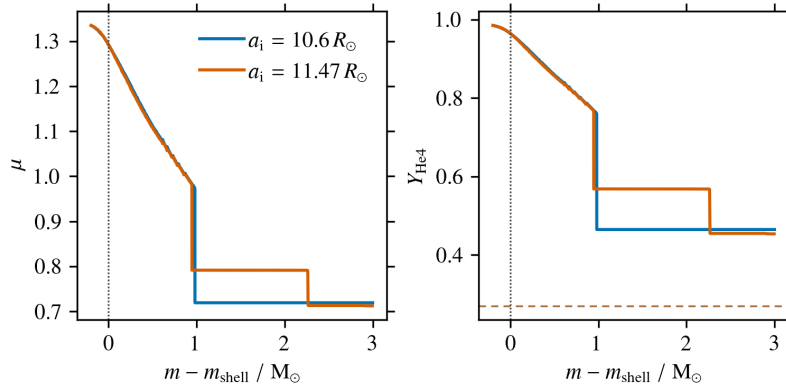


Figure 23: Mean molecular weight μ (left) and helium mass fraction Y_{He4} (right) as a function of mass above the hydrogen-burning shell, $m - m_{\text{shell}}$, at helium ignition, for the two Case A merger products of Fig. 17, $M_{1,i} = 10.16 M_\odot$, $q_i = 0.6$, with $a_i = 10.6 R_\odot$ (blue, the looping system) and $a_i = 11.47 R_\odot$ (orange, the non-looping system). The two progenitors merge to nearly the same mass, and the composition immediately above the shell is nearly identical in both, but the non-looping system carries visibly more excess helium integrated over the following few tenths of a solar mass.

This confirms that the post-merger mass alone does not determine whether a Case A merger product loops (Sect. 5.4). Two merger products of the same mass can carry substantially different amounts of excess helium above the hydrogen-burning shell, and how much mixing occurs depends on the detailed structure of the progenitors at merger rather than on their combined mass. The post-merger mass sets the range of separations and mass ratios in which looping is possible at all (Fig. 30), and the excess

helium inherited from the merger determines which systems within that range actually loop.

5.5 Distribution of merged stars across the HR diagram

Fig. 25 shows the full grid in three HR diagrams, split by mass-transfer case, with each marker sampled at 2×10^5 yr intervals. The Case A panel is dominated by green markers on the main sequence between $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.5$ and 4.3, which is the rejuvenated hydrogen-burning phase of Sect. 5.1. The single-star tracks are shown in grey for comparison, and the Case A merger products are seen to be more luminous and cooler than the single stars of comparable masses at TAMS (Terminal Age Main Sequence), whose locus has shifted to the right, parallel to the single-star TAMS, which is an indication of rejuvenation.

Figure 24 checks this directly for the $q_i = 0.5$, $a_i = 15.1 R_\odot$ merger product against a $15 M_\odot$ single star, matched to the same mass at TAMS (14.58 against $14.59 M_\odot$) so that the comparison isolates composition rather than mass. At TAMS the merger product is indeed both more luminous and cooler than the single star: $\log_{10}(L/L_\odot) = 4.766$ against 4.710 , and $\log_{10}(T_{\text{eff}}/\text{K}) = 4.383$ against 4.389 , with the corresponding radii, $\log_{10}(R/R_\odot) = 1.141$ against 1.100 , consistent with the L , R , T_{eff} via the Stefan–Boltzmann law to within 0.002 dex.

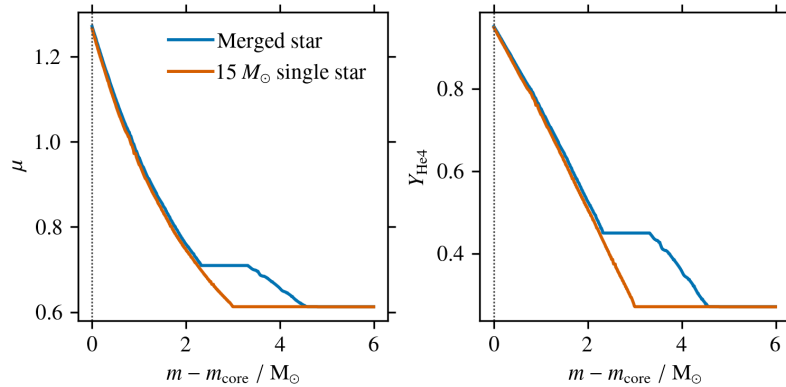


Figure 24: Mean molecular weight μ (left) and helium mass fraction Y_{He4} (right) as a function of mass above the hydrogen-burning core boundary, $m - m_{\text{core}}$, at TAMS, for the $M_{1,i} = 10.16 M_\odot$, $q_i = 0.6$, $a_i = 13.41 R_\odot$ Case A merger product (blue) and a $15 M_\odot$ single star of the same mass at TAMS (orange). Both profiles decline from the same core-envelope boundary μ value at a similar initial rate, but the merger product’s profile shows a knee from $2 M_\odot$ outside of the core, indicating a helium-rich layer above the core. This enhanced composition gradient makes the merger product cooler than the single star at matched mass, despite being more luminous.

The post-main-sequence markers (orange) of the same merger products are distributed across the diagram, with a clear population between $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.2$ and

3.8 with a $\log_{10}(L/L_{\odot}) > 4.6$ comes from the blue loops, and a pile-up on the Hayashi line.

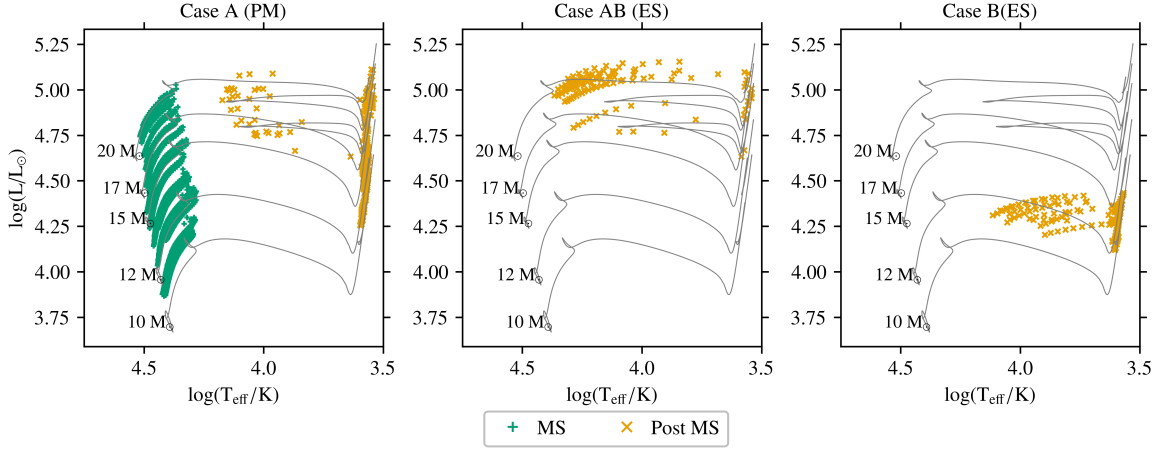


Figure 25: HR diagram of the merger products split by mass-transfer cases. *Left:* Case A merger products built with PyMMAMS. *Center:* Case AB merger products built with entropy sorting. *Right:* Case Be merger products built with entropy sorting as well. Grey curves are single-star tracks of 10, 12, 15, 17 and $20 M_{\odot}$. Green plus signs mark core-hydrogen-burning models and orange crosses post-main-sequence models, and each marker corresponds to 2×10^5 yr of evolution, so the marker density is proportional to the time spent by merger products in that part of the HR diagram. The Case A panel shows the extended rejuvenated main sequence and some merger products burning helium as a blue loop star, which translates to seeing a number of orange markers in $\log_{10}(T_{\text{eff}}/K) \approx 4.2$ to 3.8. The Case AB and Case Be panels contain no main-sequence markers at all and instead show densely populated bands of core-helium-burning yellow/blue supergiants.

The Case AB and Case Be panels contain no main-sequence markers at all. This is a direct consequence of entropy sorting retaining the post-main-sequence character of the evolved progenitors (Sect. 3.2.2), and it is the clearest illustration in the whole thesis of how much the choice of prescription matters: had we used PyMMAMS for these systems, they would have appeared on the main sequence instead (see Appendix B). What we see instead are bands of core-helium-burning supergiants at $\log_{10}(L/L_{\odot}) \approx 4.7$ –5.2 for the Case AB systems and ≈ 4.2 –4.5 for the Case Be systems, with a large fraction of the markers at $\log_{10}(T_{\text{eff}}/K) > 3.7$. These are genuine blue and yellow supergiants by the conventional definition. The luminosity difference between the two groups is a direct consequence of the mass selection: the Case AB merger products are more massive than the Case Be merger products, and more massive stars are more luminous.

The key consequence of this figure is that two physically distinct routes end up in the same place. One is hot-side helium ignition followed by a long-lived supergiant phase (Cases AB and Be mergers), and the other is cool-side helium ignition followed by a blue loop (Case A mergers). An observer does not see the ignition, only the

time-integrated distribution, and if the thermal timescale approach to helium ignition is unresolved, a blue-loop star and a long-lived hot-ignition supergiant (BSGs) of the same luminosity look the same in the HR diagram. Both therefore contribute to the same observed population, and we cannot tell them apart from their position in the HR diagram alone.

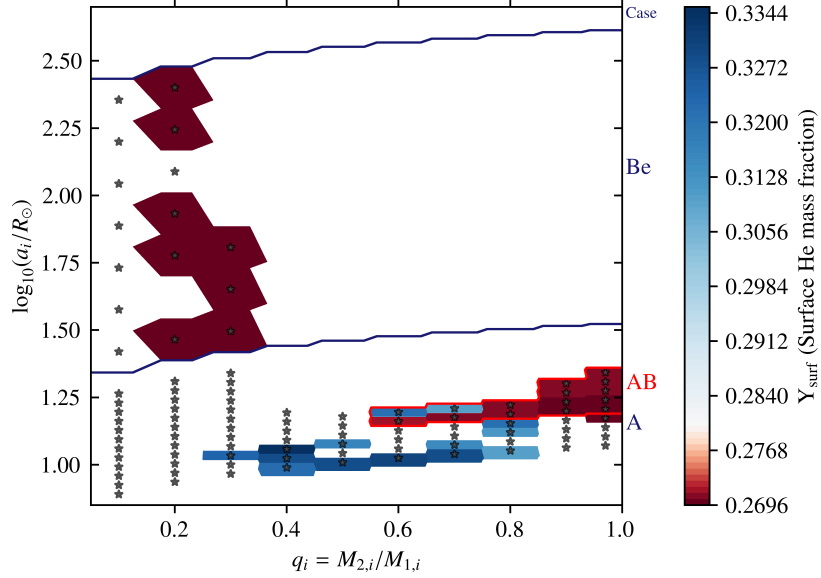


Figure 26: Surface helium mass fraction of blue-loop and BSG merger remnants at their maximum effective temperature during core helium burning, shown in the initial mass-ratio–separation plane. Red cells indicate models with $Y_{\text{surf}} < 0.28$, corresponding to surface helium abundances close to the initial value. Blue cells indicate models with enhanced surface helium abundances. Uncolored cells correspond to models that become neither blue-loop nor BSG stars. Note that the two color bars cover different ranges.

Blue-loop and BSG models differ in their surface helium mass fractions, which are affected by dredge-up. In blue-loop models, the merger remnant typically undergoes dredge-up before core helium ignition, enriching the surface in helium. In contrast, BSG models ignite helium while the near-surface layers remain radiative, so they do not experience comparable surface enrichment. Figure 26 shows the surface helium mass fraction of merger remnants at their maximum effective temperature during core helium burning, restricted to models that become either blue-loop or BSG stars. The BSG models generally retain surface helium mass fractions close to the initial value. We therefore adopt a threshold of $Y_{\text{surf}} = 0.28$ to identify models whose surface helium abundance remains similar to that of a ZAMS star. Red cells indicate models below this threshold, whereas blue cells indicate models with enhanced surface helium. Figure 26 is fully consistent with our expectations. Comparison with the classification shown in Fig. 22 demonstrates that the models identified as BSGs coincide with the low- Y_{surf} (red) cells, whereas the blue-loop models correspond almost exclusively to the helium-

enriched (blue) cells. This agreement confirms that the surface helium mass fraction effectively distinguishes the two evolutionary outcomes.

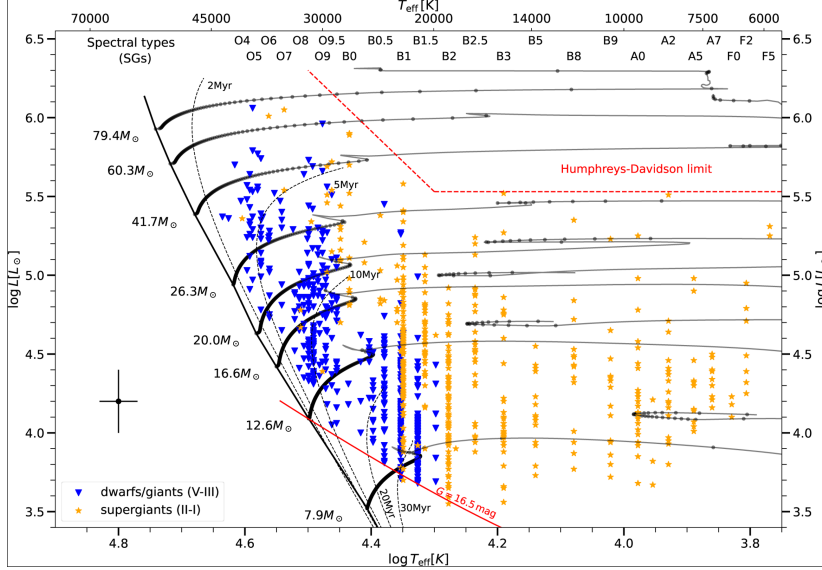


Figure 27: HR diagram of the BLOeM sample from Shenar et al. (2024). Blue triangles are dwarfs and giants (luminosity classes V–III) and orange stars are supergiants (II–I); grey curves are SMC-metallicity evolutionary tracks, with dots marking equal time intervals, and the black dashed curves are isochrones. The red dashed line is the Humphreys–Davidson limit (Humphreys and Davidson, 1979; Humphreys and McElroy, 1984) and the solid red line is the magnitude cut of the survey.

Comparison with the BLOeM survey. BLOeM is a multi-epoch spectroscopic VLT/FLAMES campaign of massive stars in the Small Magellanic Cloud, and its HR diagram (Fig. 27) contains 303 early B-Type Supergiants of luminosity class I–II and 136 late-Type Supergiants (Shenar et al., 2024). The early B-Type Supergiants sit in the part of the diagram that classical single-star evolution crosses in a thermal time, and they show much lower binary fractions than the O and B dwarfs in the same survey (Shenar et al., 2024), which is difficult to reconcile with the fact that most massive stars are born in binaries. This is the blue supergiant problem.

Comparing Fig. 27 with Fig. 25, our merger products fall in the same temperature and luminosity range as the observed blue supergiants, and they spend a nuclear rather than a thermal timescale there. The supergiants in Fig. 27 occupy the region between $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.4\text{--}3.8$ at $\log_{10}(L/L_{\odot}) \approx 3.8\text{--}5.2$, which is the same region our merger products occupy in Fig. 25. Both channels populate the blue supergiant region: post-MS + MS merger products (Case Be and Case AB) through hot-side helium ignition, and Case A merger products through blue loops. A merger is also a natural way of producing a supergiant with no companion, which is the part of the observation that single-star models struggle with.

5.6 Carbon–oxygen core masses

Our models stop at central helium exhaustion, and we measure the carbon–oxygen core mass then. Fig. 28 shows it across the grid. To interpret these values, we compare each merger product against a single star of the same mass, interpolating the single-star CO core masses of Table 2 to the mass of the merger product at the start of the evolve phase (Fig. 30). The three mass-transfer cases behave differently, and the differences follow directly from which star supplies the core.

Table 2: Carbon–oxygen core masses at central helium exhaustion for our single-star models.

Stellar mass [$M_{\odot}$]	CO core mass [$M_{\odot}$]
10	1.41
12	2.05
15	3.05
17	3.78
20	4.69

Case A. The Case A merger products return to the single-star relation. Their CO core masses span 1.61 to 4.67 $M_{\odot}$. The merger resets the nuclear clock and rebuilds a convective core appropriate to the new mass (Sect. 5.1), after which the star has no memory of having been two stars, and Schneider et al. (2024) find the same for their Case A accretors.

The widest Case A systems are those closest to the Case Be boundary, where the primary is most evolved at the time of merger and has the steepest mean molecular weight gradient above its core. These systems are least rejuvenated, so the merger product retains a core inherited from the evolved primary that is larger than the one a single star of the same total mass would build. The trend is therefore a measure of how far rejuvenation falls short across the Case A region, and it is consistent with Braun and Langer (1995) and Schneider et al. (2024), who identify the composition gradient left by the previous evolution as what limits the growth of the convective core.

Case Be. The Case Be merger products have a nearly fixed CO core mass. All 16 systems end between 1.29 and 1.48 $M_{\odot}$, essentially independent of both mass ratio and orbital separation, and close to the 1.41 $M_{\odot}$ of a 10 $M_{\odot}$ single star. In a post-MS + MS merger, the star has the helium-rich core of the 10 $M_{\odot}$ primary inside an envelope that has grown by 1–2 $M_{\odot}$, so the core continues to evolve as it would have done in the absence of the merger. This is the same behavior Schneider et al. (2024) find for their Case B accretors, whose core masses stay within about 10% of the single-star values while the envelope mass changes by a factor of a few. The merger completely changes

the star’s appearance, turning what would have been a red giant into a long-lived blue or yellow supergiant, while barely touching the core it will eventually collapse with.

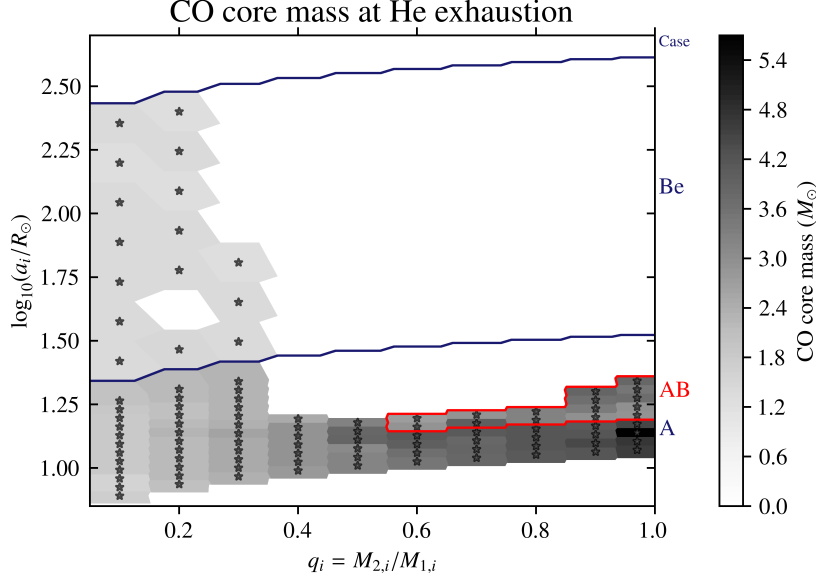


Figure 28: Carbon–oxygen core mass at central helium exhaustion for the merger products built with PyMMAMS (for Case A) and entropy sorting (for Cases Be and AB), in the initial mass ratio–separation plane. CO core mass is given by the grey color bar.

Case AB. The Case AB merger product’s CO core masses occupy a narrow band of 2.69 to $3.95 M_{\odot}$ while their post-merger masses span 13.0 to $19.4 M_{\odot}$, so the core mass is almost independent of the total mass. Inverting the single-star relation, the observed Case AB cores correspond to single stars of 13.9 to $17.6 M_{\odot}$, and the discrepancy grows with mass ratio. The reason is the mass-ratio reversal. By the time these systems merge, the original secondary is the more massive and the more evolved of the two, while the original primary has been stripped and is comparatively unevolved (Fig. 5), and almost all of these merger products are Type S (Fig. 7), i.e. the core is built from the secondary. The core that the merger product carries into helium burning is therefore the core of the evolved secondary, which is set by that star’s mass and not by the combined mass. The stripped primary contributes almost entirely to the envelope, so increasing q_i adds envelope mass without adding core mass, which is why the band stays narrow while the total mass grows and why the deficit is largest at the highest mass ratios.

The $q_i = 0.97$, $a_i = 13.77 R_{\odot}$ system’s merged star. One Case A merger product stands out with $M_{\text{CO}} = 5.69 M_{\odot}$, which is 26% larger than the single-star value expected for its post-merger mass of $19.41 M_{\odot}$ and exceeds the CO-core mass of all of our single-star models, including the $4.69 M_{\odot}$ core of the $20 M_{\odot}$ model (comparison shown

in Fig. 29). It is the only Case A system that deviates by more than 10% from the single-star relation, while neighboring systems at the same initial mass ratio remain within 1%.

This system’s binary evolution differs from that of the other Case A mergers. Following the initial mass-transfer episode, the system undergoes a mass-ratio reversal, after which a second, reverse mass-transfer phase transfers material back from the secondary onto the primary (systems with negative ΔM in Fig. 4). Consequently, the merger occurs after both stars have undergone substantial nuclear evolution and have developed significant helium cores. Instead of merging two relatively unevolved main-sequence stars, the merger combines two stars with already evolved interiors, naturally producing a merger product with an unusually large CO core.

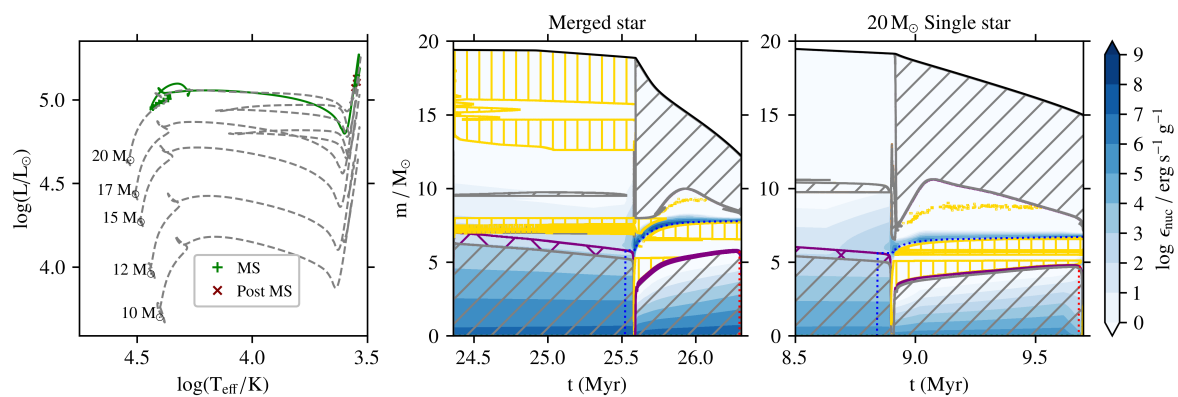


Figure 29: Evolution of the merger product of a Case A merger with $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.97$, and $a_i = 13.77 R_{\odot}$, built with PyMMAMS. *Left:* HR diagram evolution track of the merger product in green, plotted over grey dashed single-star tracks. *Center:* Kippenhahn diagram of the merged star against stellar age. *Right:* the same for a $20 M_{\odot}$ single star. Colors and hatchings as in Fig. 16. CO core mass of the merged star is significantly larger than that of the single star.

This is not the only system in the grid to experience reverse mass transfer. Two additional binary merger products, with $(q_i, a_i) = (0.90, 12.52 R_{\odot})$ and $(0.97, 12.73 R_{\odot})$, also exhibit CO-core masses that are systematically larger than those of neighboring systems with the same initial mass ratio. In these cases, however, the enhancement amounts to only $0.2\text{--}0.25 M_{\odot}$ (approximately 5%). Although these deviations are considerably smaller than in the exceptional system discussed above, they suggest that reverse mass transfer may systematically increase the final CO-core mass. Establishing this connection, however, requires a dedicated investigation beyond the scope of the present work.

Unlike the remainder of the Case A parameter space, where mergers between two main-sequence stars largely rejuvenate onto the single-star relation, the merger product of this system inherits an already well-developed stellar core. The subsequent envelope rejuvenation cannot erase this evolutionary memory. This case therefore shows that

the final CO-core mass depends not only on the post-merger mass, but also on the detailed binary interaction history that precedes the merger.

6 Discussion

In the previous sections, we followed a slice of the binary grid of Henneco et al. (2024b) at an initial primary mass of $M_{1,i} = 10.16, M_{\odot}$ from the onset of mass transfer through contact and up to when we deem the system to merge, through the construction of the merged star, and through its evolution to central helium exhaustion. This section begins by discussing when the binary models should be merged (Sect. 6.1), how the choice of merger prescription influences the resulting merger product (Sect. 6.2), and what the three-dimensional calculation reveals about that choice (Sect. 6.3). We next consider the three merger classes: MS + MS mergers and the origin of their blue loops (Sect. 6.4), post-MS + MS mergers and the blue and yellow supergiants they produce (Sect. 6.5), and post-MS + post-MS mergers (Sect. 6.6). The discussion then turns to how these outcomes are distributed across the binary grid (Sect. 6.7) and the Hertzsprung–Russell diagram in comparison with the observed supergiant population (Sect. 6.8), before examining the carbon–oxygen core masses of the merger products (Sect. 6.9) and concluding with the limitations of this work (Sect. 6.10).

6.1 Merger progenitors properties

In Sect. 3.1.2 we compared the primary’s mass at the onset of the first mass-transfer phase, $M_{p,mt,i}$, with its mass when we deem the binary to merge, $M_{p,final}$, and found differences, $\Delta M \equiv M_{p,mt,i} - M_{p,final}$, of up to $\sim 6\text{--}8 M_{\odot}$ of net mass loss, and down to $\sim -4 M_{\odot}$ of net mass *gain* from reverse mass transfer. The systems in which the difference is largest are the ones with $q_i \gtrsim 0.45$, i.e., the accretor expansion region, where contact is reached on a nuclear rather than a thermal timescale and the primary therefore has time to lose a large fraction of its envelope before the system ever gets into contact. Systems with negative ΔM would be wrongly characterized by an onset-only criterion: the primary transfers mass, the mass ratio reverses, the secondary becomes the donor, and by the time the calculation terminates, the primary has more mass than it started with.

In de Mink et al. (2013), they evaluate the mass ratio $q = M_2/M_1$ at the onset of Roche-lobe overflow and declare a merger if q falls below a critical value, adopting $q_{crit,MS} = 0.65$ for main-sequence donors and $q_{crit,HG} = 0.4$ for donors crossing the Hertzsprung gap. Schneider et al. (2014) inherit that criterion and, for main-sequence mergers, build the product by assuming that 10% of the total mass is lost and 10% of the envelope mass is mixed into the convective core. Both are onset-only merging criteria, and de Mink et al. (2013) say themselves that contact need not imply an immediate merger, since a secondary that overfills its Roche lobe by a small amount can detach again once it restores thermal equilibrium.

It is important to acknowledge that the onset-only criterion is not wrong, it is just

incomplete. It is a necessary condition for a merger, but it is not sufficient since the system can still evolve in the meantime. The difference between $M_{p,mt,i}$ and $M_{p,final}$ is therefore a measure of how much mass evolution happens between the onset of mass transfer and the point where the binary calculation is terminated. A prescription evaluated at the onset of the first mass-transfer phase does not know that a second, reverse mass-transfer phase happened, so it assigns the merger the wrong progenitor masses, the wrong mass ratio, and, since the two stars have kept evolving in the meantime, the wrong evolutionary stages as well. What we can say from this is that the input masses to the merger step is off by several solar masses for a large part of the merging parameter space if the systems are merged at the onset of the first phase of mass transfer, and that this is a larger error than most of the physics uncertainty from the merger prescriptions themselves.

The evolutionary states shown in Fig. 5 make the same point in a different variable. Every Case A system merges two main-sequence stars, and every Case Be system merges a post-main-sequence primary with a main-sequence secondary, as the mass-transfer case at the onset would suggest. In the Case AB strip, however, the roles are reversed by the preceding mass transfer: in 13 of the 15 systems the evolved star entering the merger is the secondary and not the primary, and in the remaining two both stars have left the main sequence. An onset-only criterion assigns these systems the evolutionary states of the stars that began mass transfer, not of the stars that actually merge.

6.2 Merger prescription choice

The choice between entropy sorting and PyMMAMS determines the properties of the merged stars. As Fig. 9 shows, the two prescriptions disagree about whether the Case Be and Case AB products are core-hydrogen-burning or core-helium-burning stars at the moment they are handed back to MESA, and this decision propagates into the position in the HR diagram, the time spent in each region of it, the surface properties, and the CO core mass at central helium exhaustion. Had we built the whole grid with PyMMAMS, the bands of long-lived blue and yellow supergiants in Fig. 25 would not exist, and the population would instead sit on an extended main sequence. A few of these merged stars would still reach the hot part of the HR diagram after helium ignition, but by blue-loop evolution rather than by hot-side ignition.

The physical origin of the disagreement is the treatment of shock heating, together with the implicit assumption that the mean molecular weight profiles of the two progenitors are comparable. Entropy sorting does not modify the entropy of the progenitor material at all (Sect. 2.2.2) and so corresponds to the limit of a perfectly adiabatic, infinitely slow merger. PyMMAMS raises the entropic variable of every shell according to

Eq. (2.4), calibrated by Gaburov et al. (2008) on SPH simulations of head-on collisions, which are considerably more energetic than the in-spiral driven by binary evolution. The two prescriptions therefore mark the two extremes of the structure of the merged star from either side rather than reproducing it, and Henneco, J. et al. (2025) make exactly this argument when they describe entropy sorting and PyMMAMS as the two extremes with respect to shock heating, with the physical value lying somewhere in between. Our grid clearly illustrates the two extremes, since they produce qualitatively different stars.

The head-on calibration is not equally appropriate for all progenitor pairs. For a post-MS + MS merger, the primary’s helium core lies well below the secondary’s core in the entropic variable, but the head-on amount of shock heating is nonetheless large enough to reverse that ordering, so the primary’s core is lifted above the secondary’s and displaced from the center. This is the behavior traced in Sect. 3.2.2, where PyMMAMS returns Type S and Type M cores across a region in which entropy sorting returns Type P cores.

The same reasoning explains where the two prescriptions agree. The two post-MS + post-MS systems of Sect. 5.3.2 are the only ones for which PyMMAMS returns a post-main-sequence merged star, and they are also the only ones in which the subsequent evolution is independent of the prescription: the two merged stars follow nearly identical tracks, burn helium in a convective core growing to the same $m \approx 2.9 M_{\odot}$, and end with the same $M_{\text{CO}} \approx 3.05 M_{\odot}$. When both progenitor stars are evolved and carry similar μ profiles, the shock heating is applied to both cores alike; the ordering of the entropic variable is preserved, and the two prescriptions build the same star. The same shock-heating prescription can thus be adequate for one class of progenitor pair and inadequate for another.

Heller et al. (2025) find that PyMMAMS reproduces the post-merger thermal and composition structure considerably better than entropy sorting, using a set of 39 SPH head-on collisions independent of the Gaburov et al. (2008) calibration set. That result and ours are not in conflict, because their comparison set is dominated by main-sequence collisions, the regime PyMMAMS was built for and the regime we also adopt. A further caveat is that PyMMAMS is calibrated on head-on collisions, not slow spiral-in mergers. Vynatheya et al. (2026) simulate collisions and mergers of the same 5–10 $M_{\odot}$ main-sequence stars with a magnetohydrodynamics code and find that the two channels do not produce equivalent objects, with merger products retaining core hydrogen mass fractions up to 10% higher than the corresponding collision products. Since rejuvenation, the subsequent evolution and final CO core mass depend on the core hydrogen content; this further argues that a head-on calibration is not an ideal basis for a binary-evolution-driven in-spiral. It nonetheless remains a good approximation of the merged star structure for MS + MS mergers relative to most available 1D

prescriptions.

Neither prescription mixes helium-rich core material outwards into the envelope. Menon and Heger (2017) instead accrete the secondary directly onto the primary at a fixed rate, with the secondary’s surface entropy and angular momentum, and simultaneously mix the accreted material up to a prescribed mass boundary inside the primary, so the mixing depth is a free parameter rather than one set by the post-merger entropy or composition profile. Schneider et al. (2024) apply an analogous instantaneous-mixing treatment in their accretor and merger models. Both methods raise the mean molecular weight of the envelope μ_{env} while lowering the core-to-total mass ratio, and both changes drive the star toward a blue solution. In addition, the dynamical mass loss is imposed through the same fixed expression, Eq. (2.6), in both prescriptions and for every system in the grid, so any uncertainty in it affects the merger products obtained from both codes in the same way and is not propagated. A further limitation is that neither prescription tracks angular momentum, so the merged star is non-rotating at the point of import and its rotation must be prescribed by hand, as described in Sect. 2.4.3 (see Sect. 6.10).

6.3 Three-dimensional verification of the prescriptions

The three-dimensional calculation of Sect. 4 helps us test for one binary setup: the in-spiral of a $2 M_{\odot}$ zero-age main-sequence secondary into a $10 M_{\odot}$ Hertzsprung-gap primary, which is the post-MS + MS configuration on which the two prescriptions disagree most strongly.

The secondary is disrupted, and its material is deposited no deeper than a mass coordinate of $\approx 3.3 M_{\odot}$, leaving the helium core of the primary chemically intact, so the merged star is Type P (Fig. 12). Entropy sorting returns the same classification for the same pair of progenitors, placing the secondary immediately outside the pure helium core at $\approx 2.3 M_{\odot}$ (Fig. 13). PyMMAMS places secondary material at the center and returns a Type S merged star (Fig. 14).

The residual difference between entropy sorting and SPH is in the expected direction and of the expected size. The secondary comes to rest about one solar mass further out in SPH than entropy sorting predicts. Entropy sorting redistributes material according to the specific entropy of the progenitor models and, by construction, does not account for the entropy generated during the merger itself. The shock heating that does occur in the three-dimensional calculation raises the entropy of the secondary material and places it at a larger mass coordinate than adiabatic sorting would. The 1D treatment also omits hydrodynamical mixing driven by shear between the in-spiraling secondary and the envelope. Entropy sorting is thus not exact; it disagrees with SPH by placing the secondary slightly too deep, whereas PyMMAMS fails to reproduce the SPH merger

structure altogether by placing the secondary in the wrong region of the star.

The three-dimensional calculation gives $M_{\text{unb}} \approx 0.45 M_{\odot}$, i.e. 3.75% of the total system mass, for the one configuration computed. Computing the dynamical mass loss from Eq. (2.6) for the same system gives $M_{\text{loss}} \approx 0.05 M_{\odot}$, i.e., 0.42%, an order of magnitude below the SPH value. Schneider (2025) derives the scaling of Eq. (2.6) from an energy argument in which the constant C absorbs the fraction of orbital energy used to unbind material and the binding-energy parameter λ of the donor, and is therefore not universal since it is larger for less tightly bound, more evolved stars and smaller for compact main-sequence progenitors. The value $C = 0.03$ we adopt is close to what Schneider (2025) report for the early main-sequence binary merger of Schneider et al. (2019), whereas our own configuration, with a Hertzsprung-gap primary and an extended, loosely bound envelope, falls in the regime where head-on collisions of evolved stars give $C \approx 0.24\text{--}0.33$ (Glebbeek et al., 2013). Using the fixed value calibrated for a compact main-sequence merger for our post-main-sequence progenitors is therefore the likely source of the discrepancy. We do not expect this to change the picture presented here, because the dynamical mass loss is applied identically to both prescriptions and every system in the grid, so it does not contribute to the disagreement between entropy sorting and PyMMAMS discussed in Sect. 3.2.2, and because the missing $\sim 0.4 M_{\odot}$ is added to the envelope of an already several-solar-mass merger product rather than to its core, where it would be too small to alter the core classification or the CO core masses of Sect. 5.6, though it does mean the true masses and hence luminosities of the Case Be and Case AB merger products in Fig. 25 are slightly overestimated.

The passive-scalar profiles obtained with artificial conductivity parameters $\alpha_u = 0.1$ and $\alpha_u = 1.0$ agree with one another (Fig. 15), so the Type P result is converged with respect to α_u . Only the case with the artificial conductivity switched off entirely, $\alpha_u = 0$, carries the secondary spuriously into the helium core, which is the known consequence of leaving thermal-energy discontinuities unsmoothed.

This is the evidence on which the prescription assignment of Sect. 5 rests. It is a single mass ratio at a single separation, so it does not establish that entropy sorting is the correct prescription for post-MS + MS mergers in general. What it does establish is that for the configuration tested, the head-on shock-heating calibration produces a merged star of the wrong structural class, and that the long-lived blue and yellow supergiants reported in Sect. 5 are not an artifact of the prescription chosen to produce them.

6.4 MS + MS mergers: rejuvenation and the origin of the blue loops

In the example of Sect. 5.1, a $12 M_{\odot}$ merged star keeps burning core hydrogen until ≈ 26 Myr while a $12 M_{\odot}$ single star exhausts it at ≈ 17.8 Myr, with the merger occurring at ≈ 17.63 Myr. The comparison is against a single star of the same mass evaluated at the merger age, not against a freshly formed ZAMS star, so the statement is that the nuclear clock is partly reset at the moment of the merger, not that the merged star has a longer total main-sequence lifetime than a ZAMS star of its mass.

Braun and Langer (1995) showed, and Schneider et al. (2024) confirmed for accretors, that semiconvection regulates whether the convective core can grow through the mean molecular weight gradient left by the previous evolution. We adopt $\alpha_{\text{sc}} = 10$ following Schootemeijer et al. (2019), which is a very efficient value; Schneider et al. (2024) use $\alpha_{\text{sc}} = 0.1$ and already describe their models as mostly fully rejuvenating, and note that if semiconvection is as efficient as Schootemeijer et al. (2019) argue, all Case A accretors would rejuvenate fully. Our Case A merged stars should therefore be taken as fully rejuvenated. Step overshooting above the hydrogen-burning core works in the same direction, since a sufficiently extended overshooting region mixes through the semiconvective layer and erases the composition gradient that would otherwise limit core growth.

If the Case A merged stars behave like single stars of their post-merger mass, the blue-loop island in the (q_i, a_i) plane should simply be the part of the plane that maps into a mass window in which a single star would loop anyway. Our own single-star models at 15 and $17 M_{\odot}$ loop and spend 49% and 61% of core helium burning in the loop, while the 10, 12 and $20 M_{\odot}$ models do not, so such a window exists at the physics and metallicity we adopt.

Sect. 5.4 shows that this reading is incomplete. Several merged stars fall inside the mass range spanned by our looping single stars and nonetheless do not loop, so the post-merger mass is necessary but not sufficient. The mechanism that determines the difference does not depend on mass at all. Following Lauterborn et al. (1971) and Fricke and Strittmatter (1972), a blue loop is triggered when the hydrogen-burning shell burns outwards through the composition gradient left behind by the retreating convective core, removing the excess helium above the shell that keeps the star red and extended, and Walmswell et al. (2015) showed, by isolating the opacity, the fuel supply and the mean molecular weight immediately above the shell decide whether to loop or not to loop. Whether a loop occurs depends on a competition between the shell burning out through that gradient and the growth of the core, which drives the star back toward the Hayashi line.

Both prescriptions redistribute material when they build the merged star, and both

therefore carry helium-rich material from the deep interior of the progenitors into the region that will later sit above the hydrogen-burning shell. Two merged stars of the same mass can accordingly begin core helium burning with different amounts of excess helium above the shell, and the one that carries more should be the redder of the two. The matched pair of Sect. 5.4 at $q_i = 0.6$ and post-merger masses of 16.05 and $16.01 M_\odot$, the model that spends 60.6% of core helium burning above the threshold temperature carries at least $0.97 M_\odot$ of excess helium above the shell, while the model that does not go above the threshold temperature carries at least $1.09 M_\odot$. The post-merger mass sets which part of the grid can loop, and the μ profile inherited from the merger decides which systems within that range do.

However, blue loops are among the least robust predictions a stellar evolution code makes. Their occurrence and extent depend on the treatment of semiconvection, core overshooting, opacities, and metallicity, and a different but equally defensible choice of input physics can shift the mass window or remove it.

6.5 Post-MS + MS mergers as long-lived blue and yellow supergiants

The Case AB and Case Be merger products ignite core helium at $\log_{10}(T_{\text{eff}}/\text{K}) > 3.7$ and cross the blue and yellow part of the HRD on a nuclear timescale, from $\log_{10}(T_{\text{eff}}/\text{K}) \approx 4.1$ to 3.7 , reaching the Hayashi limit only near central helium exhaustion. This behavior follows from the fractional core mass, $\xi_c = M_c/M_{\text{total}}$. A small ξ_c favors a compact, blue configuration, whereas a large ξ_c drives the star redward (Podsiadlowski et al., 1992). A merger with an evolved star efficiently reduces ξ_c because the less evolved companion is added to the envelope while the helium core of the evolved star remains unaffected. This channel has been invoked to explain long-lived blue supergiants since Podsiadlowski et al. (1990), and more recently by Menon and Heger (2017) and Schneider et al. (2024). The SPH calculation in Sect. 4 supports this picture independently of the 1D prescriptions: the helium core of the primary survives the merger with its composition profile intact, and the secondary is deposited at $m \gtrsim 3.3 M_\odot$.

The two classes differ in the structure above the helium core. In the Case AB merger products, a convective region extends from $m \approx 2.5$ to $6.7 M_\odot$ beneath a radiative envelope (Fig. 20). It persists until the star reaches the Hayashi limit at $t \approx 36.4$ Myr, when a deep convective envelope develops. In the Case Be merger products, both the hydrogen-burning shell and the envelope remain radiative throughout core helium burning (Fig. 18). Both classes are therefore blue because of their small ξ_c , but only the Case AB merger products burn hydrogen in a convective shell. This difference plausibly reflects the added envelope mass. Case AB systems merge stars of comparable mass

after mass-ratio reversal, whereas the Case Be Y/BSGs originate from systems with $q_i = 0.2\text{--}0.3$.

In neither class does the deep convective envelope develop while the star is on the hot side of $\log_{10}(T_{\text{eff}}/\text{K}) = 3.7$. This occurs after leaving the blue supergiant region in the Case AB merger products and near central helium exhaustion in the Case Be merger products. Both classes therefore retain surface helium mass fractions close to the initial value during their Y/BSG phase (Fig. 26). Blue-loop stars, by contrast, ignite core helium on the cool side after a convective envelope has dredged processed deeper material to the surface, and return to the blue with enhanced surface helium. Surface composition thus distinguishes the two routes into the blue part of the HRD. A blue supergiant with near-initial surface abundances is consistent with a merger product that has burnt helium in the blue since ignition, whereas an enriched one indicates a previous red supergiant phase.

The $q_i = 0.1$ systems (Sect. 5.2) define the lower boundary of forming Y/BSG. The secondary of $\approx 1 M_{\odot}$ reduces ξ_c too little, and the merger product burns core helium at the Hayashi limit like a single star. This provides a rough lower limit on the mass a merger must add to alter the post-main-sequence evolution.

6.6 Post-MS + post-MS mergers

As shown in Sect. 5.3.2, the merged star follows the track of a single star of its post-merger mass, executes a blue loop of comparable extent, burns helium for a comparable time, and ends with the same carbon–oxygen core mass. The merger scales the star up without changing the core-to-envelope mass ratio that decides whether it sits in the blue or the red, because both progenitors contribute a helium core and a hydrogen envelope in similar proportion, so the merged star is delivered onto the single-star relation at its new mass and leaves no fingerprint recoverable from its position in the HR diagram, its helium-burning lifetime, or its final core mass.

These are also the only systems for which the two 1D prescriptions agree (Sect. 6.2). When both progenitors are evolved and have similar μ profiles, the shock heating applies to both cores alike, and the ordering of the entropic variable is preserved. With two systems in the grid, we cannot say more than that, and whether the subset of stars is negligible in a real population depends on how much of the (q_i, a_i) plane it occupies at other primary masses.

6.7 Two routes into the same part of the HR diagram

Taken together, the three classes produce two distinct ways of ending up as a blue or yellow supergiant, and Sect. 5.4 shows where each occurs. The first is hot-side helium ignition, which is what the post-MS + MS mergers do: the Case AB strip just above

the Case A boundary is hot for essentially the whole of core helium burning, and the Case Be region at $q_i = 0.2\text{--}0.3$ behaves similarly. The second is the blue loop, confined to a compact island of Case A systems at $q_i \approx 0.35\text{--}0.85$ and the closest separations, with some cases spending up to $t_{\text{BL}}/t_{\text{He}} \approx 0.74$ of core helium burning on the hot side. Of the 100 merging systems, 20 are flagged as blue supergiants and 15 as blue-loop models.

Both populations occupy the same part of the HR diagram at any given moment, since both spend a nuclear rather than a thermal timescale there, and the ignition-based classification itself leaves no direct imprint on a star's present-day T_{eff} and L . If merger products are to be identified individually rather than statistically, the discriminant must be something other than T_{eff} and L . Menon et al. (2024) argue for a merger origin for about half of their LMC early-B supergiant sample precisely because the N/C and N/O ratios and the helium enhancement of the observed stars are reproduced by evolved-binary-merger models and not by rotating single stars or classical blue-loop models. Our two routes have different mixing histories, since one carries material through a rejuvenated convective core while the other adds the secondary to a radiative envelope, so they need not produce the same surface composition. Our own blue-loop and Y/BSG merger products separate cleanly by surface helium abundance (Fig. 26), consistent with a composition-based discriminant along the lines Menon et al. (2024) propose. Asteroseismology offers a second route, since the two differ in the extent and sharpness of the composition gradient above the core (Bellinger et al., 2024; Henneco et al., 2024a).

6.8 Distribution across the HR diagram and comparison with observations

Shenar et al. (2024) obtained multi-epoch VLT/FLAMES spectroscopy of 929 massive stars in the SMC as part of the Binaries at Low Metallicity (BLOeM) campaign, of which 303 are early B-Type Supergiants of luminosity class I–II and 136 are late-Type Supergiants. These stars sit in the part of the HR diagram that classical single-star evolution crosses on a thermal timescale, so there should not be many of them. They measure an observed spectroscopic binary fraction of $23 \pm 3\%$ for the BLOeM early-B supergiants, against $50 \pm 3\%$ for the early-B dwarfs and giants of the same survey. Most massive stars are born in binaries, so a supergiant population that is deficient in companions needs an explanation.

Our merged stars occupy the same region of the HR diagram as the observed supergiants and spend a nuclear rather than a thermal timescale there, which is one possible way a short-lived region of the diagram gets populated. A merger destroys the binary that made it, so a merged star is a natural way of producing a luminous

supergiant with no companion. Both routes of Sect. 6.7 contribute.

However, we cannot directly compare our results with the observations, for several reasons. First, our grid is computed at solar metallicity, $Z = 0.0142$, while BLOeM observes the SMC at $Z \approx 0.2 Z_\odot$. Lower metallicity changes the opacities, the wind mass-loss rates, the internal mixing, and the temperature at which core helium burning happens, and blue loops in particular are metallicity-sensitive. The overlap between the two populations in Fig. 25 and Fig. 27 should not be read as a fit, but as evidence that the mechanism can produce blue supergiants.

Second, the counts of Sect. 5.4 are counts of grid cells. They are not weighted by the initial distributions of q_i and a_i , and the grid is a single slice at $M_{1,i} = 10.16 M_\odot$ rather than an IMF-weighted population. They say that a substantial fraction of the merging parameter space produces long-lived blue and yellow supergiants. They do not say what fraction of observed supergiants are merger products.

Finally, the Case AB merged stars sit at $\log_{10}(L/L_\odot) \approx 4.7\text{--}5.2$ and the Case Be merged stars at $\approx 4.2\text{--}4.5$, while the BLOeM supergiants span $\log_{10}(L/L_\odot) \approx 3.8\text{--}5.2$. Our $10 M_\odot$ slice therefore covers the lower and middle part of the observed distribution and says nothing about the top of it. Reproducing the full observed luminosity range requires the other primary masses in the Henneco et al. (2024b) grid.

6.9 The cores of the merged stars

The Case Be merged stars all end with $M_{\text{CO}} \approx 1.39 M_\odot$, essentially independent of both mass ratio and separation, and essentially identical to the $1.41 M_\odot$ of a $10 M_\odot$ single star. This follows directly from the structure established in Sect. 6.5, and the three-dimensional calculation confirms it: the secondary goes into the envelope and the primary’s helium core continues to evolve as it would have done in the absence of the merger. The same behavior is found by Schneider et al. (2024) for their Case B accretors, whose core masses stay close to the single-star values while the envelope mass changes by a factor of a few. The merger changes the appearance of the star completely, turning what would have been a red supergiant into a long-lived blue or yellow supergiant, while barely touching the core that will eventually collapse. If these objects explode, they should do so with roughly the core they would have had as single stars, but from a very different envelope and a different position in the HR diagram.

The Case A merged stars consistently rejuvenate, and their CO core masses climb with q_i and, over the range $1.61\text{--}4.67 M_\odot$, sit alongside the single-star values of our $10\text{--}20 M_\odot$ models. This is what Schneider et al. (2024) find, namely that Case A accretors mostly fall back onto the single-star relation between final mass and CO core mass except where rejuvenation is incomplete.

The Case AB merged stars are the intermediate case. They are almost all Type S,

and their CO core masses cluster in a narrow band of $2.69\text{--}3.95 M_{\odot}$, below the Case A values. The reason is the mass-ratio reversal: by the time these systems merge, the original secondary has become the more massive and the more evolved of the two, while the original primary has been stripped and is comparatively unevolved (Fig. 5). The core that ends up at the center is the secondary’s, not the core the initial mass ratio would suggest. This effect is more pronounced in the negative ΔM region of systems in Sect. 3.1.2 since mass-ratio reversal occurs twice before merger. This illustrates why merger criteria based solely on onset of mass transfer are inadequate for these systems.

One Case A merger product, with $q_i = 0.97$ and $a_i = 13.77 R_{\odot}$, stands out. Its final core mass, $M_{\text{CO}} = 5.69 M_{\odot}$, is 26 % above the single-star value at its post-merger mass of $19.41 M_{\odot}$, and exceeds the core mass of every single-star model in our grid, including $M_{\text{CO}} = 4.69 M_{\odot}$ for the $20 M_{\odot}$ star. In this system, mass-ratio reversal is followed by a reverse mass-transfer phase, so the merger combines two stars that are both evolved and both already carry substantial helium cores. A single system cannot establish this channel, but it reminds us that the final core mass is set by the progenitor stars’ evolutionary history, not by the post-merger mass alone.

6.10 Limitations

This thesis relies on several assumptions, which must be acknowledged. The most important are as follows.

Angular momentum. Neither entropy sorting nor PyMMAMS tracks rotation, so the merged star imported into MESA is non-rotating, and we impose $\Omega = 0.05 \Omega_{\text{crit}}$ by hand at the start of the evolve phase. The value is motivated by the $\approx 0.1 \Omega_{\text{crit}}$ found post-contraction for the $17 M_{\odot}$ merged star of Schneider et al. (2019), and by the fact that merged stars that are initially rapid rotators can spin down efficiently (Schneider, 2025). A merger is a violent, angular-momentum-rich event, and the rotational history of the merged star is not something a 1D entropy-based prescription can deliver. This assumption has consequences, because rotational mixing affects surface abundances and the size of the mixed core, and a rapidly rotating merged star would experience additional mixing and could reach the blue part of the HRD (hotter T_{eff}) for a different reason than the one we identify. Determining the merged star’s angular momentum content and transport requires three-dimensional simulations. Although we performed such a simulation (Sect. 4), we did not analyze the angular momentum content and transport in detail here, and we leave this for future work.

Head-on collisions vs spiral-ins. PyMMAMS is calibrated on head-on collisions on parabolic orbits, and entropy sorting assumes a perfectly adiabatic, infinitely slow merger. The systems we model reach contact through binary evolution and spiral in

over many orbits. The three-dimensional calculation of Sect. 4 addresses this for one configuration and shows that entropy sorting is the closer of the two bounds there, but a single mass ratio at a single separation does not establish the validity of ES across the grid.

Dynamical mass loss. We impose the same fixed expression for the dynamical mass loss, Eq. (2.6), in both prescriptions and across the whole grid, so it does not appear as a source of disagreement anywhere in our results and its uncertainty is not propagated. The three-dimensional calculation gives $M_{\text{unb}} \approx 0.45 M_{\odot}$, i.e. 3.75% of the total system mass, which is an order higher than the empirical formula Eq. (2.6) (see Sec. 6.3).

The “relax” phase. The ≈ 100 yr relaxation phase is a numerical step, not a physical one. We chose its duration by requiring the transient convection zones to disappear while the chemical structure remains unchanged (Sect. 2.4.2), and we checked the energy residual and $|L_{\text{grav}}|$ along with the Kippenhahn diagrams during this phase to confirm that the model reaches a stable configuration. This is a potential source of uncertainty, because the merged star is not in thermal equilibrium and the relax phase is a numerical artifact. We do not expect it to affect the long-term evolution significantly, because it is short compared with the star’s nuclear timescale.

Coverage. We stop at central helium exhaustion, so we say nothing directly about the late evolution and final fate of the merged stars. Our mergers are restricted to a single initial primary mass and must be extended to the remaining initial primary masses in the Henneco et al. (2024b) grid.

7 Conclusions

We set out to determine what a merger involving a $\sim 10 M_{\odot}$ primary produces, and whether the products can contribute to the blue supergiant problem. Working from the Cases A, AB, and Be region of the Henneco et al. (2024b) $M_{1,i} = 10.16 M_{\odot}$ binary grid, we identified the merging systems, built the merged stars with two 1D prescriptions, verified the prescriptions against a three-dimensional SPH calculation for one configuration, and evolved the merged stars with MESA to central helium exhaustion. Our main conclusions are the following.

1. **Merger progenitors should not be defined at the onset of mass transfer.**

Between the onset of the first mass-transfer phase and the termination of the binary calculation, the primary’s mass changes by up to $\sim 6\text{--}8 M_{\odot}$, and in the reverse mass-transfer systems by up to $\sim -4 M_{\odot}$. The evolutionary states also

change: in 13 of the 15 Case AB systems, the evolved star entering the merger is the secondary rather than the primary. Onset-only criteria like those used by de Mink et al. (2013) and Schneider et al. (2014) therefore assign the merger step the wrong masses, the wrong mass ratio, and the wrong evolutionary stages across much of the merging parameter space.

2. **No single 1D prescription covers the whole grid.** PyMMAMS and entropy sorting disagree qualitatively over the wide-separation region, and the disagreement comes from the shock heating rather than from the buoyancy sorting itself. Since the Gaburov et al. (2008) calibration is built on head-on main-sequence collisions, applying it to an evolved progenitor deposits enough entropy in the helium core to lift it out of the bottom of the buoyancy ordering, so the merged star is returned rejuvenated and on the main sequence. We therefore use PyMMAMS for the Case A (MS + MS) mergers, where it agrees best with SPH (Heller et al., 2025), and entropy sorting for the Case AB and Case Be (post-MS + MS) mergers.
3. **The three-dimensional calculation.** For the in-spiral of a $2 M_{\odot}$ main-sequence secondary into a $10 M_{\odot}$ Hertzsprung-gap primary, the secondary is deposited no deeper than $m \approx 3.3 M_{\odot}$ and the helium core of the primary survives intact, giving a Type P merged star. Entropy sorting reproduces this classification, and PyMMAMS does not. The result is converged with respect to the artificial conductivity. This is one configuration, not a general proof, but it shows that the long-lived supergiants reported here are not an artifact of the prescription used to build them.
4. **Case A mergers rejuvenate, and mass alone does not decide whether they perform a blue loop.** They resume an evolution comparable to single stars of their new mass, and a compact island at $q_i \approx 0.35\text{--}0.85$ and the closest separations execute blue loops, in some cases spending up to $t_{\text{BL}}/t_{\text{He}} \approx 0.74$ of core helium burning on the hot side. Our 15 and $17 M_{\odot}$ single stars loop, and the 10, 12 and $20 M_{\odot}$ models do not, so a mass window exists at the physics we adopt. It is not sufficient: merged stars of the same mass differ in whether they loop, and the difference tracks the excess helium the merger leaves above the hydrogen-burning shell, in the sense predicted by Walmswell et al. (2015).
5. **Post-MS + MS mergers produce long-lived blue and yellow supergiants.** The Case AB and Case Be merged stars ignite helium on the hot side of $\log_{10}(T_{\text{eff}}/\text{K}) = 3.7$ and cross the blue and yellow part of the HR diagram on a nuclear timescale, with the Case AB merged stars remaining hot supergiants for essentially the whole of core helium burning. The corresponding single stars

ignite helium as red giants and stay cool. The mechanism is the classical small fractional core mass (Podsiadlowski et al., 1992): the less evolved star is added to the envelope while the helium core of the more evolved star is left untouched, and the SPH calculation confirms that the core is indeed left untouched.

6. **The Cases AB and Be post-MS + MS merger products reach the blue with different internal structures.** The Case AB merged stars burn hydrogen in a convective shell sitting directly on the helium core, with a radiative envelope above it, until they reach the Hayashi limit. Whereas, the Case Be merged stars keep both the shell and the envelope radiative throughout core helium burning. The two therefore develop their deep convective envelopes at different points along the track, which is where an observational difference between them would have to come from.
7. **Very low mass ratios do not change the outcome.** The $q_i = 0.1$ Case Be systems add only $\sim 1 M_\odot$ to the envelope, which is not enough to alter the core-to-envelope ratio meaningfully, and they evolve to the Hayashi limit and burn helium there like ordinary post-main-sequence stars. This sets a rough lower limit on how much mass a merger has to add before it changes the evolutionary outcome at all.
8. **Post-MS + post-MS mergers are indistinguishable from single stars.** The two such systems in the grid follow the single-star track at their post-merger mass in every quantity we measure, and they are the only systems for which the two prescriptions agree. The merger increases the star’s mass without changing the core-to-envelope mass ratio, and leaves no fingerprint in the HR diagram, the helium-burning lifetime, or the final core mass.
9. **BL and BSG are observationally degenerate in the HR diagram.** An observer sees only the time-integrated distribution, not the ignition, so a blue-loop star and a hot-ignition supergiant of the same luminosity are indistinguishable by position in the HR diagram alone. Separating them requires surface abundances or asteroseismology.
10. **The merger changes the star’s appearance without changing its core.** The Case Be merged stars have $M_{\text{CO}} \approx 1.39 M_\odot$ regardless of mass ratio and separation, comparable to the $1.41 M_\odot$ of a $10 M_\odot$ single star. This matches what Schneider et al. (2024) find for Case B accretors. The rejuvenated Case A merged stars instead follow the single-star CO core-mass range of $1.61\text{--}4.67 M_\odot$ at comparable masses, and the Case AB merged stars occupy a narrow band of $2.69\text{--}3.95 M_\odot$, set by the mass of the evolved secondary that ends up at the center.

11. **BL and BSG are plausible contributors to the observed supergiant population.** Our merged stars overlap the BLOeM early-B and late-Type Supergiants in T_{eff} and L and spend a nuclear timescale there, and a merger naturally produces a luminous supergiant with no companion. This is consistent with the low BLOeM multiplicity result of $23 \pm 3\%$ for the early-B supergiants against $50 \pm 3\%$ for the dwarfs and giants.

Our grid is computed at solar metallicity, whereas BLOeM targets the metal-poor SMC; as discussed in Sect. 6.8, this and the single-primary-mass coverage of our grid mean the comparison points to plausibility rather than establishing agreement.

At $M_{1,i} = 10.16 M_{\odot}$, both the post-MS + MS merger channels (Cases AB and Be) and the Case A blue-loop channel deliver core-helium-burning stars to the blue and yellow supergiant region and keep them there for a nuclear timescale. Merged stars are therefore a plausible contributor to the BLOeM supergiant population, though not necessarily the explanation for every star in it.

Outlook

The most immediate extensions follow from the limitations above. The three-dimensional comparison of Sect. 4 covers one mass ratio at one separation. Extending it to further mass ratios and to a post-MS + post-MS configuration would establish whether entropy sorting is the right prescription for post-MS + MS spiral-ins across the regime, or whether a prescription calibrated on spiral-ins rather than on collisions is needed. The same calculations already include the angular momentum distribution of the merged star, which we did not analyze here and which would remove the need to prescribe the post-merger rotation by hand.

Repeating the grid at SMC metallicity would make the comparison with BLOeM meaningful rather than indicative. Extending it to the other initial primary masses of the Henneco et al. (2024b) grid, and weighting the outcomes by the observed distributions of q_i and a_i , would turn the grid-cell counts of Sect. 5.4 into a population prediction. Continuing the models past central helium exhaustion to core collapse would connect the CO core masses of Sect. 5.6 to the supernovae these stars eventually produce.

8 Acknowledgments

யாதும் ஊரே யாவரும் கேளிர்

Yādhum Ūrē; Yāvarum Kēlir

“Every place is my home; everyone is my kin.”

Sangam literature, ca. 1st–2nd century CE

These words, from Kaniyan Pungundranar’s *Purananuru*, have stayed close to my heart throughout my time in the Stellar Evolution Theory group. Coming from another country, I often carried with me the quiet fear of being far from home. Yet, in the SET group, I never felt like an outsider. I found not only a place to work, but also warmth, kindness, and belonging. For that, I will always be grateful.

First and foremost, I would like to thank Dr. Fabian Schneider for giving me the chance of a lifetime to write my thesis in the SET group. This experience has been nothing short of life-changing. I have learned so much from your guidance, your way of thinking, and the care with which you approach both science and people. I hope that our paths cross again in the future, and that we get to work together once more. One day, I hope to become a scientist, a researcher, and a father with even a fraction of the patience, curiosity, and kindness that I have seen in you.

I would also like to express my deepest gratitude to Dr. Mike Lau, a postdoctoral researcher in the PSO group. I have enjoyed every small discussion we have had, whether scientific, technical, or simply born out of curiosity. Thank you for trusting me and for inviting me to begin a project with you on three-dimensional simulations. At a time when I was on the verge of moving away from physics, working with you reminded me of why I fell in love with this field in the first place. More than anything else, that project made me want to stay in physics and pursue academia. For this, Mike, I will always be grateful.

I thank each and every one of my colleagues in the SET group for their scientific contributions, their encouragement, and their friendship. Thank you for listening to my rants, for being patient with me, and for being there during moments when I felt far away from home. I am especially grateful to Vincent Bronner, who kindly proofread this thesis, as well as Utkarsh Jain, Raheela Sahar, Qingxuan Li, and Jerry. Your support has meant more to me than I can fully express here.

I would like to thank Max Heller for his kindness and for helping me get started with my thesis. Thank you also for helping me understand and use your code. I am grateful to Dr. Jan Henneco for the scientific discussions and for the binary grid used throughout this thesis.

Some colleagues became much more than colleagues over the past few years. For all the coffee-break conversations, the life advice, the patient listening, the badminton sessions, and the laughter that made difficult days lighter, I thank Kristian, Federico,

Veronica, Christina, Korinna, and Peter. Your friendship made this journey warmer and more memorable.

I would also like to thank Dr. Friedrich Roepke for serving as my second supervisor.

Joining this university, and especially HITS, has been one of the best decisions of my life. I will carry what I have learned and experienced here with me throughout my life.

என் குடும்பத்தினருக்கு: உபேந்திரநாத் மற்றும் கவிதா — என் வாழ்வின் ஒவ்வொரு கட்டத்திலும் அன்போடும் உறுதியோடும் எனக்குத் துணையாக நின்றமைக்கு என் மனமார்ந்த நன்றிகள்; நீங்கள் இருவரும் ஒருவருக்குக் கிடைக்கக்கூடிய மிகச் சிறந்த பெற்றோர்கள். ஜஸ்வந்தன் மற்றும் ஸ்மிதா — என் அன்பிற்குரிய சகோதரன் மற்றும் அண்ணி, வாழ்க்கையின் முக்கிய முடிவுகளில் எனக்குத் துணையாகவும் வழிகாட்டிகளாகவும் இருந்து, என் பாதையில் ஒளியாகத் திகழ்ந்தமைக்கு என் அன்பையும் நன்றியையும் தெரிவித்துக்கொள்கிறேன். நீங்கள் அனைவரும் என் வேர்கள்; நான் எங்கு சென்றாலும், எனக்குள் நான் சுமந்து செல்லும் 'வீடு' நீங்கள்தான்.

8.1 Tools used for analysis and visualization

The following software tools were used for the analysis and visualization of the results presented in this thesis: NumPy (Harris et al., 2020), PyMesaReader (Wolf and Schwab, 2017) for interacting with MESA, mkip (Marchant, 2017), Matplotlib (Hunter, 2007) for visualization, Jupyter Notebook (Kluyver et al., 2016), and mpi4py (Dalcin and Fang, 2021) for the parallelization of MESA simulations.

Grammarly and AI language models (Claude Sonnet 5 and GPT-5.5) were used during the preparation of this thesis for language editing, improving clarity of presentation, code development and debugging, and other general writing assistance. All AI-generated content and suggestions were critically evaluated and independently verified by the author prior to inclusion. The scientific concepts, methodology, analysis, interpretation of results, and conclusions presented in this thesis are entirely the author's own.

Appendices

A Figures and tables

List of Figures

- 1 Structure and evolutionary state of the primary progenitor. *Left:* Hertzsprung–Russell diagram with the evolutionary track of the $10 M_{\odot}$ primary until it reaches $10 R_{\odot}$. *Middle:* hydrogen and helium mass fractions against mass coordinate at the extraction point. The primary’s pure helium core sits within $\approx 2.3 M_{\odot}$, marked by the first vertical dashed line, and the core-envelope boundary is marked by the second vertical dashed line at $\approx 3.3 M_{\odot}$. *Right:* Kippenhahn diagram with hatching indicating convective regions. 20
- 2 *Left:* Time evolution of the absolute gravitational luminosity, $|L_{\text{grav}}|$, and the average energy residual, $\overline{E_{\text{residual}}}$, during the relaxation phase of the merged star. *Right:* Kippenhahn diagram of the same product over 10^6 yr of relaxation. Vertical reference lines mark logarithmically spaced times after the merger, separated by one dex. 25
- 3 Occurrence of contact phases in the initial mass ratio (q_i) vs. initial orbital separation ($\log_{10}(a_i/R_{\odot})$) plane for mass transfer Cases A and Be binary systems with initial primary mass $M_{1,i} = 10.16 M_{\odot}$. The dark blue quasi-horizontal lines separate the mass-transfer cases and mark, for each q_i , the initial separation at which the primary first fills its Roche lobe at the corresponding evolutionary stage; the case is labeled on the right-hand side of each boundary. The bold black contour encloses the systems that we consider to merge. Each star marks a computed binary model. 29
- 4 Difference $\Delta M = M_{\text{p,mt,i}} - M_{\text{p,final}}$ between the primary’s mass at the onset of mass transfer, $M_{\text{p,mt,i}}$, and the primary’s final mass, $M_{\text{p,final}}$ (“final” is the point at which the simulation terminates, which is also the point at which we merge the system), for the same $M_{1,i} = 10.16 M_{\odot}$ grid slice as in Fig. 3. Positive regions indicate substantial net mass loss by the primary over the mass-transfer phase(s) preceding contact; negative regions indicate net mass regain by the primary due to a reverse mass-transfer phase, discussed below. The AB contour marks a special mass transfer case explained below, and dark blue lines mark the same Case A/Case Be boundaries as in Fig. 3. 31

5	Evolutionary state of the progenitors at merger	32
6	Core classification of the merger products	34
7	Core classification (Type P/M/S; see Sect. 3.2.1) of the merger products produced by entropy sorting and PyMMAMS for the grid. Hatching marks systems with an undefined core.	35
8	Specific entropy S (in units of $\text{erg K}^{-1} \text{g}^{-1}$) as a function of enclosed mass m for the primary and secondary at the end of the mass transfer phase, for two binary systems with the same mass ratio ($q_i = 0.1$) but different initial separations, $a_i = 7.77 R_\odot$ ($\log_{10}(a_i/R_\odot) \sim 0.89$) and $a_i = 18.37 R_\odot$ ($\log_{10}(a_i/R_\odot) \sim 1.25$).	36
9	Evolutionary state (main sequence or post-main-sequence) of the merger product produced by ES (left) and PyMMAMS (right). Black hatching marks systems with an undefined core.	38
10	Face-on (top row) and edge-on (bottom row) SPH logarithmic density renders of the hydrodynamical simulation of a $10 M_\odot$ Hertzsprung-gap (HG) primary merging with a $2 M_\odot$ zero-age main-sequence (ZAMS) secondary, shown at four times spanning the simulation from the initial binary setup to the relaxed merger product. From left to right: the system at the start of the simulation (0.0 d); an early stage of the orbital inspiral (4.6 d); a render of a few snapshots before the dynamical merger (7.4 d); and the merger product after it has relaxed (9.2 d). Color bar encodes $\log_{10}(\rho/\text{g cm}^{-3})$	39
11	<i>Left:</i> Evolution of the orbital separation between the centers of the two stars during the in-spiral. The dashed vertical line marks the point at which the artificial conductivity is activated ($a = 4 R_\odot$, see Sect. 2.3.4). <i>Right:</i> Unbound mass during the in-spiral. The vertical dashed line marks the point in the simulation when the progenitor cores can no longer be distinguished.	40
12	<i>Left:</i> Passive scalar of the merged star plotted against mass coordinate. <i>Right:</i> Hydrogen (X) and helium (Y) mass fractions against mass coordinate. Dashed vertical line at $\approx 3.3 M_\odot$ on both plots is the base of the H-rich envelope of the primary.	41

13	Comparing the three-dimensional SPH calculation with ES. <i>Left:</i> Passive scalar against mass coordinate in the merger products. The passive scalar is unity for material originating from the primary and zero for material originating from the secondary. The vertical dashed lines mark the passive scalar drop given by ES (at $2.3 M_{\odot}$) and SPH (at $3.3 M_{\odot}$). <i>Right:</i> Hydrogen (X) and helium (Y) mass fractions against mass coordinate in the merger product, comparing the three-dimensional SPH calculation (presented as solid lines) with ES (presented as dashed lines).	42
14	Comparing the three-dimensional SPH calculation with PyMMAMS. <i>Left:</i> Passive scalar against mass coordinate in the merger products. The vertical dashed lines mark the passive scalar drop given by PyMMAMS (at $1.15 M_{\odot}$) and SPH (at $3.3 M_{\odot}$). <i>Right:</i> Hydrogen (X) and helium (Y) mass fractions against mass coordinate in the merger product, comparing the three-dimensional SPH calculation (presented as solid lines) with the PyMMAMS (presented as dashed lines).	43
15	Passive scalar against mass coordinate in the merger product for three values of the artificial conductivity parameter, $\alpha_u = 0, 0.1$ and 1.0 , applied after the switching point. With $\alpha_u = 0$ the secondary is carried spuriously deep into the helium core of the primary.	44
16	MS + MS merger product	45
17	Blue loop in a Case A merger product	47
18	Post-MS + MS merger product	48
19	Merger product at $q_i = 0.1$	49
20	Case AB merger product	50
21	Post-MS + post-MS merger product	51
22	Blue loops and hot-side ignition across the grid	53
23	Excess helium above the hydrogen-burning shell	54
24	Envelope μ at TAMS: merger product vs. single star	55
25	Time-weighted HR diagram of all merger products	56
26	Surface helium mass fraction	57
27	HR diagram of the BLOeM sample	58
28	CO core mass at helium exhaustion	60
29	Evolution of the merger product of a Case A merger with $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.97$, and $a_i = 13.77 R_{\odot}$	61
30	Mass of the merger product before importing into MESA.	84

31	Evolution of the $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.6$, $a_i = 15.68 R_{\odot}$ Case AB system. The only merger progenitor in our grid whose calculation is terminated by numerical failure rather than by one of the conditions of Sect. 2.1.1. <i>(a)</i> HR diagram with the tracks of the primary (blue) and secondary (orange). Markers indicate the ZAMS and the last available model. <i>(b)</i> Radii R_1 and R_2 (solid) and volume-equivalent Roche-lobe radii $R_{\text{RL},1}$ and $R_{\text{RL},2}$ (dashed) of both components against time. <i>(c)</i> Mass-transfer rate $\dot{M}_{\text{trans}}$ and $\tau_{\text{KH},1} = \frac{M_1}{\tau_{\text{KH}}}$ towards the end of computation. <i>(d) and (e)</i> Kippenhahn plots of the primary and secondary, showing the mass-ratio reversal produced by the first mass-transfer episode.	85
32	Here we present the HR and Kippenhahn diagram evolution for the merger product of the $M_p = 10.16 M_{\odot}$, $q_i = 0.7$, $a_i = 14.98 R_{\odot}$ system with varied “relax” times of 10^{-3} , 10^2 and 10^6 yr.	86

List of Tables

1	Code units adopted in the PHANTOM SPH simulation, with $u_{\text{mass}} = 1 M_{\odot}$ and $u_{\text{dist}} = 1 R_{\odot}$ as the two independent base units.	21
2	Carbon–oxygen core masses at central helium exhaustion for our single-star models.	59

B Additional figures

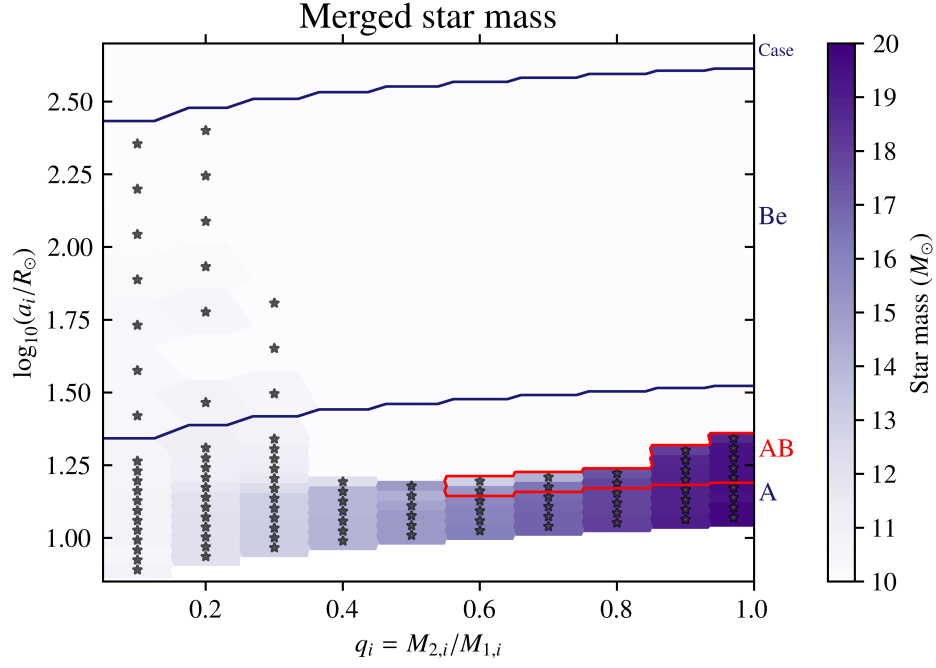


Figure 30: Mass of the merger product before importing into MESA.

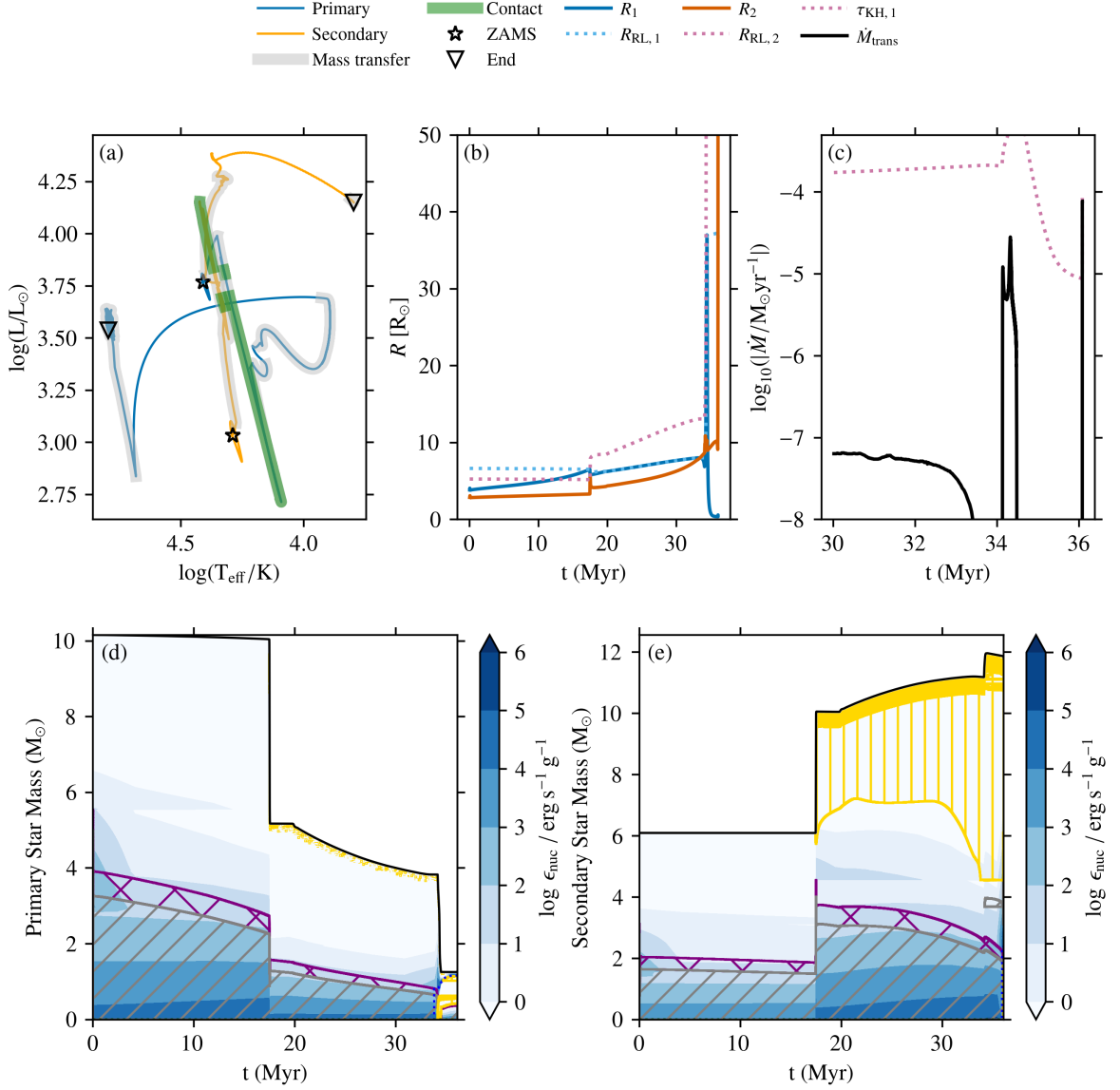


Figure 31: Evolution of the $M_{1,i} = 10.16 M_{\odot}$, $q_i = 0.6$, $a_i = 15.68 R_{\odot}$ Case AB system. The only merger progenitor in our grid whose calculation is terminated by numerical failure rather than by one of the conditions of Sect. 2.1.1. (a) HR diagram with the tracks of the primary (blue) and secondary (orange). Markers indicate the ZAMS and the last available model. (b) Radii R_1 and R_2 (solid) and volume-equivalent Roche-lobe radii $R_{\text{RL},1}$ and $R_{\text{RL},2}$ (dashed) of both components against time. (c) Mass-transfer rate $\dot{M}_{\text{trans}}$ and $\tau_{\text{KH},1} = \frac{M_1}{\tau_{\text{KH}}}$ towards the end of computation. (d) and (e) Kippenhahn plots of the primary and secondary, showing the mass-ratio reversal produced by the first mass-transfer episode.

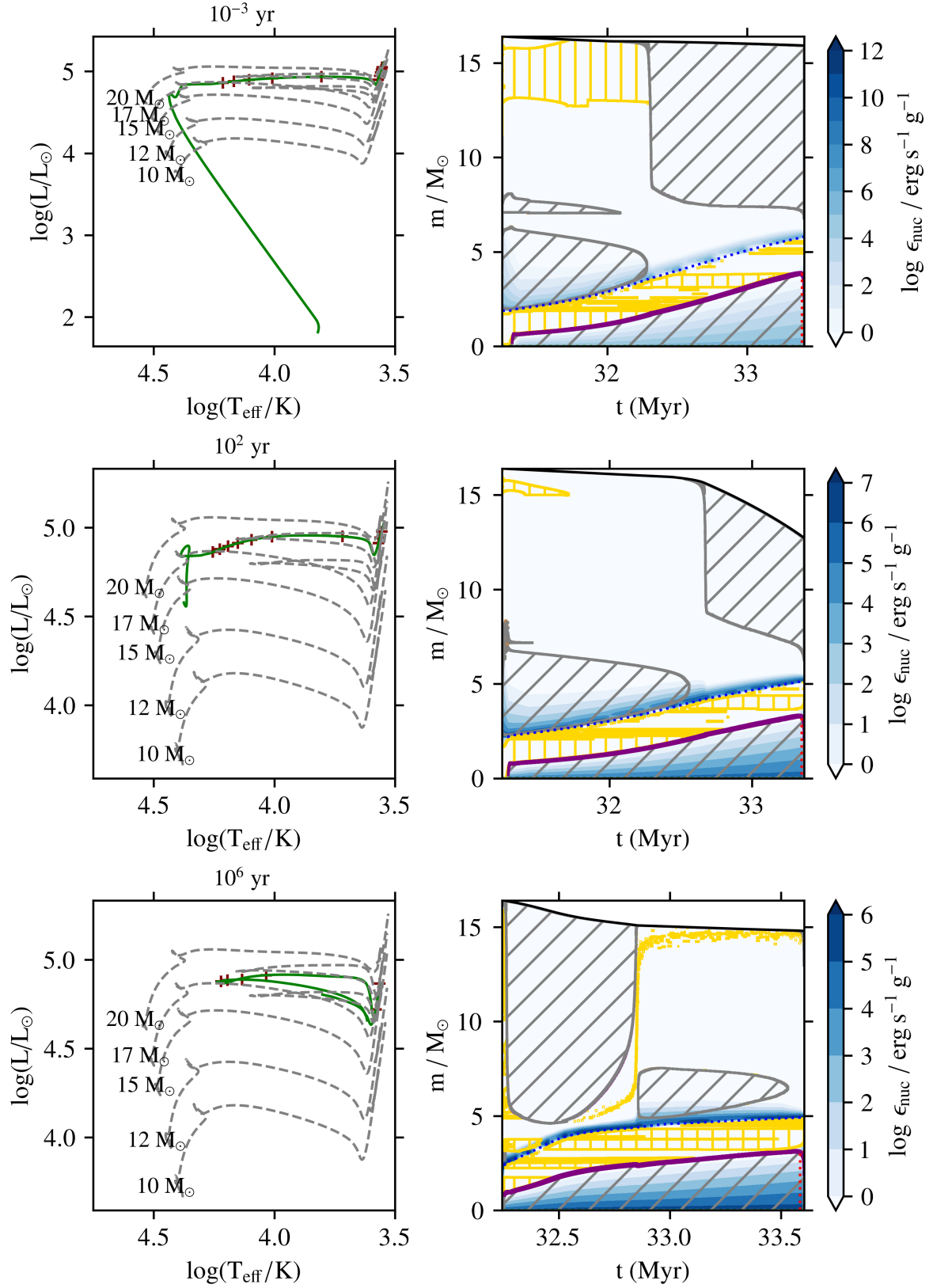


Figure 32: Here we present the HR and Kippenhahn diagram evolution for the merger product of the $M_p = 10.16 M_\odot$, $q_i = 0.7$, $a_i = 14.98 R_\odot$ system with varied "relax" times of 10^{-3} , 10^2 and 10^6 yr.

C MESA inlist

For reproducibility, the MESA Physics inlist used in this work is provided below.

```
&star_job

  ! history_columns_file
  history_columns_file = '../defaults/history_columns.list'

  ! profile_columns_file
  profile_columns_file = '../defaults/profile_columns.list'

  ! opacity tables
  kappa_file_prefix = 'a09'
  kappa_CO_prefix = 'a09_co'
  kappa_lowT_prefix = 'lowT_fa05_a09p'

/ ! end of star_job namelist

&controls
  ! Initial model
  initial_z      = 0.0142
  zams_filename = '.../zams_z0142_y2703.data'

  ! Convection
  use_Ledoux_criterion      = .true.
  mixing_length_alpha       = 2d0
  alpha_semiconvection      = 10d0
  thermohaline_coeff       = 1d0
  remove_small_D_limit      = 1d-10
  mixing_D_limit_for_log    = 1d-10
  num_cells_for_smooth_gradL_composition_term = 0

  ! Overshooting
  overshoot_scheme(1)      = 'step'
  overshoot_zone_type(1)   = 'burn_H'
  overshoot_zone_loc(1)    = 'core'
  overshoot_bdy_loc(1)     = 'top'
  overshoot_f(1)           = 0.21
  overshoot_f0(1)          = 0.01

  overshoot_scheme(2)      = 'step'
```

```

overshoot_zone_type(2) = 'nonburn'
overshoot_zone_loc(2)  = 'shell'
overshoot_bdy_loc(2)   = 'bottom'
overshoot_f(2)         = 0.055
overshoot_f0(2)        = 0.005

overshoot_scheme(3)    = 'exponential'
overshoot_zone_type(3) = 'any'
overshoot_zone_loc(3)  = 'core'
overshoot_bdy_loc(3)   = 'top'
overshoot_f(3)         = 0.01
overshoot_f0(3)        = 0.005

! Rotational mixing
am_nu_ST_factor = 1.0
D_visc_factor   = 0.0
am_nu_SH_factor = 0.0
D_ST_factor     = 0.0
D_SH_factor     = 0.0
D_GSF_factor    = 1.0
D_ES_factor     = 1.0
D_SSI_factor    = 1.0
D_DSI_factor    = 1.0
am_D_mix_factor = 0.0333d0
am_gradmu_factor = 0.1d0

! Critical rotation wind (Marchant et al. 2016)
surf_avg_tau_min      = 1 ! value adopted from Marchant et al. (2016)
surf_avg_tau          = 10 ! value adopted from Marchant et al. (2016)
max_mdot_redo_cnt     = 200 ! value adopted from Marchant et al. (2016)
min_years_dt_for_redo_mdot = 0
surf_w_div_w_crit_limit = 0.97d0
surf_w_div_w_crit_tol  = 0.025d0
mdot_revise_factor    = 1.1d0
implicit_mdot_boost    = 0.1d0

! Rotational enhancement
mdot_omega_power      = 0.43d0
rotational_mdot_boost_fac = 1d5
rotational_mdot_kh_fac  = 0.3
max_mdot_jump_for_rotation = 1d99

```

```

! Opacities
use_Type2_opacities = .true.
Zbase                = 0.0142

!!! NUMERICS

! Mass-loss timestep
delta_mdot_atol = 1d-3
delta_mdot_rtol = 0.5d0

! Nuclear burning timestep (to properly resolve the core H, He, and C
  depletion phases)
delta_lg_XH_cntr_limit      = 0.02d0
delta_lg_XH_cntr_max        = 0.0d0
delta_lg_XH_cntr_min         = -6.0d0
delta_lg_XH_cntr_hard_limit = 0.03d0

delta_lg_XHe_cntr_limit     = 0.02d0
delta_lg_XHe_cntr_max        = 0.0d0
delta_lg_XHe_cntr_min         = -6.0d0
delta_lg_XHe_cntr_hard_limit = 0.03d0

delta_lg_XC_cntr_limit      = 0.02d0
delta_lg_XC_cntr_max        = 0.0d0
delta_lg_XC_cntr_min         = -5.0d0
delta_lg_XC_cntr_hard_limit = 0.03d0

delta_HR_limit      = 0.005d0
do_adjust_J_lost = .true.

limit_for_rel_error_in_energy_conservation      = -1d0
hard_limit_for_rel_error_in_energy_conservation = -1d0
warn_when_large_rel_run_E_err                   = 1d99

! Mesh
mesh_delta_coeff = 1d0
max_dq           = 1d-3
min_dq_for_xa    = 1d-3
max_allowed_nz   = 20000

```

```

! Newton-Raphson solver
redo_limit                = -1
retry_hold                = 5
max_tries_for_retry       = 40
newton_iterations_limit   = 10
use_dedt_form_of_energy_eqn = .false.
newton_itermin_until_reduce_min_corr_coeff = 25
tiny_corr_coeff_limit     = 1000
corr_coeff_limit          = 0.2d0
ignore_too_large_correction = .true.
use_gold_tolerances       = .false.
gold_iter_for_resid_tol3  = 10
gold_tol_residual_norm3   = 1d-6
gold_tol_max_residual3    = 1d-3
tol_max_correction        = 1d-2
tol_correction_norm       = 1d-3
max_corr_jump_limit       = 1d99
max_resid_jump_limit      = 1d99
warn_when_stop_checking_residuals = .false.

! debugging
report_hydro_solver_progress = .false.
report_hydro_dt_info        = .false.
report_ierr                 = .false.
report_why_dt_limits        = .false.

/ ! end of controls namelist

```

References

- M. Asplund, N. Grevesse, A. J. Sauval, and P. Scott. The chemical composition of the sun. *Annual Review of Astron and Astrophys*, 47:481–522, September 2009. doi: 10.1146/annurev.astro.46.060407.145222.
- Earl Patrick Bellinger, Selma E. de Mink, Walter E. van Rossem, and Stephen Justham. The Potential of Asteroseismology to Resolve the Blue Supergiant Problem. *The Astrophysical Journal*, 967:L39, June 2024. ISSN 0004-637X. doi: 10.3847/2041-8213/ad4990.
- M. Bernini-Peron, W. L. F. Marcolino, A. A. C. Sander, J.-C. Bouret, V. Ramachandran, J. Saling, L. M. Oskina, and B. Kubátová. Clumping in Galactic B supergiants: Modelling optical and ultraviolet spectra. *Astronomy and Astrophysics*, 677:A50, 2023. doi: 10.1051/0004-6361/202346469.
- E. Böhm-Vitense. über die wasserstoffkonvektionszone in sternern verschiedener effektivtemperaturen und leuchtkräfte. Mit 5 textabbildungen. *Zeitschrift fuer Astrophysik*, 46:108, 1958.
- H. Braun and N. Langer. Effects of accretion onto massive main sequence stars. *Astronomy and Astrophysics*, 297:483, May 1995.
- J. P. Cox and R. T. Giuli. *Principles of Stellar Structure*, volume 2. Gordon and Breach, New York, 1968.
- Lee Cullen and Walter Dehnen. Inviscid smoothed particle hydrodynamics: Inviscid smoothed particle hydrodynamics. *Monthly Notices of the Royal Astronomical Society*, 408(2):669–683, 2010. ISSN 0035-8711. doi: 10.1111/j.1365-2966.2010.17158.x. URL <http://dx.doi.org/10.1111/j.1365-2966.2010.17158.x>.
- Lisandro Dalcin and Yao-Lung L. Fang. mpi4py: Status Update After 12 Years of Development. *Computing in Science and Engineering*, 23(4):47–54, July 2021. doi: 10.1109/MCSE.2021.3083216.
- A. de Burgos, S. Simón-Díaz, D. J. Lennon, R. Dorda, I. Negueruela, J. Maíz Apellániz, and A. Herrero. IACOB project. VII. the rotational properties of Galactic O and B stars: Blue supergiants. *Astronomy and Astrophysics*, 674:A212, 2023. doi: 10.1051/0004-6361/202346179.
- S. E. de Mink, N. Langer, R. G. Izzard, H. Sana, and A. de Koter. The Rotation Rates of Massive Stars: The Role of Binary Interaction through Tides, Mass Transfer, and Mergers. *The Astrophysical Journal*, 764:166, February 2013. ISSN 0004-637X. doi: 10.1088/0004-637X/764/2/166.

- S. E. de Mink, H. Sana, N. Langer, R. G. Izzard, and F. R. N. Schneider. The incidence of stellar mergers and mass gainers among massive stars. *The Astrophysical Journal*, 782(1):7, 2014. doi: 10.1088/0004-637X/782/1/7.
- P. P. Eggleton. Approximations to the radii of Roche lobes. *Astrophysical Journal*, 268:368, May 1983. doi: 10.1086/160960.
- K. J. Fricke and P. A. Strittmatter. *Monthly Notices of the Royal Astronomical Society*, 156:129, 1972. doi: 10.1093/mnras/156.2.129.
- E. Gaburov, J. C. Lombardi, and S. Portegies Zwart. Mixing in massive stellar mergers. *Monthly Notices of the Royal Astronomical Society*, 383:L5–L9, January 2008. doi: 10.1111/j.1745-3933.2007.00399.x.
- E. Glebbeek and O. R. Pols. Evolution of stellar collision products in open clusters. II. A grid of low-mass collisions. *Astronomy and Astrophysics*, 488:1017–1025, September 2008. doi: 10.1051/0004-6361:200809931.
- E. Glebbeek, O. R. Pols, and J. R. Hurley. Evolution of stellar collision products in open clusters. I. Blue stragglers in N-body models of M 67. *Astronomy and Astrophysics*, 488:1007–1015, September 2008. doi: 10.1051/0004-6361:200809930.
- E. Glebbeek, E. Gaburov, S. Portegies Zwart, and O. R. Pols. Structure and evolution of high-mass stellar mergers. *Monthly Notices of the Royal Astronomical Society*, 434:3497–3510, October 2013. doi: 10.1093/mnras/stt1268.
- Charles R. Harris, K. Jarrod Millman, Stéfan van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with numpy. *CoRR*, abs/2006.10256, 2020. URL <https://arxiv.org/abs/2006.10256>.
- Max Heller, Fabian R. N. Schneider, Jan Henneco, Vincent A. Bronner, and Mike Y. M. Lau. 1d stellar mergers: entropy sorting and pymmams, 2025. URL <https://arxiv.org/abs/2512.14669>.
- J. Henneco, F. R. N. Schneider, S. Hekker, and C. Aerts. Merger seismology: Distinguishing massive merger products from genuine single stars using asteroseismology. *Astronomy and Astrophysics*, 690, October 2024a. ISSN 0004-6361. doi: 10.1051/0004-6361/202450508.

- J. Henneco, F. R. N. Schneider, and E. Laplace. Contact tracing of binary stars: Pathways to stellar mergers. *Astronomy and Astrophysics*, 682(A169):A169, February 2024b. ISSN 0004-6361. doi: 10.1051/0004-6361/202347893.
- Henneco, J., Schneider, F. R. N., Heller, M., Hekker, S., and Aerts, C. Asteroseismic predictions for a massive main-sequence merger product. *A&A*, 698:A49, 2025. doi: 10.1051/0004-6361/202453533. URL <https://doi.org/10.1051/0004-6361/202453533>.
- F. Herwig. The evolution of AGB stars with convective overshoot. *Astronomy and Astrophysics*, 360:952–968, 2000.
- J. G. Hills and C. A. Day. Stellar collisions in globular clusters. *Astrophysics Letters*, 17:87, February 1976.
- R. M. Humphreys and K. Davidson. Studies of luminous stars in nearby galaxies. III - Comments on the evolution of the most massive stars in the Milky Way and the Large Magellanic Cloud. *Astrophysical Journal*, 232:409–420, September 1979. doi: 10.1086/157301.
- R. M. Humphreys and D. B. McElroy. The initial mass function for massive stars in the galaxy and the Magellanic clouds. *Astrophys. J.*, 284:565–577, September 1984. doi: 10.1086/162439.
- John D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi: <http://dx.doi.org/10.1109/MCSE.2007.55>.
- Adam S. Jermyn, Evan B. Bauer, Josiah Schwab, R. Farmer, Warrick H. Ball, Earl P. Bellinger, Aaron Dotter, Meridith Joyce, Pablo Marchant, Joey S. G. Mombarg, William M. Wolf, Tin Long Sunny Wong, Giulia C. Cinquegrana, Eoin Farrell, R. Smolec, Anne Thoul, Matteo Cantiello, Falk Herwig, Odette Toloza, Lars Bildsten, Richard H. D. Townsend, and F. X. Timmes. Modules for Experiments in Stellar Astrophysics (MESA): Time-dependent Convection, Energy Conservation, Automatic Differentiation, and Infrastructure. *The Astrophysical Journal Supplement Series*, 265(1):15, March 2023. ISSN 0067-0049, 1538-4365. doi: 10.3847/1538-4365/acae8d.
- R Kippenhahn, A Weigert, and E Hofmeister. Methods in computational physics. *Methods in Computational Physics*, 7:129, 1967.
- R. Kippenhahn, G. Ruschenplatt, and H.-C. Thomas. The time scale of thermohaline mixing in stars. *Astronomy and Astrophysics*, 91:175–180, 1980.

- Rudolf Kippenhahn, Alfred Weigert, and Achim Weiss. *Stellar Structure and Evolution*. Springer-Verlag Berlin Heidelberg, 2 edition, 2012. doi: 10.1007/978-3-642-30304-3.
- Thomas Kluyver, Benjamin Ragan-Kelley, Fernando Pérez, Brian Granger, Matthias Bussonnier, Jonathan Frederic, Kyle Kelley, Jessica Hamrick, Jason Grout, Sylvain Corlay, Paul Ivanov, Damián Avila, Safia Abdalla, Carol Willing, and Jupyter Development Team. Jupyter notebooks – a publishing format for reproducible computational workflows. In F. Loizides and B. Schmidt, editors, *Positioning and Power in Academic Publishing: Players, Agents and Agendas*, pages 87–90. IOS Press, 2016. doi: 10.3233/978-1-61499-649-1-87.
- U. Kolb and H. Ritter. A comparative study of the evolution of a close binary using a standard and an improved technique for computing mass transfer. *Astron. Astrophys.*, 236:385–392, September 1990.
- N. Langer. Presupernova evolution of massive single and binary stars. *Annual Review of Astronomy and Astrophysics*, 50:107–164, September 2012. ISSN 0066-4146. doi: 10.1146/annurev-astro-081811-125534.
- N. Langer, M. F. El Eid, and K. J. Fricke. Semiconvective diffusion and energy transport. *Astronomy and Astrophysics*, 145:179–184, 1985.
- Mike Y. M. Lau, Ryosuke Hirai, Miguel González-Bolívar, Daniel J. Price, Orsola De Marco, and Ilya Mandel. Common envelopes in massive stars: Towards the role of radiation pressure and recombination energy in ejecting red supergiant envelopes. *Monthly Notices of the Royal Astronomical Society*, 512:5462–5480, June 2022. ISSN 0035-8711. doi: 10.1093/mnras/stac049.
- D. Lauterborn, S. Refsdal, and A. Weigert. Stars with Central Helium Burning and the Occurrence of Loops in the H-R Diagram. *Astron. Astrophys.*, 10:97, January 1971.
- P. Ledoux. Stellar models with convection and with discontinuity of the mean molecular weight. *The Astrophysical Journal*, 105:305, 1947. doi: 10.1086/144905.
- Peter J. T. Leonard and Mario Livio. The rotational rates of blue stragglers produced by physical stellar collisions. *The Astrophysical Journal*, 447(2):L121, jul 1995. doi: 10.1086/309581. URL <https://doi.org/10.1086/309581>.
- J. C. Lombardi, Jr., F. A. Rasio, and S. L. Shapiro. Collisions of main-sequence stars and the formation of blue stragglers in globular clusters. *Astrophysical Journal*, 468: 797, September 1996. doi: 10.1086/177736.

- James C. Lombardi, Jr., Jessica S. Warren, Frederic A. Rasio, Alison Sills, and Aaron R. Warren. Stellar Collisions and the Interior Structure of Blue Stragglers. *The Astrophysical Journal*, 568:939–953, April 2002. ISSN 0004-637X. doi: 10.1086/339060.
- Pablo Marchant. mkip: Kippenhahn diagram plotter for MESA. <https://github.com/orlox/mkip>, 2017.
- Pablo Marchant and Julia Bodensteiner. The Evolution of Massive Binary Stars. *Annual Review of Astronomy and Astrophysics*, 62:21–61, September 2024. ISSN 0066-4146. doi: 10.1146/annurev-astro-052722-105936.
- Pablo Marchant, Norbert Langer, Philipp Podsiadlowski, Thomas M. Tauris, and Takashi J. Moriya. A new route towards merging massive black holes. *Astronomy and Astrophysics*, 588(A50):A50, April 2016. doi: 10.1051/0004-6361/201628133.
- Pablo Marchant, Kaliroë M. W. Pappas, Monica Gallegos-Garcia, Christopher P. L. Berry, Ronald E. Taam, Vicky Kalogera, and Philipp Podsiadlowski. The role of mass transfer and common envelope evolution in the formation of merging binary black holes. *Astronomy and Astrophysics*, 650(A107):A107, June 2021. doi: 10.1051/0004-6361/202039992.
- A. Menon and A. Heger. The quest for blue supergiants: Binary merger models for the evolution of the progenitor of SN 1987A. *Monthly Notices of the Royal Astronomical Society*, 469:4649–4664, August 2017.
- Athira Menon, Andrea Ercolino, Miguel A. Urbaneja, Daniel J. Lennon, Artemio Herero, Ryosuke Hirai, Norbert Langer, Abel Schootemeijer, Emmanouil Chatzopoulos, Juhan Frank, and Sagiv Shiber. Evidence for Evolved Stellar Binary Mergers in Observed B-type Blue Supergiants. *The Astrophysical Journal*, 963(2):L42, March 2024. ISSN 0004-637X. doi: 10.3847/2041-8213/ad2074.
- Maxwell Moe and Rosanne Di Stefano. Mind your ps and qs: The interrelation between period (P) and mass-ratio (Q) distributions of binary stars. *Astrophysical Journal, Supplement*, 230(15):15, June 2017. doi: 10.3847/1538-4365/aa6fb6.
- T. Morris and Ph. Podsiadlowski. The triple-ring nebula around SN 1987A: Fingerprint of a binary merger. *Science*, 315:1103–, February 2007. doi: 10.1126/science.1136351.
- S. S. R. Offner, M. Moe, K. M. Kratter, S. I. Sadavoy, E. L. N. Jensen, and J. J. Tobin. The origin and evolution of multiple star systems. In S. Inutsuka, Y. Aikawa, T. Muto, K. Tomida, and M. Tamura, editors, *Protostars and Planets VII*, volume 534 of *Astronomical Society of the Pacific Conference Series*, page 275, July 2023. doi: 10.48550/arXiv.2203.10066.

- B. Paczyński. Evolutionary processes in close binary systems. *Annual Review of Astronomy and Astrophysics*, 9:183–208, 1971. doi: 10.1146/annurev.aa.09.090171.001151.
- B. Paczyński. Common envelope binaries. In Peter Eggleton, Simon Mitton, and John Whelan, editors, *Structure and Evolution of Close Binary Systems*, volume 73 of *IAU Symposium*, page 75, January 1976.
- B. Paxton, L. Bildsten, A. Dotter, F. Herwig, P. Lesaffre, and F. Timmes. Modules for experiments in stellar astrophysics (MESA). *Astrophysical Journal, Supplement*, 192(3):3, January 2011. doi: 10.1088/0067-0049/192/1/3.
- B. Paxton, M. Cantiello, P. Arras, L. Bildsten, E. F. Brown, A. Dotter, C. Mankovich, M. H. Montgomery, D. Stello, F. X. Timmes, and R. Townsend. Modules for experiments in stellar astrophysics (MESA): Planets, oscillations, rotation, and massive stars. *Astrophysical Journal, Supplement*, 208(4):4, September 2013. doi: 10.1088/0067-0049/208/1/4.
- B. Paxton, P. Marchant, J. Schwab, E. B. Bauer, L. Bildsten, M. Cantiello, L. Dessart, R. Farmer, H. Hu, N. Langer, R. H. D. Townsend, D. M. Townsley, and F. X. Timmes. Modules for experiments in stellar astrophysics (MESA): Binaries, pulsations, and explosions. *Astrophysical Journal, Supplement*, 220(15):15, September 2015. doi: 10.1088/0067-0049/220/1/15.
- B. Paxton, J. Schwab, E. B. Bauer, L. Bildsten, S. Blinnikov, P. Duffell, R. Farmer, J. A. Goldberg, P. Marchant, E. Sorokina, A. Thoul, R. H. D. Townsend, and F. X. Timmes. Modules for experiments in stellar astrophysics (MESA): Convective boundaries, element diffusion, and massive star explosions. *Astrophysical Journal, Supplement*, 234(34):34, February 2018. doi: 10.3847/1538-4365/aaa5a8.
- Bill Paxton, R. Smolec, Josiah Schwab, A. Gautschy, Lars Bildsten, Matteo Cantiello, Aaron Dotter, R. Farmer, Jared A. Goldberg, Adam S. Jermyn, S. M. Kanbur, Pablo Marchant, Anne Thoul, Richard H. D. Townsend, William M. Wolf, Michael Zhang, and F. X. Timmes. Modules for experiments in stellar astrophysics (MESA): Pulsating variable stars, rotation, convective boundaries, and energy conservation. *Astrophysical Journal, Supplement*, 243(10):10, July 2019. doi: 10.3847/1538-4365/ab2241.
- F. I. Pelupessy, A. van Elteren, N. de Vries, S. L. W. McMillan, N. Drost, and S. F. Portegies Zwart. The astrophysical multipurpose software environment. *Astronomy & Astrophysics*, 557:A84, September 2013. ISSN 1432-0746. doi: 10.1051/0004-6361/201321252. URL <http://dx.doi.org/10.1051/0004-6361/201321252>.

- P. Podsiadlowski, P. C. Joss, and S. Rappaport. A merger model for SN 1987 A. *Astronomy and Astrophysics*, 227:L9–L12, January 1990.
- P. Podsiadlowski, P. C. Joss, and J. J. L. Hsu. Presupernova evolution in massive interacting binaries. *Astrophysical Journal*, 391:246–264, May 1992. doi: 10.1086/171341.
- S. Portegies Zwart, S. L. W. McMillan, E. van Elteren, I. Pelupessy, and N. de Vries. Multi-physics simulations using a hierarchical interchangeable software interface. *Computer Physics Communications*, 184(3):456–468, March 2013. doi: 10.1016/j.cpc.2012.09.024.
- S. F. Portegies Zwart, J. Makino, S. L. W. McMillan, and P. Hut. Star cluster ecology III: Runaway collisions in young compact star clusters. *Astronomy and Astrophysics*, 348:117–126, 1999.
- Simon Portegies Zwart and Steve McMillan. *Astrophysical Recipes*. 2514-3433. IOP Publishing, 2018. ISBN 978-0-7503-1320-9. doi: 10.1088/978-0-7503-1320-9. URL <https://doi.org/10.1088/978-0-7503-1320-9>.
- Simon Portegies Zwart, Steve McMillan, Stefan Harfst, Derek Groen, Michiko Fujii, Breannán Ó. Nualláin, Evert Glebbeek, Douglas Heggie, James Lombardi, Piet Hut, Vangelis Angelou, Sambaran Banerjee, Houria Belkus, Tassos Fragos, John Fregeau, Evghenii Gaburov, Rob Izzard, Mario Jurić, Stephen Justham, Andrea Sottoriva, Peter Teuben, Joris van Bever, Ofer Yaron, and Marcel Zemp. A multiphysics and multiscale software environment for modeling astrophysical systems. *New Astronomy*, 14(4):369–378, May 2009. doi: 10.1016/j.newast.2008.10.006.
- Simon Portegies Zwart, Arjen van Elteren, Inti Pelupessy, Steve McMillan, Steven Rieder, Nathan de Vries, Marcell Marosvolgyi, Alfred Whitehead, Joshua Wall, Niels Drost, Lucie Jilkova, Carmen Martinez Barbosa, Edwin van der Helm, Jeroen Beedorf, Patrick Bos, Tjarda Boekholt, Ben van Werkhoven, Thomas Wijnen, Adrian Hamers, Daniel Caputo, Guilherme Ferrari, Silvia Toonen, Evghenii Gaburov, Jan-Pieter Paardekooper, Jurgen Janes, Davide Punzo, Chael Kruip, and Gabriel Al-tay. AMUSE: the Astrophysical Multipurpose Software Environment, 2023. URL <https://doi.org/10.5281/zenodo.8409512>.
- D. J. Price and J. J. Monaghan. An energy-conserving formalism for adaptive gravitational force softening in smoothed particle hydrodynamics and N-body codes. *Monthly Notices of the Royal Astronomical Society*, 374:1347–1358, February 2007. doi: 10.1111/j.1365-2966.2006.11241.x.

- Daniel J. Price, James Wurster, Terrence S. Tricco, Chris Nixon, Stéven Toupin, Alex Pettitt, Conrad Chan, Daniel Mentiplay, Guillaume Laibe, Simon Glover, Clare Dobbs, Rebecca Nealon, David Liptai, Hauke Worpel, Clément Bonnerot, Giovanni Dipierro, Giulia Ballabio, Enrico Ragusa, Christoph Federrath, Roberto Iaconi, Thomas Reichardt, Duncan Forgan, Mark Hutchison, Thomas Constantino, Ben Ayliffe, Kieran Hirsh, and Giuseppe Lodato. Phantom: A smoothed particle hydrodynamics and magnetohydrodynamics code for astrophysics. *Publications of the Astron. Soc. of Australia*, 35(e031):e031, September 2018. doi: 10.1017/pasa.2018.25.
- Friedrich K. Röpke and Orsola De Marco. Simulations of common-envelope evolution in binary stellar systems: Physical models and numerical techniques. *Living Reviews in Computational Astrophysics*, 9(1):2, 2023. doi: 10.1007/s41115-023-00017-x.
- Taeho Ryu, Alison Sills, Ruediger Pakmor, Selma de Mink, and Robert Mathieu. Magnetic Field Amplification during Stellar Collisions between Low-Mass Stars, September 2024.
- H. Sana, S. E. de Mink, A. de Koter, N. Langer, C. J. Evans, M. Gieles, E. Gosset, R. G. Izzard, J.-B. Le Bouquin, and F. R. N. Schneider. Binary interaction dominates the evolution of massive stars. *Science*, 337:444, July 2012. doi: 10.1126/science.1223344.
- F. R. N. Schneider, R. G. Izzard, S. E. de Mink, N. Langer, A. Stolte, A. de Koter, V. V. Gvaramadze, B. Hußmann, A. Liermann, and H. Sana. Ages of young star clusters, massive blue stragglers, and the upper mass limit of stars: Analyzing age-dependent stellar mass functions. *Astrophysical Journal*, 780(117):117, January 2014. doi: 10.1088/0004-637X/780/2/117.
- F. R. N. Schneider, Ph. Podsiadlowski, N. Langer, N. Castro, and L. Fossati. Rejuvenation of stellar mergers and the origin of magnetic fields in massive stars. *Monthly Notices of the Royal Astronomical Society*, 457:2355–2365, April 2016. doi: 10.1093/mnras/stw148.
- F. R. N. Schneider, Ph. Podsiadlowski, and E. Laplace. Pre-supernova evolution and final fate of stellar mergers and accretors of binary mass transfer. *Astronomy and Astrophysics*, 686(A45):A45, June 2024. doi: 10.1051/0004-6361/202347854.
- Fabian R. N. Schneider. Theory, simulations and observations of stellar mergers. *Annual Review of Astronomy and Astrophysics*, 2025. URL <https://arxiv.org/abs/2509.18421>.
- Fabian R. N. Schneider, Sebastian T. Ohlmann, Philipp Podsiadlowski, Friedrich K. Röpke, Steven A. Balbus, Rüdiger Pakmor, and Volker Springel. Stellar mergers as

- the origin of magnetic massive stars. *Nature*, 574(7777):211–214, October 2019. doi: 10.1038/s41586-019-1621-5.
- A. Schootemeijer, N. Langer, N. J. Grin, and C. Wang. Constraining mixing in massive stars in the Small Magellanic Cloud. *Astron. Astrophys.*, 625:A132, May 2019. doi: 10.1051/0004-6361/201935046.
- Martin Schwarzschild. *Structure and Evolution of the Stars*. Princeton University Press, Princeton, 1958.
- T. Shenar, J. Bodensteiner, H. Sana, P. A. Crowther, D. J. Lennon, M. Abdul-Masih, L. A. Almeida, F. Backs, S. R. Berlanas, M. Bernini-Peron, J. M. Bestenlehner, D. M. Bowman, V. A. Bronner, N. Britavskiy, A. de Koter, S. E. de Mink, K. Deshmukh, C. J. Evans, M. Fabry, M. Gieles, and G. Gilkis. 690:A289, 2024. doi: 10.1051/0004-6361/202451586.
- A. Sills, J. C. Lombardi, Jr., C. D. Bailyn, P. Demarque, F. A. Rasio, and S. L. Shapiro. Evolution of stellar collision products in globular clusters. I. Head-on collisions. *Astrophysical Journal*, 487:290, September 1997. doi: 10.1086/304588.
- Pavan Vynatheya, Taeho Ryu, Chen Wang, Alison Sills, and Rüdiger Pakmor. The collision and merger products of stars do not look alike: A magnetohydrodynamics comparison. *The Astrophysical Journal*, 999(1):64, feb 2026. doi: 10.3847/1538-4357/ae40ba. URL <https://doi.org/10.3847/1538-4357/ae40ba>.
- J. J. Walmswell, C. A. Tout, and J. J. Eldridge. On the blue loops of intermediate-mass stars. *Monthly Notices of the Royal Astronomical Society*, 447(3):2951–2960, 2015. doi: 10.1093/mnras/stu2666.
- R. F. Webbink. Double white dwarfs as progenitors of R Coronae Borealis stars and type I supernovae. *The Astrophysical Journal*, 277:355–360, February 1984. ISSN 0004-637X. doi: 10.1086/161701.
- Bill Wolf and Josiah Schwab. wmwolf/py_mesa_reader: Interact with mesa output, July 2017. URL <https://doi.org/10.5281/zenodo.826958>.